



GE HealthCare

Vivid™ Explorer/Vivid™ Advanced/Vivid™ Focus General Service Manual

6007618-1EN Rev. 3

Software v211

General Service Documentation.

©2026 GE HealthCare. GE is a trademark of General Electric Company used under trademark license.

Reproduction and/or distribution is prohibited.



This manual is a reference for the Vivid Explorer/Vivid Advanced/Vivid Focus ultrasound systems (hereafter listed as Vivid Explorer/Vivid Advanced/Vivid Focus). All information provided in this manual is relevant for all systems unless otherwise specified.



GE VINGMED ULTRASOUND AS
Strandpromenaden 45
3191 Horten
Norway
Tel: (+47) 3302 1100
<http://www.gehealthcare.com>

Revision History

Table -1 Revision History

Revision	Date (YYYY-MM-DD)	Reason for change
1	2026-03-11	Initial release
2	2026-05-11	1. Update wheels contents to remove directional wheel 2. Update charge box labels
3	2026-09-09	1. Update power cord pictures 2. Add a warning 'If the system fails to start normally due to battery replacement not being performed in the correct order specified on the label, please disconnect the 220V power supply for no less than 60 seconds and then reconnect it. If the system still cannot start properly, please contact service personnel.' 3. Update contents for ICE RS-DLP adapter 4. Add QR code information for quick card 5. Update windows label location 6. Update description for DICOM/TLS

Please verify that you are using the latest revision of this document. Information pertaining to this document is maintained on MyWorkshop. If you need to know the latest revision, contact your distributor, local GE HealthCare Sales Representative or in the USA call the GE HealthCare Ultrasound Clinical Answer Center at 1 800 682 5327 or 1 262 524 5698.

Language Disclaimers

WARNING

- (English)
(EN)
- This Service Manual is available in English only.
- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
 - Do not attempt to service the equipment unless this Service Manual has been consulted and is understood.
 - Failure to heed this Warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

AVERTISSEMENT

- (Français)
(FR)
- Ce manuel de maintenance est disponible en anglais uniquement.
- Si un client de la personne responsable de la maintenance demande une langue autre que l'anglais, il est de la responsabilité du client de fournir les services de traduction.
 - N'essayez pas d'effectuer vous-même la maintenance de l'équipement avant d'avoir préalablement lu et compris le manuel de maintenance.
 - Le non-respect cet avertissement peut entraîner des blessures dues à un choc électrique, une défaillance mécanique ou à d'autres éléments dangereux chez la personne en charge de la maintenance, l'opérateur ou le patient.

ADVERTENCIA

- (Español)
(ES)
- Este Manual de servicio está disponible en idioma inglés únicamente.
- Si un proveedor de servicio del cliente requiere un idioma distinto, es responsabilidad del cliente ofrecer servicios de traducción.
 - No intente reparar el equipo a menos que haya consultado y comprendido este Manual de servicio.
 - Si no presta atención a esta Advertencia, se pueden ocasionar lesiones al proveedor de servicio, al operador o al paciente por descarga eléctrica, por riesgos mecánicos o de otra índole.

WARNUNG

Deutsch
(DE)

Dieses Wartungshandbuch ist nur auf Englisch verfügbar.

- Wenn der Kundendiensttechniker eines Kunden eine andere Sprache als Englisch benötigt, unterliegt es der Verantwortung des Kunden eine Übersetzung anfertigen zu lassen.
- Warten Sie das Gerät nur, wenn Sie dieses Wartungshandbuch gelesen und verstanden haben.
- Die Nichtbeachtung dieses Warnhinweises kann zu Verletzungen des Kundendiensttechnikers, Anwenders oder Patienten durch Stromschläge, mechanische oder andere Gefahren führen.

AVVERTENZA

italiano
(IT)

Il presente Manuale di assistenza è disponibile solo in inglese.

- Se il fornitore di servizi di un cliente ne richiede una copia in una lingua diversa dall'inglese, è responsabilità del cliente fornire il servizio di traduzione.
- Non tentare di riparare l'apparecchio se questo Manuale di assistenza non è stato letto e compreso.
- Il mancato rispetto di questa avvertenza può comportare il rischio di lesioni al fornitore di servizi, all'operatore o al paziente causate da scosse elettriche o da pericoli di origine meccanica o di altro tipo.

WAARSCHUWING

Nederlands
(NL)

Deze servicehandleiding is alleen beschikbaar in het Engels.

- Als de serviceleverancier van een klant vraagt om een andere taal dan Engels, is het de verantwoordelijkheid van de klant om een vertaalde versie te bieden.
- Probeer geen onderhoud aan de apparatuur uit te voeren tenzij deze servicehandleiding is geraadpleegd en begrepen.
- Het niet opvolgen van deze waarschuwing kan bij de serviceleverancier, de operator of de patiënt leiden tot letsel door elektrische schokken, mechanische of andere gevaren.

ADVERTÊNCIA

Português
(PT-BR)

Este Manual de Manutenção está disponível apenas em Inglês.

- Caso um prestador de serviços do cliente solicite o manual em idioma diferente do inglês, é de responsabilidade do cliente o fornecimento de serviços de tradução.
- Não tente realizar a manutenção do equipamento antes de ler e compreender este Manual de manutenção.
- O não cumprimento desta advertência pode resultar em danos por choque elétrico e riscos mecânicos para o prestador de serviços, operador ou paciente.

VIÐVÖRUN

Íslenska
(IS)

Þessi þjónustuhandbók er aðeins fáanleg á ensku.

- Ef þjónustuaðili viðskiptavinar þarf annað tungumál en ensku er það á ábyrgð viðskiptavinarins að veita þýðingarþjónustu.
- Ekki reyna að þjónusta búnaðinn nema þessir þjónustuhandbækur hafi verið skoðaðir og skilið.
- Ef ekki er gætt að þessari viðvörðun getur það valdið meiðslum á þjónustuaðila, stjórnanda eða sjúklingi vegna raflosts, vélrænni hættu eða annarri hættu.

HOIATUS!

Eesti
(ET)

Service Manual (Hooldusjuhend) on saadaval ainult ingliskeelsena.

- Kui kliendi teenusepakkuja nõue on, et juhend oleks mõnes muus keeles, korraldab juhendi tõlkimise klient.
- Tutvuge enne seadme hooldustööde tegemist kindlasti juhendiga Service Manual (Hooldusjuhend).
- Selle nõude eiramise korral võib teenindaja, kasutaja või patsient saada elektrilöögi, samuti võivad kaasneda muud ohud.

OPOZORILO

Slovenščina
(SL)

Ta servisni priročnik je na voljo samo v angleščini.

- Če ponudnik servisnih storitev za stranko potrebuje navodila v drugem jeziku, mora stranka sama poskrbeti za prevajanje.
- Ne poskušajte servisirati opreme, ne da bi prej prebrali in razumeli servisni priročnik.
- Če tega opozorila ne upoštevate, obstaja nevarnost električnega udara, mehanskih ali drugih nevarnosti in posledičnih poškodb ponudnika servisnih storitev, uporabnika opreme ali pacienta.

警告

日本語
(JA)

- このサービスマニュアルは英語版のみ提供されています。
- お客様の保守担当者が英語以外のマニュアルを必要とされる場合は、お客様の負担にて翻訳サービスをご利用ください。
 - 装置の保守を行う前に、必ずサービスマニュアルを読み、内容を理解してください。
 - この警告に注意を払わない場合、保守担当者やオペレータ、患者に対して、電気ショック、機械またはその他の危険による傷害が発生する恐れがあります。

警告

繁体中文
(ZH-CN)

- 本维修手册仅提供英文版。
- 如果客户需要其它语种版本，请自行翻译。
 - 在维修机器前，请务必阅读并完全理解本维修手册。
 - 若违反本警告，有可能会给维修提供商、操作员或患者带来电击伤害、机械损伤或其它危害。

VARNING

Svenska
(SV)

- Den här servicehandboken finns endast på engelska.
- Om en kunds servicetekniker kräver ett annat språk än engelska är det kundens ansvar att tillhandahålla en översatt version.
 - Försök inte att utföra service på utrustningen om du inte har läst igenom och förstått den här servicehandboken.
 - Om du inte tar hänsyn till den här varningen kan serviceteknikern, operatören eller patienten utsättas för elektriska stötar eller mekaniska eller andra faror, vilket kan leda till personskador.

警告

繁體中文
(ZH-TW)

- 此服務手冊僅推出英文版。
- 若客戶的維修人員需要英文以外的其他語言版本，客戶需自行負責提供翻譯服務。
 - 在詳閱此服務手冊並充分理解其內容之前，請勿試圖開始維修設備。
 - 若忽視此警告，可能導致維修人員、操作人員或病患因為觸電、機械問題或其他危險而受傷。

경고

- 이 서비스 설명서는 영어로만 제공됩니다.
- 고객의 서비스 공급자가 영어 이외의 언어를 요구하는 경우 번역 서비스를 제공할 책임은 고객에게 있습니다.
 - 이 서비스 설명서를 참조 및 이해하지 못한 경우 장비를 만지지 마십시오.
 - 이 경고를 무시한 경우 서비스 공급자, 오퍼레이터 또는 환자가 감전, 기계적 위험 또는 기타 위험으로 인한 부상을 입을 수 있습니다.

ПРЕДУПРЕЖДЕНИЕ

Данное руководство по обслуживанию доступно только на английском языке.

- Если специалисту по техническому обслуживанию клиента требуется документация на каком-либо другом языке, ответственность за выполнение перевода возлагается на клиента.
- Приступайте к обслуживанию оборудования только после того, как изучите данное руководство по обслуживанию и полностью поймете его содержание.
 - Несоблюдение данного требования может привести к травмированию специалиста по техническому обслуживанию, пользователя или пациента вследствие поражения электрическим током, механических и прочих повреждений.

OSTRZEŻENIE

Niniejszy podręcznik serwisowy jest dostępny wyłącznie w języku angielskim.

- Jeżeli dostawca usług klienta posługuje się językiem innym niż angielski, za zapewnienie usług tłumaczeniowych odpowiada klient.
- Przed przystąpieniem do czynności serwisowych należy zapoznać się z informacjami zawartymi w niniejszym podręczniku serwisowym i je zrozumieć.
- W przeciwnym wypadku dostawca usług, operator lub pacjent mogą odnieść obrażenia spowodowane porażeniem prądem elektrycznym, działaniem elementów mechanicznych lub innymi zagrożeniami.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ

Ελληνικά
(EL)

Το παρόν Εγχειρίδιο σέρβις διατίθεται μόνο στα Αγγλικά.

- Εάν ο πάροχος σέρβις του πελάτη απαιτεί γλώσσα εκτός των Αγγλικών, η παροχή μεταφραστικών υπηρεσιών αποτελεί ευθύνη του πελάτη.
- Μην επιχειρήσετε να επισκευάσετε τον εξοπλισμό εάν πρώτα δεν συμβουλευτείτε και κατανοήσετε το παρόν Εγχειρίδιο σέρβις.
- Σε περίπτωση μη τήρησης της παρούσας προειδοποίησης, ενδέχεται να προκληθεί τραυματισμός στον πάροχο σέρβις, το χειριστή ή τον ασθενή εξαιτίας ηλεκτροπληξίας καθώς και μηχανικών ή άλλων κινδύνων.

FIGYELMEZTETÉS

Magyar
(HU)

A szervizkézikönyv kizárólag angol nyelven érhető el.

- Amennyiben az ügyfél szolgáltatójának nem felel meg az angol nyelvű dokumentáció, úgy a fordításról az ügyfélnek kell gondoskodnia.
- Kizárólag úgy lásson hozzá a berendezés karbantartásához, hogy elolvasta és megértette a szervizkézikönyvben foglaltakat.
- Ezen figyelmeztetés figyelmen kívül hagyása esetén a szolgáltató, a kezelő vagy a páciens áramütést, mechanikus sérülést vagy más veszély által okozott személyi sérülést szenvedhet.

VAROVANIE

Slovenčina
(SK)

Táto servisná príručka je dostupná iba v anglickom jazyku.

- Ak poskytovateľ služieb zákazníkovi vyžaduje iný jazyk ako anglický jazyk, jeho povinnosťou je zabezpečiť prekladateľské služby.
- Zariadenie nepoužívajte bez prečítania a porozumenia tejto servisnej príručky.
- Nedodržanie tejto výstrahy môže viesť k zraneniu poskytovateľa služieb, operátora alebo pacienta spôsobeného elektrickým šokom, mechanickým alebo iným nebezpečenstvom.

VÝSTRAHA

česky
(CZ)

Tato servisní příručka je k dispozici pouze v angličtině.

- Pokud poskytovatel služby zákazníkovi požaduje jiný jazyk než angličtinu, je odpovědností zákazníka poskytnout služby překladu.
- Nepokoušejte se provádět servis zařízení, dokud si neprostudujete a neporozumíte servisní příručce.
- Nevěnování pozornosti této výstraze může způsobit poskytovateli služeb, obsluze nebo pacientovi úraz elektrickým proudem, mechanická nebo jiná nebezpečí.

UYARI

Türkçe
(TK)

Servis Kılavuzu yalnızca İngilizce olarak mevcuttur.

- Müşterinin servis sağlayıcısı için kılavuzun İngilizce dışında başka bir dile çevrilmesi gerekiyorsa çeviri hizmeti sağlamak müşterinin sorumluluğudur.
- Bu Servis Kılavuzu'na bakıp talimatları anlamadan ekipmanı kullanmaya çalışmayın.
- Bu Uyarının göz ardı edilmesi servis sağlayıcısının, operatörün veya hastanın, elektrik çarpması, mekanik arıza ya da diğer tehlikeler nedeniyle yaralanmasına neden olabilir.

ADVARSEL

Dansk
(DA)

Denne servicemanual fås kun på engelsk.

- Hvis en kundes tjenesteudbyder kræver et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelsesydelse.
- Forsøg ikke at udføre service på udstyret, medmindre denne servicemanual er læst og forstået.
- Manglende overholdelse af denne advarsel kan medføre skade på serviceudbyderen, operatøren eller patienten som følge af elektrisk stød, mekaniske eller andre farer.

ADVARSEL

Norsk
(NO)

Denne servicehåndboken er bare tilgjengelig på engelsk.

- Hvis en kundes tjenestetilbyder krever et annet språk enn engelsk, er det kundens ansvar å tilby oversettelsestjenester.
- Ikke forsøk å utføre service på utstyret før denne servicehåndboken er lest og forstått.
- Dersom det ikke tas hensyn til denne advarselen, kan det føre til skader på tjenestetilbyderen, operatøren eller pasienten fra elektrisk støt, mekaniske eller andre farer.

VAKAVA VAROITUS

Suomi
(FI)

Tämä huolto-opas on saatavana vain englanniksi.

- Jos asiakkaan palveluntarjoaja tarvitsee oppaan jollain muulla kielellä, käännöspalveluiden hankkiminen on asiakkaan vastuulla.
- Laitetta ei saa huoltaa ellei huolto-oppaaseen ole sitä ennen tutustuttu huolellisesti.
- Jos tätä varoitusta ei noudateta, palveluntarjoaja, käyttäjä tai potilas saattaa saada sähköiskun, ja saattaa aiheutua mekaanisia tai muita vaurioita.

ПРЕДУПРЕЖДЕНИЕ

Български
(BG)

- Настоящото Сервизно ръководство се предлага само на английски език.
- Ако доставчикът на сервизни услуги на клиента изисква ръководство на език, който се различава от английския, клиентът има отговорност да осигури адекватен превод.
 - Не правете опити за сервиз на оборудването, без да проверите и да разберете съветите в Сервизното ръководство.
 - Неспазването на това предупреждение може да доведе до нараняване на доставчика на сервизни услуги, оператора или пациента вследствие на токов удар, механична или други опасности.

AVERTISMENT

Română
(RO)

- Acest manual de service este disponibil doar în engleză.
- Dacă furnizorul de servicii al unui client solicită altă limbă decât engleza, este responsabilitatea clientului să ofere servicii de traducere.
 - Nu încercați să efectuați lucrări de service asupra echipamentului, în afară de cazul când ați consultat acest manual de service și l-ați înțeles.
 - Nerespectarea acestui avertisment poate avea ca rezultat rănirea furnizorului de servicii, a operatorului sau a pacientului ca urmare a electrocutării, pericolelor mecanice sau a altor pericole.

UPOZORENJE

Hrvatski
(HR)

- Ovaj servisni priručnik dostupan je samo na engleskom jeziku.
- Ako klijentov serviser zahtijeva jezik koji nije engleski, odgovornost klijenta je pružiti usluge prijevoda.
 - Nemojte pokušavati servisirati opremu ako niste pročitali i razumjeli servisni priručnik.
 - Ako ne poštujete ovo upozorenje, može doći do ozljede serviser, operatera ili pacijenta prouzročene strujnim udarom, mehaničkim i drugim opasnostima.

ĮSPĖJIMAS

Lietuvių k.
(LT)

- Šis priežiūros vadovas galimas tik anglų kalba.
- Jei kliento paslaugų teikėjas reikalauja kitos kalbos nei anglų, klientas atsako už vertimo paslaugos teikimą.
 - Atlikite įrangos priežiūrą tik gerai susipažinę su priežiūros vadovu ir jį supratę.
 - Nesilaikant šio įspėjimo galimas paslaugos teikėjo, operatoriaus ar paciento sužeidimas dėl elektros šoko, mechaninio ar kito pavojaus.

BRĪDINĀJUMS

- (LV) Latviski
- Šī apkalpes rokasgrāmata ir pieejama tikai angļu valodā.
- Ja klienta pakalpojumu sniedzējam ir nepieciešama cita valoda, kas nav angļu valoda, klienta pienākums ir nodrošināt tulkojumu.
 - Nemēģiniet apkalpot aprīkojumu, ja apkalpes rokasgrāmata nav izlasīta un izprasta.
 - Ja šis brīdinājums netiek ievērots, pakalpojumu sniedzējs, operators vai pacients var gūt traumas no elektrošoka vai var rasties mehānisks vai cita veida apdraudējums.

UPOZORENJE

- (SR) Srpski
- Ovaj priručnik za servisiranje dostupan je samo na engleskom jeziku.
- Ako klijentov serviser zahteva jezik koji nije engleski, odgovornost je na klijentu da pruži usluge prevođenja.
 - Nemojte da pokušavate da servisirate opremu ako prethodno niste pročitali i razumeli ovaj priručnik.
 - Ako ne poštujete ovo upozorenje, može doći do povređivanja serviser, operatera ili pacijenta uzrokovanog električnim udarom, mehaničkim i drugim opasnostima.

AVISO

- (PT-PT) Português (Portugal)
- Este manual de assistência está disponível apenas em inglês.
- Se o prestador de serviços de assistência do cliente necessitar do manual noutra língua, a disponibilização dos serviços de tradução é da responsabilidade do cliente.
 - Não tente reparar o equipamento se não tiver consultado e compreendido este manual de assistência.
 - O não cumprimento das instruções constantes neste aviso pode resultar em ferimentos no prestador de serviços de assistência, no operador ou no paciente devido a choques eléctricos, perigos mecânicos ou outros problemas.

ПОПЕРЕДЖЕННЯ

Українська
(UK)

Цей посібник із технічного обслуговування доступний лише англійською мовою.

- Якщо постачальнику послуг із технічного обслуговування потрібна інформація мовою, відмінною від англійської, відповідальність за надання послуг перекладу несе користувач.
- Технічне обслуговування обладнання можна виконувати лише після ознайомлення з посібником із технічного обслуговування та усвідомлення його змісту.
- Недотримання цього попередження може призвести до травм постачальника послуг, оператора або пацієнта, спричинених дією електричного струму, механічних або інших пошкоджень.

PERINGATAN

Bahasa
Indonesia
(ID)

Panduan Servis ini hanya tersedia dalam Bahasa Inggris.

- Jika penyedia layanan pelanggan memerlukan bahasa di luar Bahasa Inggris, maka pelanggan bertanggung jawab untuk memberikan layanan tersebut.
- Jangan mencoba menyervis peralatan ini, kecuali Panduan Servis ini telah dijadikan rujukan dan dipahami dengan baik.
- Kelalaian memperhatikan Peringatan ini dapat menyebabkan cedera terhadap penyedia layanan, operator, atau pasien akibat bahaya kejutan listrik, mekanik, dan bahaya lainnya.

คำเตือน

ไทย
(TH)

คู่มือซ่อมบำรุงนี้เฉพาะภาษาอังกฤษเท่านั้น

- หากผู้ให้บริการของลูกค้าต้องการจับภาษาอื่นนอกเหนือจากภาษาอังกฤษ ลูกค้าต้องเป็นผู้รับผิดชอบในการจัดเตรียมคู่มือซ่อมบำรุงฉบับแปล
- โปรดอย่าซ่อมอุปกรณ์โดยไม่ศึกษา และทำความเข้าใจคู่มือซ่อมบำรุงนี้
- หากไม่ปฏิบัติตามคำเตือนนี้อาจส่งผลให้ผู้ให้บริการ ผู้ใช้งานอุปกรณ์ หรือผู้ป่วยได้รับบาดเจ็บจากไฟฟ้าช็อต อันตรายจากกลไกของอุปกรณ์ หรืออันตรายอื่น ๆ

CẢNH BÁO

(VI) Tiếng Việt

Hướng dẫn sử dụng dịch vụ này chỉ sẵn dùng bằng tiếng Anh.

- Nếu nhà cung cấp dịch vụ của khách hàng yêu cầu ngôn ngữ khác ngoài tiếng Anh, thì khách hàng phải có trách nhiệm cung cấp các dịch vụ dịch thuật.
- Không được tìm cách sửa chữa thiết bị trừ khi đã tham khảo và hiểu rõ Hướng dẫn sử dụng dịch vụ này.
- Bỏ qua lời cảnh báo này có thể gây thương tích cho nhà cung cấp dịch vụ, nhân viên vận hành hoặc bệnh nhân do sốc điện, những nguy hiểm về máy móc hoặc yếu tố khác.

ЕСКЕРТУ

(KK) Қазақ тілінде

Осы қызмет көрсету нұсқаулығы тек ағылшын тілінде қолжетімді.

- Егер тұтынушылардың қызметтер жеткізушісі ағылшын тілінен басқа тілді талап етсе, аудару қызметтерімен қамтамасыз ету тұтынушының жауапкершілігіне кіреді.
- Осы қызмет көрсету нұсқаулығын түсініп, ол туралы кеңес алмайынша жабдыққа қызмет көрсетуге тырыспаңыз.
- Осы ескертуді орындамау электр тогының соғуы, механикалық немесе басқа да қауіптер салдарынан қызметтер жеткізушісінің, оператордың немесе емделушінің жарақаттануына алып келуі мүмкін.

BABALA

(TL) Tagalog

Available lamang sa Ingles ang Manwal ng Serbisyo ng ito.

- Kung ang kailangan lamang ng tagabigay ng serbisyo ng kustomer ng wika maliban sa Ingles, responsibilidad ng kustomer na magbigay ng serbisyo sa pagsasalin wika nito.
- Huwag subukan na iserbisyo ang mga kasangkapan maliban kung nakonsulta ang nauunawaan itong Manwal ng Serbisyo.
- Ang pagkabigong maunawaan ang Babalang ito ay maaring maging resulta ng pinsala sa tagabigay ng serbisyo, nagpapagana o pasyente mula sa pagkakakoryente, mekanikal o iba pang peligro.

Legal Notes

The contents of this publication may not be copied or duplicated in any form, in whole or in part, without prior written permission of GE HealthCare.

GE HealthCare makes no representations or warranties with respect to the information herein. In addition, the information is subject to change without notice. Every precaution has been taken in the preparation of this document. Nevertheless, GE HealthCare assumes no responsibility for errors, omissions, or any damages, including special or consequential, resulting from the use of this information. GE HealthCare will issue updates to this information periodically, as needed. If there are any questions regarding the information contained in this manual, please contact your GE HealthCare Representative.

Damage in Transportation

Follow this procedure if damage is apparent:

1. Write "Damage in Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE HealthCare representative or hospital receiving agent.
2. Report the damage to the carrier.
 - Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier.
 - A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

Certified Electrical Contractor Statement - For USA Only

All electrical installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations, and testing shall be performed by qualified GE HealthCare personnel. In performing all electrical work on these products, GE HealthCare will use its own specially trained field engineers. All of GE HealthCare's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE HealthCare equipment shall only utilize qualified personnel (i.e. GE HealthCare field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

Omission and Errors

If there are any omissions, errors or suggestions for improving this documentation, contact the GE HealthCare Ultrasound Global Documentation Group with specific information listing the system type, manual title, part number or direction number, revision number, page number and suggestion details.

Mail the information to:

GE Medical Systems (China) Co., Ltd.

Service Documentation

No.19, Changjiang Road,

WuXi National Hi-Tech Dev.Zone,

214028 Jiangsu

P.R. China

TEL: +86 510 85225888

FAX: +86 510 85226688

GE Vingmed Ultrasound AS

Service Documentation

Strandpromenaden 45

3191 Horten

NORWAY

GE HealthCare employees should use Post-Market Quality Management (PQM) to report service documentation issues.

Table of Contents

Chapter 1 - Safety and General Information

1.1 Safety	28
1.1.1 Hazard Definition	28
1.1.2 Safety and Regulatory Procedures	28
1.1.3 Software Configuration Checks	34
1.1.4 Mechanical Function Checks	34
1.1.5 Site Log	34
1.1.6 Contraindications	35
1.1.7 Prescription Device Statement	35
1.1.8 Product and Packaging Information	35
1.1.9 UDI	43
1.1.10 System Manufacturer Name and Address	43
1.1.11 Security Updates	43
1.2 General Information	44
1.2.1 Product Description	44
1.2.2 Principles of operation	44
1.2.3 Model Designations	44
1.3 Scope	45
1.3.1 Typical Users of the Service Documentation	45
1.3.2 Purpose of the Operator Manual(s)	45
1.3.3 System Naming	45
1.4 Illustrations of the Ultrasound System (Front/Rear/Right/Left Convention)	46
1.5 Publication Conventions	48
1.5.1 Conventions Used in this Manual	48
1.5.2 Label and Icon Conventions	48
1.6 Manuals	50
1.6.1 Where to Find the User Manuals	50
1.6.2 Where to Find the Privacy and Security Manual	50
1.6.3 Where to Find the General Service Manual	50
1.6.4 Where to Find the Service Notes	50
1.7 Site Log	51
1.8 Customer Assistance	52
1.8.1 Contact Information	52
1.8.2 Contacting GE HealthCare Ultrasound	52

Chapter 2 - Site Preparations (Pre-Installation)

2.1 Preparing the Site	60
2.1.1 Purchaser Responsibilities	60
2.1.2 Required Facility Needs	60
2.1.3 Desirable Features	61

2.1.4 Minimal Floor Plan Suggestion	61
2.1.5 Recommended Floor Plan Suggestion	62
2.1.6 Suggested Floor Plan with ULS and Workstation in Same Room	63
2.1.7 Seismic	63
2.2 Environmental Requirements	64
2.2.1 Environmental Dangers	64
2.2.2 System Acclimation Time	66
2.2.3 Environmental Specifications	66
2.2.4 Heat Output (Cooling)	66
2.3 Electrical Requirements	67
2.4 Power Requirements	68
2.4.1 Site circuit breaker	68
2.4.2 Site power outlets	68
2.4.3 Mains power cable	68
2.4.4 Lighting	68
2.5 Networking Requirements	69
2.5.1 Stand-alone System (without Network Connection)	69
2.5.2 System Connected to Hospital's Network	69
2.5.3 Purpose of the DICOM Network Function	69
2.5.4 DICOM Option Pre-installation Requirements	69
2.5.5 Connectivity Setup Worksheet	70
2.5.6 Wireless Network Settings	71
2.6 Electromagnetic Interference Survey	73
2.6.1 Electromagnetic Compatibility (EMC)	73
2.6.2 Interference caution	73
2.6.3 Electromagnetic emissions	74
2.6.4 Electromagnetic interferences	75
2.6.5 EMI limitations	75
2.6.6 EMI prevention/abatement	75
2.7 Environmental Requirements for Probes	77
2.8 Time and Manpower Requirements	78
2.9 Environmental protection	79
2.9.1 System and battery disposal	79
2.10 Appendix	80
2.10.1 EMC Compliance	80
2.10.2 How to Access Supplier Manuals for Product Options	80
2.10.3 Option Catalog Numbers	80

Chapter 3 - Tools

3.1 Tools Available via the Service Desktop, InSite etc.	82
--	----

Chapter 4 - System Setup

4.1 System Setup	84
4.1.1 Average Setup Time	84
4.1.2 Installation Warnings	84

4.1.3 Safety Reminders	86
4.1.4 Service Form	97
4.1.5 Mechanical Installation	97
4.2 Upgrades	105
4.3 Installation Instruction for Vivid Explorer Update Kit	106
4.3.1 Tools	106
4.3.2 Manpower	106
4.3.3 Preparations	106
4.3.4 Install the Optional Update Kit	106
4.4 Options	110
4.4.1 Software Options Configuration	110
4.4.2 View-X	110
4.5 Accessories	112
4.5.1 Accessory List	112
4.5.2 Connection of Accessories	112
4.6 Product Locator Installation Card	126
4.6.1 Paperwork	126
4.6.2 Instructions for Use	126

Chapter 5 - Alignment, Setup, Calibration

5.1 Mechanical Alignment Procedures	128
5.1.1 Height Adjustment of the Operator Panel	128
5.1.2 Position Controls for the Main Monitor	128
5.1.3 Castor Controls	129
5.2 Central Listing	130
5.2.1 LED Descriptions	130
5.2.2 Adjustable Components Description	131
5.3 Configuration	133
5.3.1 System Configuration	133
5.3.2 Connectivity tab	135
5.3.3 Security tab	181
5.3.4 System tab	194
5.3.5 About tab	196
5.3.6 Admin tab	197
5.3.7 Service tab	203
5.4 Calibration	209
5.4.1 Monitor and Touch Screen Alignment	209
5.4.2 Touch Screen Calibration	210
5.5 InSite	213
5.5.1 Introduction	213
5.5.2 Request for Service (RFS)	213

Chapter 6 - Functional Checks and Checkout Procedures

6.1 Functional Checks	216
6.1.1 Basic Control Functional Checks	216

6.1.2 Mechanical Functions Checks	216
6.2 Ultrasound Image/Equipment Quality Check (IQC and EQC)	218
6.2.1 Image Quality Check (IQC) preset for service	218

Chapter 7 - Software

7.1 General Information	220
7.1.1 Customer Provided Prerequisites	220
7.1.2 Tools Provided with the System at Delivery or After an Software Installation	220
7.1.3 Space Management - Moving All Images	220
7.1.4 Backing up the Patient Archive and System Configurations	221
7.2 eDelivery - Software update	222
7.2.1 Introduction	222
7.2.2 Software Update via Insite Remote Service Platform (RSvP)	222
7.2.3 Software download and installation	222
7.2.4 Software update via End-User Portal	229
7.2.5 Installing the Software from USB stick	231
7.2.6 Reload Software	234
7.3 Setup after software loading	240
7.3.1 Procedure for Setup After Software Loading	240
7.4 Backup and Restore	241

Chapter 8 - Theory

8.1 General Information	244
8.1.1 Getting to know the Vivid Explorer System	244
8.1.2 Getting to know the Vivid Advanced system	245
8.1.3 Getting to know the Vivid Focus system	246
8.1.4 Significant Features	246
8.1.5 Major System Components	247
8.2 Advanced Security Features	248
8.2.1 Disk Encryption	248
8.2.2 Whitelisting	249
8.2.3 Secure Wipe	249
8.3 Peripherals and/or optional Equipment	250
8.3.1 Printer	250
8.3.2 DVD Drive	250
8.3.3 Foot Switch	250
8.3.4 ECG Module	250
8.3.5 Wireless Network (WLAN) Adapter	250

Chapter 9 - System Diagrams

9.1 System Diagram	252
------------------------------	-----

Chapter 10 - Troubleshooting

10.1 Collect Vital System Information	254
10.1.1 Screen Captures and Logs	254
10.1.2 Needed Data in Case of System Crash or Errors	257
10.1.3 Capturing Network Logs	257
10.2 Service desktop	259
10.2.1 Disruptive mode	259
10.2.2 Color statuses	259
10.2.3 Licenses	259
10.2.4 Home	260
10.2.5 Diags	277
10.2.6 DICOM	282
10.2.7 Utilities	282
10.2.8 Options	299
10.2.9 Configuration Steps	300
10.2.10 Configuration Steps	304
10.3 Request for Service (RFS)	307
10.4 Check Point Voltages	309
10.5 Battery Check	310
10.5.1 Battery safety	310
10.5.2 Power state indications	310
10.5.3 Battery information	311
10.5.4 Battery Scan Mode	311
10.5.5 Shipping and Long-term Storage	312
10.5.6 Battery replacement needed	313
10.5.7 Battery failure is detected	314
10.6 Monitor Troubleshooting	315
10.7 Printer Malfunction	316

Chapter 11 - Parts

11.1 Parts List Groups	318
11.2 Power Cords	320
11.3 CRU parts	323
11.4 Optional Peripherals and Accessories	327
11.4.1 Optional Peripherals and Accessories	331
11.5 System Manuals	332
11.6 Probes	333

Chapter 12 - Parts Replacement

12.1 Service Safety Considerations	342
12.2 Returning/Shipping System, Probes and Repair Parts	343
12.2.1 Repacking the Ultrasound System	343
12.3 Operator Panel Top Cover Rear Replacement	344

12.3.1 Operator Panel Top Cover Rear - Removal Procedure	344
12.3.2 Operator Panel Top Cover Rear - Installation Procedure	344
12.4 Trolley Left Cover Replacement	345
12.4.1 Trolley Left Cover - Removal Procedure	345
12.4.2 Trolley Left Cover - Installation Procedure	345
12.5 Trolley Right Cover Replacement	346
12.5.1 Trolley Right Cover - Removal Procedure	346
12.5.2 Trolley Right Cover - Installation Procedure	346
12.6 Air Filter Replacement	347
12.6.1 Air Filter - Removal Procedure	347
12.6.2 Air Filter - Installation Procedure	347
12.7 Battery Replacement	348
12.7.1 Battery - Removal Procedure	348
12.7.2 Battery - Installation Procedure	349
12.7.3 Functional Checks for Battery Replacement	351
12.8 Charge Box Module Replacement	352
12.8.1 Charge Box Module - Removal Procedure	352
12.8.2 Charge Box Module - Installation Procedure	353
12.8.3 Functional Checks for Charge Box Module Replacement	354
12.9 Trolley Mounted Dual Storage Bin Replacement	355
12.9.1 Trolley Mounted Dual Storage Bin - Removal Procedure	355
12.9.2 Trolley Mounted Dual Storage Bin - Installation Procedure	355
12.10 Trolley Rear Basket Replacement	356
12.10.1 Trolley Rear Basket - Removal Procedure	356
12.10.2 Trolley Rear Basket - Installation Procedure	356
12.11 OPIO Upper Deck Storage Tray	357
12.11.1 OPIO Upper Deck Storage Tray - Installation Procedure	357
12.11.2 Operator Rear Tray - Removal Procedure	357
12.12 Gel Cup Replacement	358
12.12.1 Gel Cup - Removal Procedure	358
12.12.2 Gel Cup - Installation Procedure	358
12.13 Trolley Lower Front Tray Replacement	359
12.13.1 Trolley Lower Front Tray - Removal Procedure	359
12.13.2 Trolley Lower Front Tray - Installation Procedure	359
12.14 AN Keyboard Protective Film Replacement	361
12.14.1 AN Keyboard Protective Film - Installation Procedure	361
12.14.2 AN Keyboard Protective Film - Removal Procedure	362
12.15 Encoder collection Replacement	363
12.15.1 Encoder collection - Removal Procedure	363
12.15.2 Encoder collection - Installation Procedure	363
12.15.3 Functional Checks after Encoder Collection Replacement	366
12.16 Monitor Sticker Replacement	367
12.16.1 Monitor Sticker - Installation Procedure	367
12.17 Nameplate kit Replacement	368
12.17.1 Nameplate Kit - Installation Procedure	368

Chapter 13 - Care and Maintenance

13.1 Introduction to Care and Maintenance	370
13.2 Warnings	371
13.3 Why do Maintenance	372
13.3.1 Preventive Maintenance Inspections	372
13.3.2 Quality Assurance	372
13.4 Maintenance Task Schedule	373
13.4.1 How often should Care and Maintenance Tasks be performed?	373
13.5 Tools Required	375
13.6 System Maintenance	376
13.6.1 Preliminary Checks	376
13.6.2 Functional Checks	376
13.6.3 Physical Inspection	378
13.6.4 Cleaning	378
13.6.5 Probe Maintenance	379
13.6.6 Using a Phantom	380

Chapter 14 - Optional Features

14.1 Optional Features for Vivid Explorer/Vivid Advanced/Vivid Focus	382
--	-----

Chapter 15 - OEM and Other Manuals

15.1 OEM Manuals	384
15.2 Product Manuals for Vivid Explorer/Vivid Advanced/Vivid Focus	385
15.2.1 Product Manuals - Software v211	385
15.2.2 Product Manual Language codes	385

Chapter 16 - Favorite Links

16.1 Documentation Web Site Links	388
16.2 Other links	389

Chapter 17 - Glossary

17.1 Glossary	392
-------------------------	-----

Chapter 1

Safety and General Information

In this chapter

Safety.	28
General Information.	44
Scope.	45
Illustrations of the Ultrasound System (Front/Rear/Right/Left Convention).	46
Publication Conventions.	48
Manuals.	50
Site Log.	51
Customer Assistance.	52

1.1 Safety

1.1.1 Hazard Definition

1.1.1.1 Definition of Safety Messages

Important information will usually be preceded by the exclamation point (!) contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could cause harm. Even if a symbol isn't used in this manual, it is included for your reference.

- **Danger:** the most severe label, indicates a hazard with a high level of risk which, if not avoided, will result in death or serious injury.
- **Warning:** indicates a hazard with a medium level of risk which, if not avoided could result in death or serious injury
- **Caution:** indicates a hazard with a low level of risk which, if not avoided, could result in minor or moderate injury.
- **Notes** are used to provide important information about an item or a procedure.

Be sure to read the notes; the Information contained in a note can often save you time or effort.

1.1.2 Safety and Regulatory Procedures

The following safety precautions must be observed during all phases of operation, service and repair of this equipment. Failure to comply with these precautions or with specific warnings elsewhere in this manual violates safety standards of design, manufacture, and intended use of the equipment.

1.1.2.1 Human Safety

- Operating personnel must not remove the system covers.
- Servicing should be performed by authorized personnel only.
- Only personnel who have participated in the product's Training Seminar are authorized to service the equipment.



DANGER

Do not attempt to disassemble or alter any part of the ultrasound system including the probe, the AC/DC adapter and accessories.

Disassembly or modification may result in electrical shock.

NOTE

Local laws may restrict this device for sale or use by or on the order of a physician.

NOTE

Because of the limited access to cabinets and equipment in the field, placing people in awkward positions, GE HealthCare has limited the lifting weight for one person in the field to 16 KG (35 LBS). Anything over 16 KG (35 LBS) requires two people.

NOTE

Have two people available to deliver and unpack the ultrasound system.

Attempts to move the ultrasound system considerable distances or on an incline by one person could result in injury or damage or both.

NOTE

DO NOT substitute parts or modify equipment.

Because of the danger of introducing additional hazards, ONLY install GE HealthCare approved parts. DO NOT perform any unauthorized modification of the equipment.

NOTE

Tilting the console requires two people in order to avoid injury to service personnel and damage to the equipment.

NOTE

Ensure that the ultrasound system is turned off and unplugged.

Wait for at least 20 seconds for capacitors to discharge as there are no test points to verify isolation. The amber light on the OP panel ON/OFF button will turn off.

Ultrasound system components may be energized. Always refer to the ultrasound system's service manual for LOTO warnings and cautions.

NOTE

Risk of electrical shock, the ultrasound system must be turned off and disconnected from power source. Cord must be controlled at all times.

Wait for at least 20 seconds for capacitors to discharge and the on/off button (white light on the operator panel) to turn off, as there are no test points to verify isolation.

Disconnect the batteries from the backplane.

Beware that the main power supply, the power assist batteries or power module may be energized even if the power is turned off if the cord is still plugged into the AC outlet.

NOTE

Use all Personal Protection Equipment (PPE) such as gloves, safety shoes, safety glasses, and kneeling pads, to reduce the risk of injury.

NOTE

Beware of possible sharp edges on all mechanical parts. If sharp edges are encountered, the appropriate PPE should be used to reduce the risk of injury.

NOTE

Wear all PPE including gloves as indicated in the chemical MSDS.

1.1.2.2 Mechanical Safety

NOTE

Ultrasound probes are highly sensitive medical instruments that can easily be damaged by improper handling. Use care when handling and protect from damage when not in use. Do not use a damaged or defective probe. Failure to follow these precautions can result in serious injury and equipment damage.

NOTE

Never use a probe that has fallen to the floor. Even if it looks OK, it may be damaged, possibly resulting in misdiagnosis.

Before returning the probe to use, perform the following tests:

- Perform safety testing per this service manual or per IEC62353.
- Test that the probe function as intended.

If the probe fails any of the tests, contact your GE HealthCare representative.

NOTE

The ultrasound system should not be moved with the Operator Panel extended. Move the Operator Panel to its centered and locked position. Lower the Operator Panel as much as possible before moving the ultrasound system.

NOTE

When the ultrasound system is raised for a repair or moved along any incline, use extreme caution since it may become unstable and tip over.

NOTE

Before you move or transport the ultrasound system, make sure to lock the Monitor Arm firmly and flip down the monitor to prevent damage to the ultrasound system.

NOTE

To avoid injury when you move the main monitor and the monitor arm, do not put your finger, hand, or object on the joint of the monitor or the monitor arm.

NOTE

Ensure that nobody touches the console arm/frogleg when moving the Operating Panel.

NOTE

Keep the heat venting holes on the monitor unobstructed to avoid overheating.

NOTE

Vivid Explorer/Vivid Advanced/Vivid Focus weighs 90 kg (198 lb.) or more, depending on installed peripherals, when ready for use. Care must be used when moving it or replacing its parts.

Failure to follow the precautions listed below could result in injury, uncontrolled motion and costly damage.

- Be sure the pathway is clear.
- Use slow, careful motions.
- Use two people when moving on inclines or lifting more than 16 kg (35 lbs).

NOTE

Do not transport Vivid Explorer/Vivid Advanced/Vivid Focus in a vehicle without locking the casters (wheels) and securing it.

NOTE

Special care should be taken when transporting the ultrasound system in a vehicle:

- Before transporting, place the ultrasound system in its special storage case.
- Ensure that the ultrasound system is firmly secured while inside the vehicle.
- Secure ultrasound system with straps or as directed otherwise to prevent motion during transport.
- Prevent vibration damage by driving cautiously. Avoid unpaved roads, excessive speeds, and erratic stops or starts.

NOTE

Use protective glasses during drilling, filing smooth surfaces, and during all other work where eyes need protection.

NOTE

Use safety shoes when doing work where there is any chance of foot injury.

NOTE

Use protective gloves when working with sharp edges or when directed to wear PPE during a removal/ replacement procedure.

NOTE

Be careful not to pinch any of the cables.

1.1.2.3 Electrical Safety



DANGER - ELECTRICAL HAZARD

Do not attempt to disassemble or alter any part of the ultrasound system including the probe, the AC/DC adapter and accessories.

Disassembly or modification may result in electrical shock.

1.1.2.4 Safe practices

Follow these guidelines to minimize shock hazards whenever you are using the ultrasound system:

- To minimize shock hazard, the equipment chassis must be connected to an electrical ground.
- The ultrasound system is equipped with a three-conductor AC power cable. This must be plugged into an approved electrical outlet with safety ground.
- The power outlet used for this equipment should not be shared with other types of equipment.
- Both the ultrasound system power cable and the power connector must meet international electrical standards.



WARNING

Connecting the ultrasound system to the wrong voltage level will most likely destroy it.

1.1.2.5 Probes

Follow these guidelines before connecting a probe to the ultrasound system:

- Inspect the probe prior to each use for damage or degradation to the:
 - housing
 - cable strain relief
 - lens
 - seal
 - connector pins
 - locking mechanism
- Do not use a damaged or defective probe.
- Never immerse the probe connector or adapter into any liquid.
- The Vivid Explorer/Vivid Advanced/Vivid Focus has more than one type of probe port. Use the appropriate probe port designed for the probe you are connecting.

1.1.2.6 Peripherals

Please see: [Environmental Dangers](#) on page 64.

1.1.2.7 Battery Safety

This section applies for Vivid Explorer/Vivid Advanced/Vivid Focus ultrasound systems with batteries. The batteries are based on lithium-ion chemistry.



DANGER

Do not disassemble or alter the Batteries.

Only use Batteries approved by GE HealthCare as suitable for use with the ultrasound system.

Replace all batteries at the same time to avoid mix-up with used and new battery packs.

Remove the batteries before performing service.

NOTE

Replace the batteries every two years, or earlier if the system informs you that the battery needs replacement.

NOTE

If equipped with Batteries, the ultrasound system shall be set to Shipping / Storage charging mode and checked for storage readiness prior to its long-term storage.

NOTE

If equipped with Batteries, the ultrasound system shall be set to Shipping / Storage charging mode and checked for shipping readiness prior to its transportation from one premise to another.

NOTE

If you have any questions about the Batteries, consult your local GE HealthCare representative.

NOTE

Used batteries should not be placed with common household waste products. Contact local authorities for the location of a chemical waste collection program nearest you.

Regulations vary for different countries. Dispose of used batteries in accordance with local regulations.

Store the Battery packs between -10 °C (14 °F) and 50 °C (122 °F).

NOTE

The battery shall be removed from the system when in storage for more than one year.



WARNING

If the system is not being used on a monthly basis, the battery needs to be removed during the lengthy non-use period.



WARNING

Keep batteries out of the reach of children.



CAUTION

Do not remove battery from its original packaging until required for use.



WARNING

In the event of a cell in battery leaking, do not allow the liquid to come in contact with the skin or eyes. If contact has been made, wash the affected area with copious amounts of water and seek medical advice.



CAUTION

Always keep battery clean and dry.



CAUTION

Wipe battery terminals with a clean dry cloth if they become dirty.



CAUTION

After extended periods of storage, it may be necessary to charge and discharge the battery several times to obtain maximum performance.



CAUTION

Use the battery only in the application for which it was intended.



CAUTION

Observe the rotation of the battery and host to ensure correct assembly.



WARNING

Once end of service life notification is received, all rechargeable batteries shall be removed within 3 years and no new battery shall be installed in the device. Since there is no longer availability for GEHC authorized batteries, all batteries placed in the system are not in compliance with this system after this time frame.



WARNING

When replacing a battery, all batteries must be replaced at the same time.



WARNING

Do not refurbish / rejuvenate the batteries.

**WARNING**

Do not mix battery types.

**WARNING**

If batteries are leaking they need to be replaced.

1.1.2.8 Auxiliary Devices Safety

Commercial devices such as laser cameras, printers, VCRs and external monitors, usually exceed allowable leakage current limits and, when plugged into separate AC outlets, are in violation of patient safety standards. Suitable electrical isolation of such external AC outlets, or providing the device with extra protective earth, will be required in order to meet IEC/EN 60601-1 Clause 16 standards for electrical leakage.

Patient Environment IEC/EN 60601-1

Sub Clause 3.79 and figure A.9 from IEC/EN 60601-1

It is difficult for this standard to define dimensions for the volume in which diagnosis, monitoring or treatment occurs. The dimensions for the PATIENT ENVIRONMENT given in the figure below have been justified in practice.

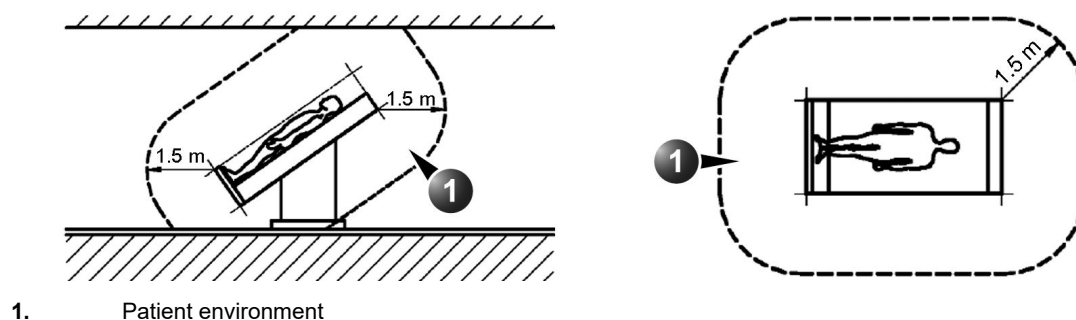


Figure 1-1 Patient environment

1.1.2.9 Functional Hazards

There are no known functional hazards.

1.1.2.10 Lockout/Tagout (LOTO) Requirements

Read and understand the lockout/tagout (LOTO) requirements before performing any replacement procedures. Keep these instructions for future reference. Follow all of the safety measures and procedures listed in this service manual. Lockout/tagout requirements ensure you are in total control of the AC power plug at all times during the service process. For step-by-step instructions to lock out and tag out the Vivid Explorer/Vivid Advanced/Vivid Focus, see:

- [Complete Power Down](#) on page 92

1.1.2.11 Cybersecurity Precautions

**CAUTION**

To ensure protection against cybersecurity threats, do not alter Windows security settings such as security policies, registry settings, firewall configuration or BIOS/UEFI settings on the ultrasound system.

It is essential to comply with relevant data privacy laws, including but not limited to HIPAA, GDPR, and PIPEDA, to ensure the protection of sensitive information and maintain data security.

1.1.2.12 List of chemicals associated with servicing and installing

There are currently no chemicals associated with servicing and installing.

1.1.3 Software Configuration Checks

Verify that all the options that the customer has ordered are active and that they are working.

1.1.4 Mechanical Function Checks

1.1.4.1 Control Console Positioning

The control console can be rotated and adjusted in height.

- height adjustment: 34 cm (13.39 inch)
- rotation adjustment: +/- 45°

1.1.4.2 Wheels

The ultrasound system is driven by four free-swivel wheels, controlled by black brake pedals situated above the wheel.



1. Locked Wheels
2. Unlocked Wheels

Figure 1-2 Free-Swivel wheel

Examine the wheels frequently for defects to avoid breaking or jamming.

Do not move the system when the brakes are engaged.



CAUTION

Always place the system on horizontal ground and engage the caster brakes.

Always ensure complete brake engagement on all 4 casters by firmly stepping on the brake pedals of the casters.

Failure to follow these precautions could result in injury, uncontrolled motion and damage.

1.1.5 Site Log

The Site Log is available at the end of this chapter. See: [Site Log](#) on page 51.

1.1.6 Contraindications

The Vivid Explorer/Vivid Advanced/Vivid Focus is not intended for ophthalmic use or any use causing the acoustic beam to pass through the eye.

NOTE

Vivid Explorer/Vivid Advanced/Vivid Focus should be used in compliance with law. Some jurisdictions restrict certain uses, such as gender determination.

1.1.7 Prescription Device Statement

NOTE

For USA only: Federal law restricts this device to sale by or on the order of a physician.

1.1.8 Product and Packaging Information

1.1.8.1 Tilt Watch Indicators

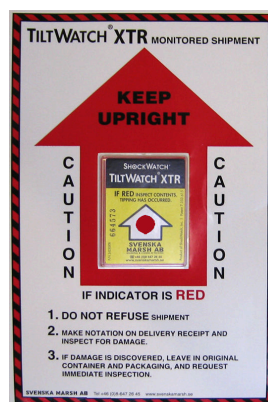
Improper handling during transportation may harm the equipment inside the package even if the package itself is undamaged.

To make it easier to detect improper handling during transportation, a set of Tilt indicators have been attached to the transportation box.

The purpose of the tilt indicator label is to alert package handlers that the package content is sensitive to tipping and that it must remain upright at all times. It functions as an active “Up Arrow” that will change color if the package is tilted 89 degrees or more from its upright position. On rare occasions these labels have been falsely activated if the package was tipped less than 89 degrees, but shocked or vibrated at the same time. This is considered uncommon.



OK (Not Triggered)



Triggered

Figure 1-3 Tilt Indicator

Related reference

[If Tilt Indicator has Triggered or is Missing](#) on page 98

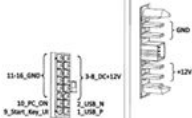
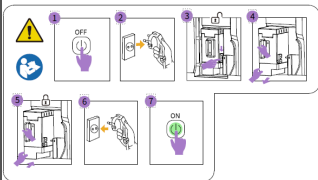
[Position of the Tilt Indicators](#) on page 98

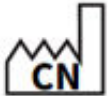
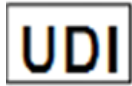
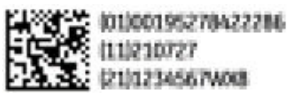
1.1.8.2 Symbols and Labels

1.1.8.3 Symbols

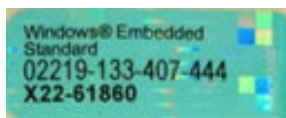
Label	Purpose	Location	Standard
Identification Plate	Manufacturer's name and address. Model. Device Listing/Certification Labels.	Rear of the system	N/A- by GE HealthCare
	GE HealthCare Logo	Trolley OPIO	N/A- by GE HealthCare (GE is a trademark of General Electric Company used under trademark license.)
	A mark to guide the operator panel up/down and swivel direction	Front Handle	N/A- by GE HealthCare
	USB-C port	OPIO	N/A- by GE HealthCare
	Standard USB mark	OPIO Trolley	N/A- by GE HealthCare
	HDMI mark	Trolley	N/A- by GE HealthCare
	Mark for network connection	Trolley	N/A- by GE HealthCare
	RS type probe mark	Trolley for Vivid Focus	N/A- by GE HealthCare
	Mark of Lock and unlock to indicate the probe status	Trolley for Vivid Focus	N/A- by GE HealthCare
Vivid	"Vivid" Logo.	Trolley	N/A- by GE HealthCare
	On/off button Warning: System shutdown using the On/Off button does not disconnect the ultrasound system from mains voltage. For disconnecting the ultrasound system from mains voltage after system shutdown, please set the circuit breaker close to the mains inlet to OFF.	OPIO	IEC 60601-1; IEC 60417-5009

Label	Purpose	Location	Standard
	Defibrillator-proof Type CF equipment.	Trolley	IEC 60601-1; IEC 60417-5336
	Equipment Type Body Floating (BF), in which protection against electric shock does not rely on basic insulation only. Provides additional safety precautions such as double insulation or reinforced insulation, because there is no provision for protective earthing or reliance upon installation conditions.	Trolley for Vivid Explorer and Vivid Advanced Accessory Label	IEC 60417-5333
	Equipment Type CF indicates equipment containing a floating applied part with a degree of protection suitable for direct cardiac contact.	Accessory Label	IEC 60417-5335
	Max. loading Symbol	Basket and Tray	N/A- by GE HealthCare
	Avoid crushing of hands.	Trolley Fix Arm	ISO 7010-W025
	CAUTION - Do not push the system sideways when casters are in brake position. Instability may occur.	Monitor	IEC60601-1, ISO 7010-P017
	Main monitor and operation panel lock label: main monitor and operation panel shall be locked in transport position during transportation.	Monitor	N/A- by GE HealthCare
	A symbol for warning do not put heavy objects on the monitor	Monitor	N/A- by GE HealthCare
	A symbol for warning do not press the monitor	Monitor	N/A- by GE HealthCare
	A symbol for pinch warning	Monitor	N/A- by GE HealthCare

Label	Purpose	Location	Standard
	Pinch point Watch your hands and fingers when adjusting the monitor. Keep hands clear of openings.	Flexible Arm	GE HealthCare crafted symbol based on IEC60601-1; ISO7010-P001 and ISO 3864-1.
	The power on and off positions of the mains switch	AC-Box	IEC 60417-5007 and 60417-5008
	Equipotentiality: Indicates terminal to be used for connecting equipotential conductors when interconnecting (grounding) with other equipment as described in IEC/EN60601-1.	AC-Box	IEC 60601-1, IEC 60417-5021
INPUT: 100-240V~, 50/60Hz, 600VA	The power supply information includes Supply Voltage, Frequency, Power Consumption	AC-Box	N/A- by GE HealthCare
FUSE: T10AL, 250VAC	The type and full rating of the fuse (voltage, current, operating speed and breaking capacity)	AC-Box	N/A- by GE HealthCare
	A label to describe the pin define of connectors to E-cage and charge box for FE quick diagnosis.	AC-Box	N/A- by GE HealthCare
	A label to guide users how to assemble and disassemble charge box: (1) Press on/off button on the ultrasound system to turn off the system. (2) Disconnect the power source to the ultrasound system. (3)~(4) Use one hand to press down the lock lever of the rotary push-to-close latch while at the same time use the other hand to pull out the charge box. (5) Carefully push battery box into main frame with both hands. When the Charge Box is fully installed on the system, the latch will automatically lock. (6) Connect the power source to the ultrasound system. (7) Press on/off button on the ultrasound system to turn on the system.	Charge Box	N/A- by GE HealthCare
	A warning label to warn people not to stretch hand into the dock.	Power Box	N/A- by GE HealthCare
	Device part number identifier.	Rating plate	N/A- by GE HealthCare

Label	Purpose	Location	Standard
	Manufacturer name and address.	Rating Plate Accessory Label	EN ISO 15223-1, ISO 7000-3
	Date of manufacture: The date could be a year, year and month, or year, month and day, as appropriate.	Accessory Label	EN ISO 15223-1, ISO 7000-2497
 Or 	This symbol represents two meanings: (1)Date of manufacture: The date could be a year, year and month, or year, month and day, as appropriate. (2)Country of manufacture: CN represents China is the country of manufacture; AT represents Austria is the country of manufacture.	Rating Plate	EN ISO 15223-1, IEC 60417-6049
	Device Catalog Number.	Rating Plate Accessory Label	EN ISO 15223-1, ISO 7000-2493
	Serial number. NOTE Device serial number is available both in the device label and in the SW application, and shall always be checked both places when needed.	Rating Plate Accessory Label	EN ISO 15223-1, ISO 7000-2498
	Symbol indicates the item is a medical device.	Rating Plate Accessory Label	EN ISO 15223-1
	Unique Device Identification (UDI). Every system has a unique marking for identification. Scan or enter the UDI information into the patient health record as required by governing laws.	Rating Plate Accessory Label	EN ISO 15223-1
	Every system has a unique marking for identification, the Unique Device Identification (UDI) Label. The UDI label consists of a series of alpha-numeric characters and barcode which uniquely identify the ultrasound system as a medical device manufactured by General Electric. Scan or enter the UDI information into the patient health record as required by country-specific laws.	Rating Plate Accessory Label	EN ISO 15223-1(only for UDI symbol)
	This symbol indicates the authorised representative in Switzerland.	Rating Plate	ISO15223-1
	Max Weight: xx kg with all hardware and mechanical configurations connected.	Rating Plate	EN 60601-1:2006/ A1:2013 (IEC60601-1:2005/ A1:2012)
100-240V~,50/60Hz,600VA	Power supply information	Rating Plate	

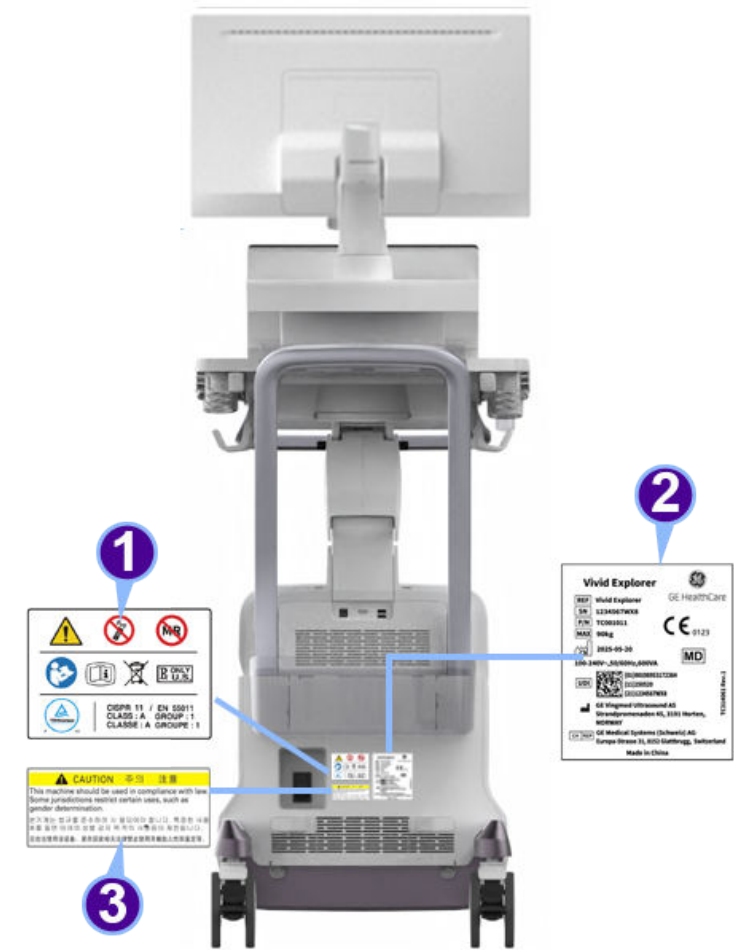
Label	Purpose	Location	Standard
	Indicates that the product is in compliance with all relevant European Directives and Regulations, and under surveillance by Notified Body 0123.	Rating Plate	REGULATION (EU) 2017/745, REGULATION (EC) No 765/2008
	The CE Mark of Conformity indicates this equipment conforms with the applicable directives/regulations for EU market.	Accessory label	REGULATION (EU) 2017/745, REGULATION (EC) No 765/2008
Made in China, Assembled in Austria	Country of Original information	Rating Plate	N/A- by certification body
	Follow instructions for use. Read and understand all instructions in the User's Manual before attempting to use the ultrasound system.	Safety Label	IEC 60601-1, ISO 7010-M002
	Warning Indicates a hazard with a medium level of risk which, if not avoided, could result in death or serious injury. Caution Indicates a hazard with a low level of risk which, if not avoided, could result in minor or moderate injury.	Safety Label	IEC 60601-1; ISO 7010 -W001
	Caution Consult accompanying documents is intended to alert the user to refer to the operator manual or other instructions when complete information cannot be provided on the label.	Probe	ISO 7000-0434A
	WARNING - The system is MR Unsafe and should remain outside the MRI scanner room.	Safety Label	ASTM F2503-20
	Do not use the following devices near this equipment: cellular phone, radio receiver, mobile radio transmitter, radio controlled toy, etc. Use of these devices near this equipment could cause this equipment to perform outside the published specifications. Keep power to these devices turned off when near this equipment.	Safety Label	ISO 7010-P013
	Symbol indicating that the Instructions for Use are supplied in electronic form	Safety Label	ISO 7000-3500

Label	Purpose	Location	Standard
	This symbol indicates that the waste of electrical and electronic equipment must not be disposed as unsorted municipal waste and must be collected separately. Please contact the manufacturer or other authorized disposal company to decommission your equipment.	Safety Label Accessory Label	EN 50419
	For USA only: Caution: Federal law restricts this device to sale by or on the order of a physician.	Safety Label	USA Regulation: 21 CFR 801.109
CISPR 11 / EN 55011 CLASS : A GROUP : 1 CLASSE : A GROUPE : 1	EMC class level label: CISPR CAUTION: The ultrasound system conforms to the CISPR11, Group 1, Class A of the international standard for Electromagnetic disturbance characteristics.	Safety Label	CISPR 11/EN 55011
	NRTL Listing and Certification Mark is used to designate conformance to nationally recognized product safety standards. The Mark bears the name and/or logo of the testing laboratory, product category, safety standard to which conformity is assessed and a control number.	Safety Label	NRTL Listing and Certification Mark
	Windows sticker	Inside of the trolley left cover	N/A- by GE HealthCare
	Charge box label	Charge box	N/A- by GE HealthCare
IPX7	Protected against water immersion - Immersion for 30 minutes at a depth of 1 meter.	Applied parts of probes	IEC 60529
IPX8	Protected against water submersion - The equipment is suitable for continual submersion in water under conditions which are identified by the manufacturer.	Applied parts of probes	IEC 60529
XDclear™	Product Marketing indicator information.	Probes	N/A- by GE HealthCare

Label	Purpose	Location	Standard
	This is the QR code label which users can scan to download the Vivid Explorer/Vivid Advanced/Vivid Focus Quick Card.	Probe Holder	N/A- by GE HealthCare

1.1.8.4 Label Location

The following figure describes the location of labels attached to the rear cover of the system.



- 1. Safety Label
- 2. Rating Plate includes MAX. weight label
- 3. Gender Caution Label (For China Only)

Figure 1-4 Label Location on the Rear Cover (Take Vivid Explorer as an example)

NOTE
The label content may be different for different regions and systems. Please refer to the labels on the system for the actual content.

NOTE

There is an additional product label for specific country located on the right side of the rating label and not shown in the above figure, and its content may vary by country.

The Windows sticker is pasted on the front pillar of the main frame inside the left side cover of the trolley.



Figure 1-5 Windows Label Location

1.1.9 UDI

The UDI information is printed on a label on the rear of the ultrasound system.

1.1.10 System Manufacturer Name and Address



GE Vingmed Ultrasound AS
Strandpromenaden 45
3191 Horten, Norway
Tel.: (+47) 3302 1100

1.1.11 Security Updates

There are no security updates for the Vivid Explorer/Vivid Advanced/Vivid Focus.

1.2 General Information

1.2.1 Product Description

The Vivid Explorer/Vivid Advanced/Vivid Focus ultrasound system contains three cardiovascular ultrasound products Vivid Explorer, Vivid Advanced and Vivid Focus. It is intended for many diagnostic Ultrasound applications for both adult and pediatric patients. It offers innovative ergonomic design, superb image quality, advanced connectivity, productivity tools and advanced technology. The devices are especially indicated for cardiac and vascular diagnostic ultrasound applications. The ultrasound systems produce cross-sectional images of the body by computer reconstruction of pulse-echo data. The device may include signal analysis and display equipment, software and hardware for data and image processing and analysis, hand-held probes, and other accessories. The device output is a valuable tool to assist with medical diagnosis of diseases, identification of indeterminate abnormalities and/or support for follow-up imaging.

1.2.2 Principles of operation

Medical ultrasound images are created by computer and digital memory from the transmission and reception of mechanical high-frequency waves applied through a transducer. The mechanical ultrasound waves spread through the body, producing an echo where density changes occur. For example, in the case of human tissue, an echo is created where a signal passes from an adipose tissue (fat) region to a muscular tissue region. The echoes return to the transducer where they are converted back into electrical signals.

These echo signals are highly amplified and processed by several analog and digital circuits having filters with many frequency and time response options, transforming the high-frequency electrical signals into a series of digital image signals which are stored in memory. Once in memory, the image can be displayed in real-time on the image monitor. All signal transmission, reception and processing characteristics are controlled by the main computer. By selection from the system control panel, the operator can alter the characteristics and features of the system, allowing a wide range of uses, from obstetrics to peripheral vascular examinations.

Transducers are accurate, solid-state devices, providing multiple image formats. The digital design and use of solid-state components provides highly stable and consistent imaging performance with minimal required maintenance. Sophisticated design with computer control offers a system with extensive features and functions which is user-friendly and easy to use.

1.2.3 Model Designations

List of ultrasound models covered by this manual.

Table 1-1 Software v211 Ultrasound System Model and Software Compatibility

Model Number	Description	System SW	Appl. SW	Can be upgraded to:
TC001011	Vivid Explorer	v211.xx	v211.xx	v211.xx
TC001012	Vivid Advanced	v211.xx	v211.xx	v211.xx
TC001013	Vivid Focus	v211.xx	v211.xx	v211.xx

1.3 Scope

The information needed to service the the Vivid Explorer/Vivid Advanced/Vivid Focus series is collected in this Service Manual.

This Service Manual applies to all models unless otherwise specified.

1.3.1 Typical Users of the Service Documentation

Typical users of the General Service Manual

- service personnel (setup, maintenance, etc.)
- hospital's service personnel
- architectural planners/installation planners (some parts of the Site Preparations chapter.)

1.3.2 Purpose of the Operator Manual(s)

The operator manuals should be fully read and understood before operating the ultrasound system. The online versions of the operator manuals are available via the Help function on Vivid Explorer/Vivid Advanced/Vivid Focus's Operating Panel.

The User Manuals must be installed as part of the set-up (at the customer).

All the user manuals are available as PDF files on a separate Manual Media Usb Flash Drive (UFD), delivered with the ultrasound system.

1.3.3 System Naming

Vivid Explorer/Vivid Advanced/Vivid Focus is available in these configurations:

- Vivid Explorer Dimension
- Vivid Explorer
- Vivid Explorer 2D
- Vivid Advanced
- Vivid Focus

1.4 Illustrations of the Ultrasound System (Front/Rear/Right/Left Convention)

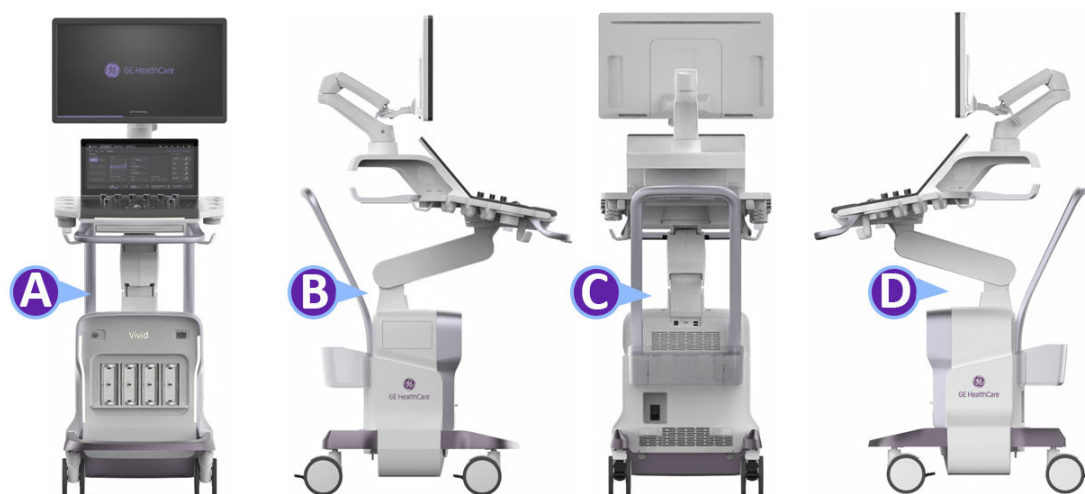


Figure 1-6 Vivid Explorer Console Views of FRONT Side (A) - LEFT (B) - BACK/REAR (C) - RIGHT Side (D)

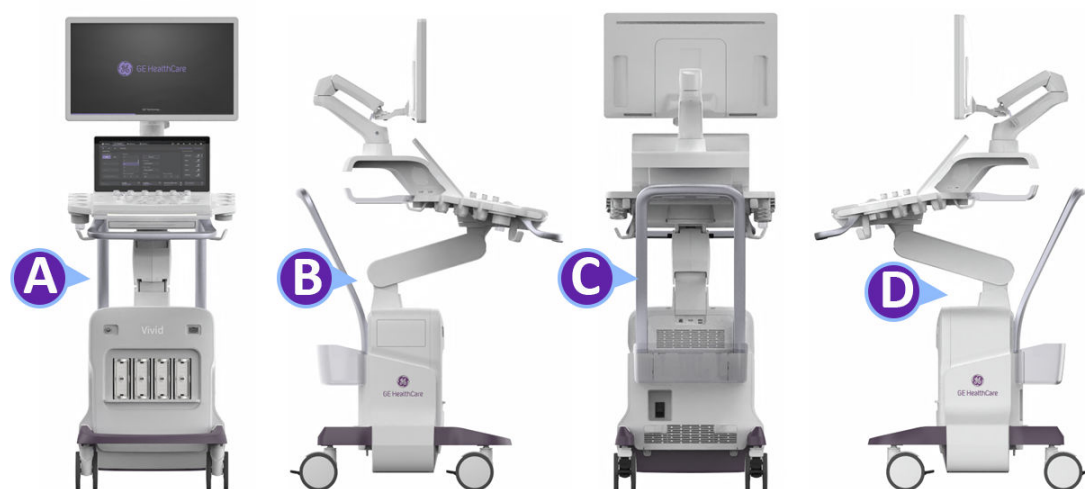


Figure 1-7 Vivid Advanced Console Views of FRONT Side (A) - LEFT (B) - BACK/REAR (C) - RIGHT Side (D)



Figure 1-8 Vivid Focus Console Views of FRONT Side (A) - LEFT (B) - BACK/REAR (C) - RIGHT Side (D)

1.5 Publication Conventions

1.5.1 Conventions Used in this Manual

Model Designation

NOTE

The screen graphics and photo illustrations in this Service Manual are for illustrative purposes only. Actual screen displays and device appearances (e.g., color) may vary slightly

Icons

Pictures, or icons are used wherever they reinforce the printed message. The icons, labels and conventions used on the product and in the service information are described in this chapter.

Safety Precaution Messages

Various levels of safety precaution messages may be found on the equipment and in the service information. The different levels of concern are identified by a flag word that precedes the precautionary message. Known or potential hazards to personnel are labeled in one of following ways:

- **Danger**
- **Warning**
- **Caution**

1.5.2 Label and Icon Conventions

1.5.2.1 Standard Hazard Icons

Table 1-2 Standard hazard icons

Icon	Meaning
	Biological
	Electrical
	Moving
	Acoustic output
	Explosion

Icon	Meaning
	Smoke/fire
	Heat
 or 	Pinch point

**DANGER**

Danger is used to indicate the presence of a hazard that will cause severe personal injury or death, or substantial property damage if the instructions are ignored.

1.5.2.2 Product Labels and Icons

For an overview of the product labels and icons, please refer to: [Symbols and Labels](#) on page 36.

1.5.2.3 Photos

Photos and drawings are used to illustrate the descriptions and procedures in this manual.

1.5.2.4 Screenshots

Screenshots are used to illustrate the options and parameters that are available.

1.5.2.5 Block Diagrams

Block Diagrams are used to illustrate the main functional blocks of the ultrasound system.

1.6 Manuals

For details, please refer to the OEM and Other Manuals chapter in this manual.

1.6.1 Where to Find the User Manuals

The User Manuals (multiple languages) are delivered on a manual media for viewing on a personal computer.

An onboard version of the User Manual is installed at setup of the product so it can be accessed via the help function.

Publicly released user manuals are available for downloading from the GE HealthCare Customer Documentation Portal, <https://www.gehealthcare.com/support/manuals>.

1.6.2 Where to Find the Privacy and Security Manual

The Privacy and Security Manual is delivered on a manual media for viewing on a personal computer.

The Privacy and Security Manual is only available to Customers and other non-GE HealthCare parties otherwise authorized by GE HealthCare under a specific agreement with GE HealthCare.

Such users can access the Privacy and Security Manual via the **Product Security Portal** on <https://securityupdate.gehealthcare.com/>.

1.6.3 Where to Find the General Service Manual

The General Service Manual is included on a manual media for viewing on a personal computer.

Publicly released service manuals are available for downloading from the GE HealthCare Customer Documentation Portal, <https://www.gehealthcare.com/support/manuals>.

1.6.4 Where to Find the Service Notes

GE HealthCare employees and Channel Partners with access to the SIMS Content Viewer can view the Service Notes.

1.8 Customer Assistance

1.8.1 Contact Information

If this equipment does not work as indicated in this service manual or in the user manual, or if you require additional assistance, please contact the local distributor or appropriate support resource, as listed below.

Before you call, identify the following information and acquire image (Alt+D) to send to the Customer Care team:

1. Ultrasound system ID serial number.
2. Software version.
3. Date and time of occurrence.
4. Sequence of events leading to issue.
5. Is the issue repeatable?
6. Imaging mode, probe, preset/application.
7. Media brand, speed, capacity, type.
8. Save secondary image capture, cine loop, 4D multi-volume loop.
9. Detailed description of any problem encountered.
10. Where applicable, save the appropriate log files.

Remember to save the log files for each day on a separate media, labelled accordingly.

NOTE

Restart the application before resuming clinical scanning.

1.8.2 Contacting GE HealthCare Ultrasound

For additional information or assistance, please contact your local distributor or the appropriate support resource listed on the following pages:

INTERNET	https://www.gehealthcare.com https://cleaning.gehealthcare.com
Clinical Questions	For information in the United States, Canada, Mexico, and parts of the Caribbean, call the Customer Answer Center. TEL: (1) 800-682-5327 or (1) 262-524-5698 In other locations, contact your local Applications, Sales, or Service Representative.
Service Questions	For service in the United States, call GEHC CARES. TEL: (1) 800-437-1171 In other locations, contact your local Service Representative.
Information Requests	To request technical product information in the United States, call GE HealthCare. TEL: (1) 800-643-6439 In other locations, contact your local Applications, Sales, or Service Representative.
Placing an Order	To order accessories, supplies, or service parts in the United States, call the GE Technologies Contact Center. TEL: (1) 800-558-5102 In other locations, contact your local Applications, Sales, or Service Representative.

1.8.2.1 AMERICAS

ARGENTINA	GE Healthcare TEL: 11-5298-2400 Nicolas Vedia 3616, piso 5 Buenos Aires
------------------	---

BRAZIL
 GE Healthcare do Brasil Comércio e Serviços para Equipamentos Médico- Hospitalares Ltda
 Av. Magalhães de Castro, 4800, Andar 11 Conj. 111 e 112, Andar 12 Conj. 121 e 122, Torre 3 - Cidade Jardim
 CEP: 05676-120 - São Paulo/SP - Brasil
 C.N.P.J.: 00.029.372/0001-40
 TEL: 3004 2525 (Capitais e regiões metropolitanas) / 08000 165 799 (Demais regiões)

CANADA
 GE Healthcare
 Ultrasound Service Engineering
 9900 Innovation Drive
 Wauwatosa, WI 53226
 TEL: (1) 800-668-0732
 Customer Answer Center TEL: (1) 262-524-5698

LATIN & SOUTH AMERICA
 GE Healthcare
 Ultrasound Service Engineering
 9900 Innovation Drive
 Wauwatosa, WI 53226
 TEL: (1) 262-524-5300
 Customer Answer Center TEL: (1) 262-524-5698

MEXICO
 GE Sistemas Medicos de Mexico S.A. de C.V.
 Rio Lerma #302, 1º y 2º Pisos
 Colonia Cuauhtemoc FAX: (5) 211-4631
 06500-Mexico, D.F.
 TEL: (5) 228-9600

USA
 GE Healthcare
 Ultrasound Service Engineering
 9900 Innovation Drive
 Wauwatosa, WI 53226
 TEL: (1) 800-437-1171 FAX: (1) 414-721-3865

1.8.2.2 ASIA

ASIA PACIFIC JAPAN
 GE Healthcare Asia Pacific
 4-7-127, Asahigaoka
 Hinoshi, Tokyo
 191-8503, Japan
 TEL: +81 42 585 5111

AUSTRALIA
 GE Healthcare
 32 Philip Street
 Parramatta, NSW 2150
 TEL: 1800 659 465

CHINA
 GE Healthcare - China
 No. 1, Yongchang North Road
 Beijing Economic & Technology Development Area
 Beijing 100176, China
 TEL: (8610) 5806 8888 FAX: (8610) 6787 1162

INDIA
 Wipro GE Healthcare Pvt Ltd
 No. 4, Kadugodi Industrial Area
 Sadaramangala, Whitefield
 Bangalore, 56067
 TEL: +(91) 1-800-425-8025

KOREA
 GE Healthcare Korea
 15F, 416 Hangang Dae ro, Chung-gu
 Seoul, 04637, Korea
 TEL: +82 1544 6119

NEW ZEALAND
 Level 7 Vero Centre
 48 Shortland St, Auckland 1010
 New Zealand
 TEL: 0800 659 465

SINGAPORE
 GE HEALTHCARE PTE ASEAN (Singapore)
 11 North Buona Vista Drive
 #11-07 The Metropolis Tower 2
 Singapore 138589
 TEL: +65 6291 8528 FAX: +65 6291 7006

1.8.2.3 AFRICA

EGYPT
 GE Healthcare
 32 Philip Street
 Parramatta, NSW 2150
 TEL: 1800 659 465

KENYA
 GE East Africa Services Limited
 General Mathenge Drive, Courtyard Building
 Westlands
 Nairobi 30, 00100 KE
 TEL: +254 719 093 044

NIGERIA
 GE International Operations (Nig) Ltd
 Bishop Aboyade Cole Street No. 927/928
 Mansard Place, PO Box 54255
 Victoria Island, Lagos LA, NG
 TEL: +234 (01) 4607101, +234 (01) 4607102

KINGDOM OF SAUDI ARABIA
 GE Healthcare Arabia Co. Ltd
 Platinum Centre, Building 1
 Salahuddin Ayoubi Road
 Riyadh-12811, Kingdom of Saudi Arabia
 TEL: +966 (11) 494 5779, FAX: +966 (11) 207 3946

SOUTH AFRICA

General Electric South Africa (Pty) Ltd.
 60 Glenhove Road
 Green on Glenhove Customer Innovation Centre
 Johannesburg GP, 2196 ZA
 TEL: +27 0100725000, FAX: +27 0862958385

1.8.2.4 EUROPE and MIDDLE EAST

For all other European countries not listed, please contact your local GE HealthCare distributor or the appropriate support resource listed on www.gehealthcare.com.

AUSTRIA

GE Healthcare Austria GmbH & Co OG
 EURO PLAZA, Gebäude E
 Technologiestr. 10, A-1120 Wien
 TEL: (+43) 1 97272 0 FAX: (+43) 1 97272 2222

BELGIUM & LUXEMBURG

GE Healthcare BVBA
 Eagle Building
 Kouterveldstraat 20
 1831 DIEGEM
 TEL: (+32) 2 719 7204 FAX: (+32) 2 719 7205

CZECH REPUBLIC

GE Medical Systems Česká Republika, s.r.o.
 Bucharova 2641/14
 158 00 Praha 5
 Česká Republika
 TEL: (+420) 224 446 162, FAX: (+420) 224 446 161

DENMARK

GE Medical Systems Ultrasound
 Park Alle 295, 2605 Brøndby
 TEL: (+45) 43 295 400 FAX: (+45) 43 295 399

ESTONIA & FINLAND

GE Healthcare Finland Oy
 Kuortaneenkatu 2, 000510 Helsinki
 P.O.Box 330, 00031 GE Finland
 TEL: (+358) 10 39 48 220 FAX: (+358) 10 39 48 221

FRANCE

GE Medical Systems SCS
 24 Avenue de l'Europe - CS20529
 78457 Velizy Villacoublay Cedex
 TEL: (+33) 13 449 50 00 FAX: (+33) 13 44 95 202

GERMANY

GE Healthcare GmbH
 Beethovenstrasse 239
 42655 Solingen
 TEL: (+49) 212-28 02-0 FAX: (+49) 212-28 02 380

GREECE

GE Healthcare
 8-10 Sorou Str. Marousi
 Athens 15125 Hellas
 TEL: (+30) 210 8930600 FAX: (+30) 210 9625931

HUNGARY	GE Hungary Zrt. Bence utca 3 Budapest BU, 1138 HU TEL: (+36) 23 410 314, FAX: (+36) 23 410 390
IRELAND	GE Healthcare 3050 Lake Drive, Citywest Business Campus Dublin 24 TEL: (+353) 1 4605500
ITALY	GE Medical Systems Italia spa Via Galeno, 36, 20126 Milano TEL: (+39) 02 2600 1111 FAX: (+39) 02 2600 1599
NETHERLANDS	GE Healthcare B.V. De Wel 18, 3871 MV Hoevelaken PO Box 22, 3870 CA Hoevelaken TEL: (+31) 33-25 41 222
NORWAY	GE Healthcare Vitaminveien 1A, 0485 Oslo TEL: (+47) 23 18 50 50 GE Healthcare Strandpromenaden 45, P.O. Box 141, 3191 Horten TEL: (+47) 33 02 11 16
POLAND	GE Medical Systems Polska Sp. z o.o., ul. Woloska 9 02-583 Warszawa, Poland TEL: (+48) 22 330 83 00 FAX: (+48) 22 330 83 83
PORTUGAL	General Electric Portuguesa SA SA. Avenida do Forte, n° 6-6A Edificio Ramazzotti, 2790-072 Carnaxide TEL: (+351) 21 425 1300 FAX: (+351) 21 425 1343
RUSSIA	GE Healthcare 12th floor, 10C, Presnenskaya nab. Moscow 123317 Russia TEL: (+7) 495 739 69 31 FAX: (+7) 495 739 69 32
SPAIN	GE Healthcare España C/ Gobelos 35-37 28023 Madrid TEL: (+34) 91 663 2500 FAX: (+34) 91 663 2501
SWEDEN	GE Healthcare Sverige AB FE 314, SE-182 82 Stockholm, Sweden TEL: (+46) 8 559 50010

SWITZERLAND	GE Medical Systems (Schweiz) AG Europastrasse 31, 8152 Glattbrugg TEL: (+41) 1 809 92 92 FAX: (+41) 1 809 92 22
TURKIYE	GE Healthcare Türkiye Istanbul Office Levent Ofis Esentepe Mah. Harman Sok. No:8 Sisli-Istanbul TEL: +90 212 398 07 00 FAKS: +90 212 284 67 00
UNITED ARAB EMIRATES (UAE)	GE Healthcare Holding ME SA Dubai Internet City, Building No. 18 P. O. Box # 11549, Dubai U.A.E TEL: (+971) 4 429 6161 FAX (+971) 4 429 6200/01/02
UNITED KINGDOM	GE Medical Systems Ltd Pollards Wood Nightingales Lane Chalfont St Giles, Buckinghamshire HP8 4SP TEL: (+44) 1494 544000, FAX: (+44) 1707 289742

Chapter 2

Site Preparations (Pre-Installation)

In this chapter

Preparing the Site.	60
Environmental Requirements.	64
Electrical Requirements.	67
Power Requirements.	68
Networking Requirements.	69
Electromagnetic Interference Survey.	73
Environmental Requirements for Probes.	77
Time and Manpower Requirements.	78
Environmental protection.	79
Appendix.	80

2.1 Preparing the Site

2.1.1 Purchaser Responsibilities

The work and materials needed to prepare the site is the responsibility of the purchaser. Delay, confusion, and waste of manpower can be avoided by completing pre-installation work before delivery. Purchaser responsibilities include:

- Procuring the materials required.
- Completing the preparations before delivery of the ultrasound system.
- Paying the costs for any alterations and modifications not specifically provided in the sales contract.

NOTE

All electrical installations that are preliminary to the positioning of the equipment at the site prepared for the equipment must be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations, and testing must also be performed by qualified personnel. The products involved (and the accompanying electrical installations) are highly sophisticated and special engineering competence is required. All electrical work on these products must comply with the requirements of applicable electrical codes. The purchaser of GE HealthCare equipment must only utilize qualified personnel to perform electrical servicing on the equipment.

The desire to use a non-listed or customer provided product or to place an approved product further from the Ultrasound system than the interface kit allows, presents challenges to the installation team. To avoid delays during installation, such variances should be made known to the individuals or group performing the installation at the earliest possible date (preferably prior to the purchase).

The ultrasound suite must be clean prior to delivery of the ultrasound system. Carpet is not recommended because it collects dust and creates static. Potential sources of EMI (electromagnetic interference) should also be investigated before delivery. Dirt, static, and EMI can negatively impact ultrasound system reliability.

2.1.2 Required Facility Needs

NOTE

GE HealthCare requires a dedicated power and ground for the proper operation of its ultrasound equipment. This dedicated power shall originate at the last distribution panel before the ultrasound system.

The ultrasound system will function on voltages from 100-240 Volts and 50 or 60 Hz. However, if using 220-volt power in North America, then a center tapped power source is required.

Sites with a mains power system with defined Neutral and Live:

The dedicated line shall consist of one phase, a neutral (not shared with any other circuit), and a full-size ground wire from the distribution panel to the ultrasound outlet.

Sites with a mains power system without a defined Neutral:

The dedicated line shall consist of one phase (two lines), not shared with any other circuit, and a full-size ground wire from the distribution panel to the ultrasound outlet.

- Dedicated single branch power outlet of adequate amperage, meeting all local and national codes, which is located less than 2.5 m (8 ft.) from the ultrasound system's proposed location
- Door opening is at least 76 cm (30 in) wide
- Proposed location for ultrasound system is at least 0.5 m (1.5 ft.) from the wall for cooling
- Power outlet and place for any external peripheral are within 2 m (6.5 ft.) of each other with peripheral within 1 m of the ultrasound system to connect cables
- Power outlets for other medical equipment
- Power outlets for test equipment within 1 m (3.2 ft.) of ultrasound system
- Clean and protected space to store probes (in their cases or on a rack)
- Material to safely clean probes (done with a plastic container, never metal)
- An active network outlet in the vicinity of the ultrasound system
- A network cable of appropriate length (regular Pin-to-Pin network cable).

- An IT administrator who will assist in configuring the system to work with your local network. A fixed IP address is required

NOTE

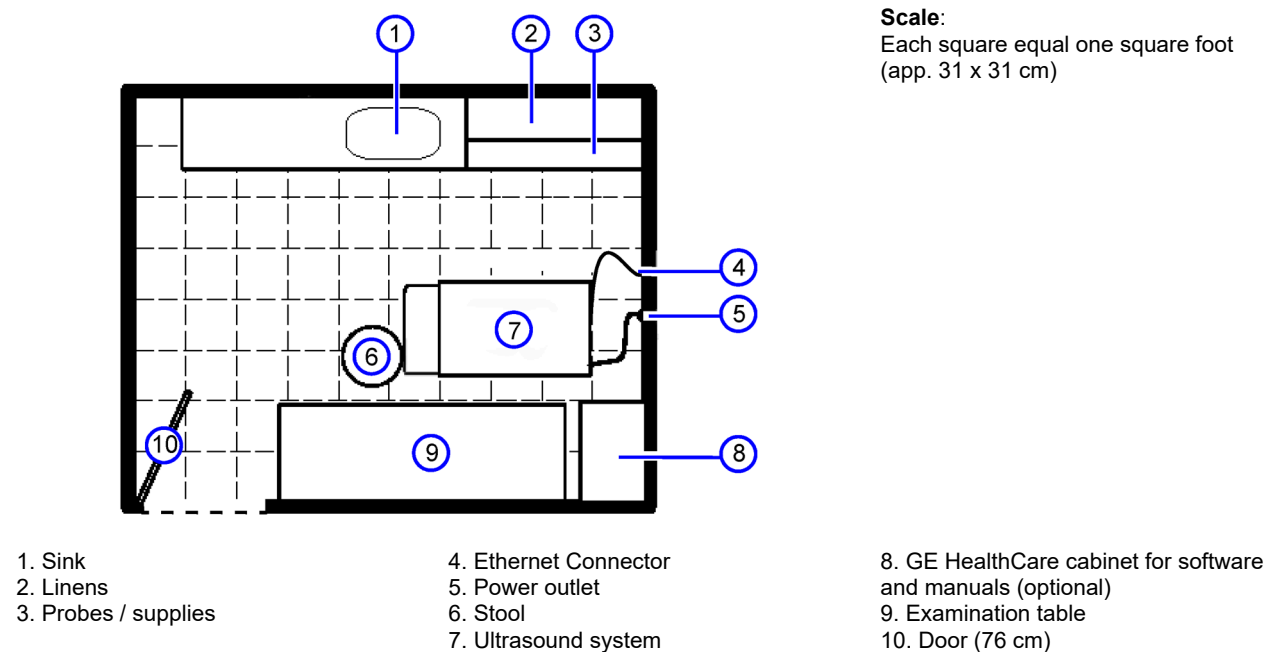
All relevant preliminary network port installations at the prepared site must be performed by authorized contractors. The purchaser of GE HealthCare equipment must utilize only qualified personnel to perform servicing on the equipment.

2.1.3 Desirable Features

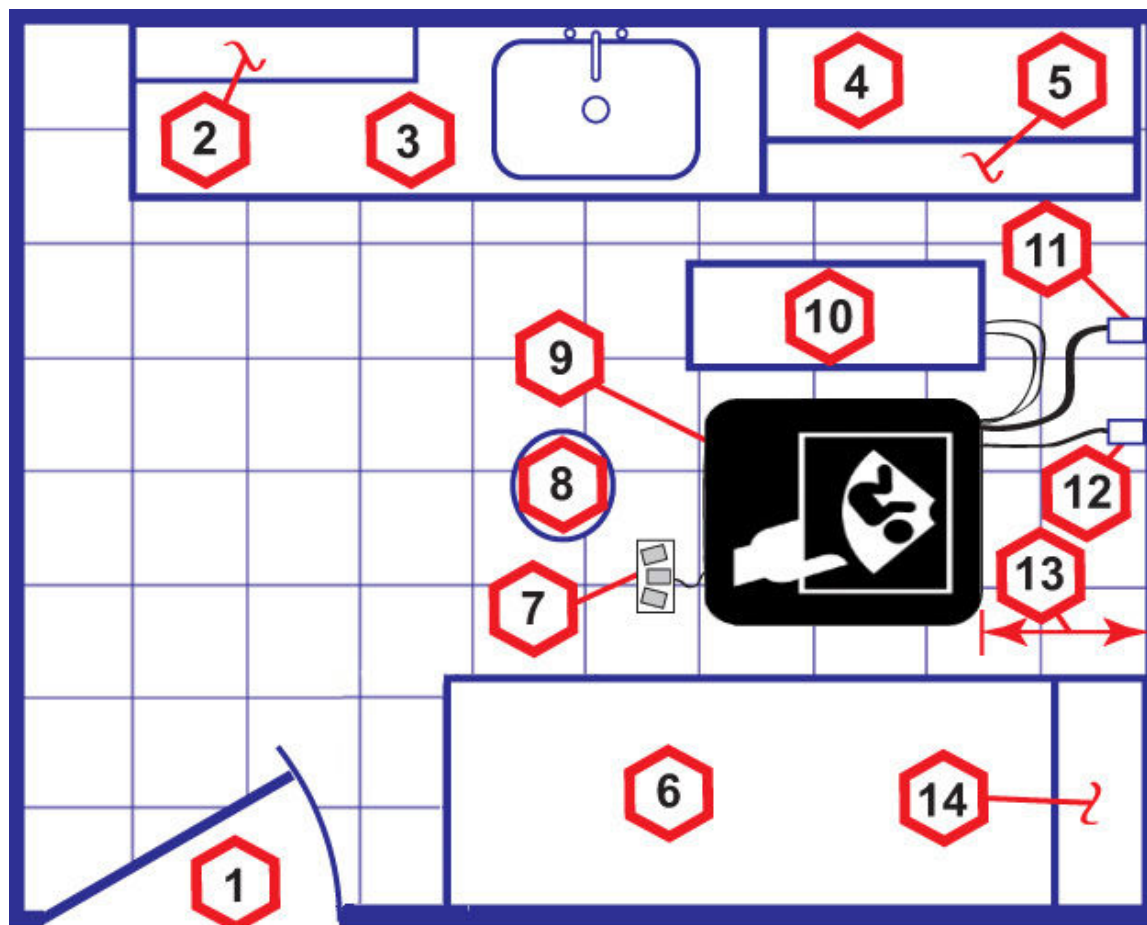
- door is at least 92 cm (3 ft.) wide
- circuit breaker for dedicated power outlet is easily accessible
- sink with hot and cold water
- receptacle for bio-hazardous waste, like used probe sheaths
- storage for linens and equipment
- nearby waiting room, lavatory, and dressing room
- dual level lighting (bright and dim)
- lockable cabinet ordered by GE HealthCare for its software and proprietary manuals

2.1.4 Minimal Floor Plan Suggestion

Table 2-1 Minimal floor plan, 2.5 m x 3 m (8 by 10 foot)



2.1.5 Recommended Floor Plan Suggestion



Scale: Each square equal one square foot (app. 31 x 31 cm)

1. Door – at least 762 mm (30 inches)	6. Examination Table – 1930 x 610 mm (76 x 24 inches)	11. Dedicated Power Outlet - Circuit Breaker protected and easily accessible
2. Film Viewer	7. Footswitch	12. Network Interface
3. Film Supplies	8. Stool	13. 457 mm (18 inches) distance of ultrasound system from wall or objects
4. Storage for Linens and Equipment	9. Ultrasound system	
5. Counter Top and Sink with hot and cold water	10. External Peripherals	
		14. Red checkmark

		14. File Cabinet
--	--	------------------

Figure 2-1 Recommended floor plan, 4.27 x 5.18 m (14 x 17 foot)

2.1.6 Suggested Floor Plan with ULS and Workstation in Same Room

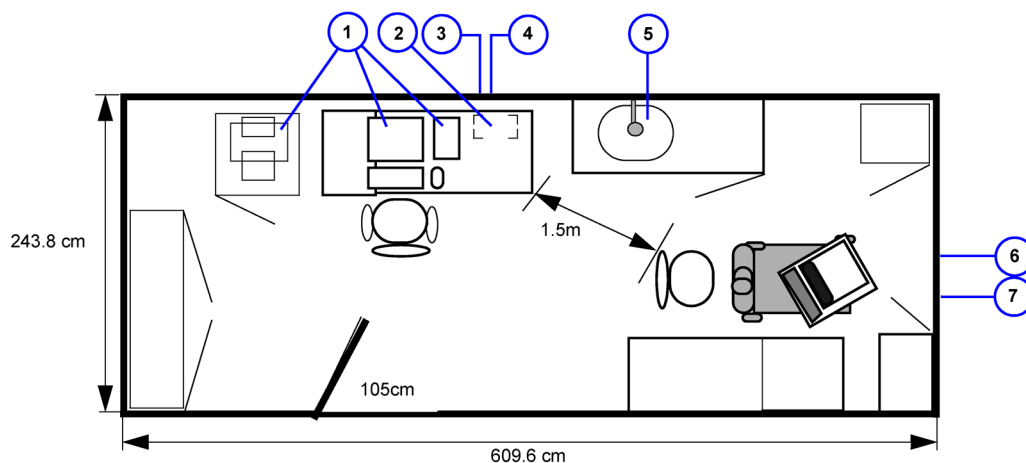


Figure 2-2 Suggested Room with workstation and ultrasound system

- | | | |
|---------------------------------|---------------------------------|---------------------------------|
| 1. Workstation parts | 4. 3x mains power outlets | 7. Ethernet network wall outlet |
| 2. UPS | 5. Hot and Cold water | |
| 3. Ethernet network wall outlet | 6. Dedicated mains power outlet | |

2.1.7 Seismic

Seismic requirements are determined and specified by the hospital design professional of record and may require approval by the specific state or country agency. Seismic attachment hardware shown on seismic calculations may differ from hardware supplied with system. Any additional hardware that is required will be the responsibility of the institution and/or their contractor. Contact your local GE HealthCare Project Manager of Installations or Installation Representative to obtain seismic calculations. Seismic calculations are per California Building Code (CBC) and International Building Code (IBC).

2.2 Environmental Requirements

2.2.1 Environmental Dangers

Commercial devices such as laser cameras, printers, VCRs and external monitors, usually exceed allowable leakage current limits and, when plugged into separate AC outlets, are in violation of patient safety standards. Suitable electrical isolation of such external AC outlets, or providing the device with extra protective earth, will be required in order to meet IEC/EN 60601-1 Clause 16 standards for electrical leakage.

2.2.1.1 Peripheral devices

**DANGER**

If externally powered equipment supplied by GE HealthCare is electrically connected to the scanner system, a medical system is formed. Only items that have been specified as part of the system (that is, the ultrasound system and all its electrically connected electrical devices) or have been specified as compatible with the system should be connected.

**DANGER**

Use only GE HealthCare approved equipment when replacing an peripheral.

External peripheral equipment not verified by GE HealthCare must be in compliance with related IEC/EN standards (eg. 60601/62368 series) for safety. Meanwhile, the whole system's conformance to IEC/EN60601-1 Clause 16 must be verified.

All non-medical electrical equipment not verified by GE HealthCare must be kept outside of the patient environment, unless it is equipped with necessary protective earth or extra separating transformer and conformance to IEC/EN 60601-1 is verified. Commercial devices such as laser cameras, printers, VCRs and external monitors, usually exceed allowable leakage current limits per IEC/EN60601-1 and, when plugged into separate AC outlets, are in violation of patient safety standards. Suitable electrical isolation of such external AC outlets, or providing the device with extra protective earth, will be required in order to meet IEC/EN60601-1 Clause 16 standards for electrical leakage currents.

2.2.1.2 External Connection of other peripheral devices

**DANGER**

External devices not verified by GE HealthCare can be used only if in compliance with related IEC or ISO Safety standards (eg. 60601/62368 series). Meanwhile, conformance to IEC/EN60601-1 Clause 16 requirements for the whole system must be verified.

Accessory equipment connected to the analog and digital interfaces must be certified according to the respective IEC or ISO Safety standards (e.g. IEC/EN62368-1 for data processing equipment or IEC/EN60601-1 for medical equipment). Furthermore all complete configurations shall comply with the requirements for Medical Electrical systems of IEC/EN60601-1. Anybody connecting additional equipment to the signal input part or signal output part of the ultrasound system configures a medical electrical system, and is therefore responsible that the system complies with the requirements of the IEC/EN60601-1 Clause 16. If in doubt, consult the technical service department or your local GE HealthCare representative.

**WARNING**

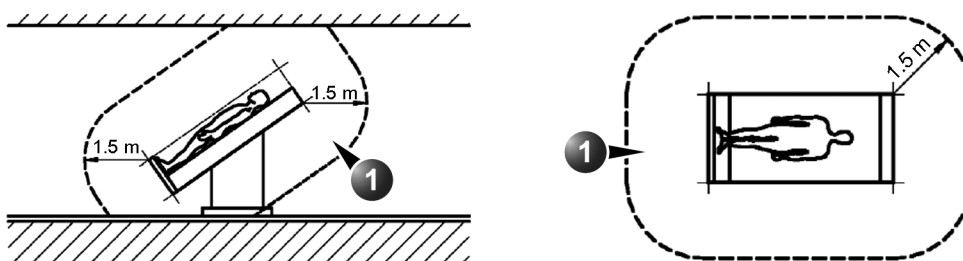
Any devices or cables, other than those sold with the ultrasound system, connected to the Peripheral /accessory connector panel or to an USB port on the system may result in an increase of the electromagnetic emission from the system, or a decrease of the electromagnetic immunity of the system.

**CAUTION**

When using peripheral device, observe all warnings and cautions given in peripheral operator manuals.

2.2.1.3 Patient environment

To learn about the patient environment, please see [Figure 2-3 Example of patient environment](#) on page 65.



1. Patient environment

Figure 2-3 Example of patient environment

The Vivid Explorer/Vivid Advanced/Vivid Focus ultrasound system has been verified for overall safety, compatibility and compliance with probes and their accessories listed in Probes Chapter and Peripherals Chapter.

The Vivid Explorer/Vivid Advanced/Vivid Focus ultrasound system has also been verified for compatibility, and compliance for connection to a local area network (LAN) via the rear panel Ethernet connection, provided the LAN components are IEC/EN 62368 compliant.

Connection may also be made to a CE Marked and IEC/EN 60601/62368 series standard compliant modem using one of the serial or USB ports on the system provided that the combination conforms to IEC/EN 60601-1.

The Vivid Explorer/Vivid Advanced/Vivid Focus may also be used safely while connected to devices other than those recommended above if the devices and their specifications, installation, and interconnection with the system conform to the requirements of IEC/EN 60601-1.

However, not all peripherals verified with system according to IEC/EN 60601-1 are suitable to be used in patient environment, such as GE Healthcare provided View X, externally powered printers, and those peripherals that are not approved by GE Healthcare. Please verify applicable requirements defined in IEC/EN 60601-1 if you do want to use them in patient environment, or consult GE Healthcare local representative for technical support.

General precautions for installing an alternate on-board device would include:

1. The added device must have appropriate safety standard conformance and CE Marking.
2. The total power consumption of the added devices, which connect to the ultrasound system and are used simultaneously, must be less than or equal to the rated supply of the ultrasound system.
3. There must be adequate heat dissipation and ventilation to prevent overheating of the device.
4. There must be adequate mechanical mounting of the device and stability of the combination.
5. Risk and leakage current of the combination must comply with IEC/EN 60601-1.
6. Electromagnetic emissions and immunity of the combination must conform to IEC/EN 60601-1-2.

General precautions for installing an alternate off-board, remote device or a network would include:

1. The added device(s) must have appropriate safety standard conformance and CE Marking.
2. The added device(s) must be used for their intended purpose having a compatible interface.
3. Signal or mains isolation devices and additional protective earth may be needed to assure compliance with IEC/EN 60601-1.



CAUTION

The connection of equipment or transmission networks other than as specified in the user instructions can result in an electric shock hazard or equipment malfunction. Substitute or alternate equipment and connections requires verification of compatibility and conformity to IEC/EN 60601-1 by the installer. Equipment modifications and possible resulting malfunctions and electromagnetic interference are the responsibility of the owner.

2.2.1.3.1 Unapproved Devices



CAUTION

Please use unapproved devices with caution.

If devices are connected without the approval of GE HealthCare, the warranty will be INVALID.

Any device connected to the Vivid Explorer/Vivid Advanced/Vivid Focus must conform to the requirements for IEC or equivalent standards appropriate to devices.

2.2.1.3.2 Accessories, Options, Supplies



CAUTION

Unsafe operation or malfunction may result. Use only the accessories, options and supplies approved or recommended in these instructions for use.

2.2.2 System Acclimation Time



CAUTION

Possible equipment damage. Turning the system on without acclimation after arriving at site may cause the system to be damaged.

After unpacking, allow the temperature of the ultrasound system to stabilize before powering up. The following table describes guidelines for reaching operational temperatures from storage or transport temperatures.

Table 2-2 Ultrasound system acclimation time

°C	-40	-35	-30	-25	-20	-15	-5	0	5	10	15	20	25	30	35	40	45	50	55	60	65	70
°F	-40	-31	-22	-13	-4	5	14	32	41	50	59	68	77	86	95	104	113	122	131	140	149	158
hrs	20	18	16	14	12	8	6	4	2	0	0	0	0	0	0	2	4	6	8	10	12	14

2.2.3 Environmental Specifications

Table 2-3 Environmental specifications for Vivid Explorer/Vivid Advanced/Vivid Focus

Conditions	Temperature limits	Humidity limits	Air pressure limits	Heat dissipation
Operation:	3–40 °C(37 to 104°F) (system without battery) 3-30 °C(37 to 86°F) (system with battery)	30 - 80%	700-1060 hPa	2390 BTU/h
Storage and transport:	-20 to 50 °C (-68 to +122 °F)	10 - 90%	700-1060 hPa	N/A

2.2.4 Heat Output (Cooling)

The cooling requirement for the Vivid Explorer/Vivid Advanced/Vivid Focus with monitor and on board peripherals, is up to 2390 BTU/h This figure does not include cooling needed for lights, people, or other equipment in the room.

NOTE

Each person in the room places an additional 300 BTU/h demand on the cooling system.

2.3 Electrical Requirements

General Requirements

NOTE

GE HealthCare requires a dedicated power and ground for the proper operation of its ultrasound equipment. This dedicated power shall originate at the last distribution panel before the ultrasound system.

The ultrasound system will function on voltages from 100-240 Volts and 50 or 60 Hz. However, if using 220-volt power in North America, then a center tapped power source is required.

Sites with a mains power system with defined Neutral and Live:

The dedicated line shall consist of one phase, a neutral (not shared with any other circuit), and a full-size ground wire from the distribution panel to the ultrasound outlet.

Sites with a mains power system without a defined Neutral:

The dedicated line shall consist of one phase (two lines), not shared with any other circuit, and a full-size ground wire from the distribution panel to the ultrasound outlet.

NOTE

Please note that image artifacts can occur, if at any time within the facility, the ground from the main facility's incoming power source to the ultrasound system is only a conduit.

Electrical Requirements for Vivid

In the table below, the electrical specifications for the ultrasound system includes monitor and on-board peripherals.

Table 2-4 Models Delivered with Software 211

Model Number	Description	Electrical specifications
TC001011	Vivid Explorer	- Voltage: 100-240 V~ - Tolerances: $\pm 10\%$ - Power Consumption: 600 VA - Frequency: 50/60 Hz The current drain will vary depending on the mains voltage. - At 230 VAC the current may be up to 2.5 A - At 110 VAC the current may be up to 5.0 A
TC001012	Vivid Advanced	
TC001013	Vivid Focus	

Inrush Current

Inrush current: < 50A peak

2.4 Power Requirements

2.4.1 Site circuit breaker



CAUTION

Power outage may occur. The Ultrasound system requires a dedicated single branch circuit. To avoid circuit overload and possible loss of critical care equipment, make sure you do not have any other equipment operating on the same circuit.

It is recommended that the branch circuit breaker for the Ultrasound system be readily accessible.

2.4.2 Site power outlets

A dedicated AC power outlet must be within reach of the Ultrasound system without extension cords. Other outlets adequate for the external peripherals, medical and test equipment needed to support this Ultrasound system must also be present within 1 m (3.2 ft.) of the Ultrasound system. Electrical installation must meet all current local, state, and national electrical codes.

2.4.3 Mains power cable

If the Ultrasound system arrives without a mains power cable, or with the wrong mains power cable, you must contact your GE HealthCare dealer or the installation engineer must supply what is locally required.

2.4.4 Lighting

Bright light is needed for Ultrasound system installation, updates and repairs. However, operator and patient comfort may be optimized if the room light is subdued and indirect. Therefore a combination lighting system (dim/bright) is recommended. Keep in mind that lighting controls and dimmers can be a source of EMI which could degrade image quality. These controls should be selected to minimize possible interference.

2.5 Networking Requirements

2.5.1 Stand-alone System (without Network Connection)

None

2.5.2 System Connected to Hospital's Network

Supported networks:

- Ethernet network connection
- Wireless LAN (option)

To be able to use the network functions when connected to a hospital network, the system must have a proper network address. Typically, the source for this information is the hospital network administrator.

2.5.3 Purpose of the DICOM Network Function

DICOM^{®1} (Digital Imaging and Communications in Medicine) services provide the operator with clinically useful features for moving images and patient information over a hospital network. Examples of DICOM services include the transfer of images to workstations for viewing or transferring images to remote printers. As an added benefit, transferring images in this manner frees up the on-board monitor and peripherals, enabling viewing to be done while scanning continues. With DICOM, images can be archived, stored, and retrieved faster, easier, and at a lower cost.

2.5.4 DICOM Option Pre-installation Requirements

To configure the ultrasound system to work with other network connections, the network administrator must provide some necessary information.

Use the [Connectivity Setup Worksheet](#) on page 70 to record required information that must include:

- Details: DICOM network details for the ultrasound system, including the host name, local port, IP address, AE title and net mask.
- Routing Information: IP addresses for default gateway and other routers in use at site.
- DICOM Application Information: Details of DICOM devices in use at the site, including the DICOM host name, AE title, DICOM port number and IP addresses.

NOTE

For further details refer to the User Manual.

¹ DICOM is the registered trademark of the National Electrical Manufacturers Association for its standards publications relating to digital communications of medical information.

2.5.5 Connectivity Setup Worksheet

Table 2-5 Site System Information

Site System Information					
Site:	Floor:	Comments: 			
Dept:	Room:				
Voluson E_	Type:				
		REV:			
CONTACT INFORMATION					
Name	Title	Phone	E-Mail Address		

TCP/IP Settings	Remote Archive Setup
System IP Settings Name - AE Title: IP Address: Subnet Mask: Default Gateway:	Name - AE Title: IP Address: Subnet Mask: Default Gateway: Server Name: Remote DB User Name:

Services (Destination Devices)						
	Device Type	Manufacturer	Name	IP Address	Port	AE Title
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						

Table 2-6 Worksheet for DICOM Network Information

Voluson E__		Local Port		IP Address	
Host Name					
AE Title		Net Mask			

ROUTING INFORMATION				GATEWAY IP Addresses				
	Destination IP Addresses				Default			
ROUTER1								
ROUTER2								
ROUTER3								

DICOM APPLICATION INFORMATION					
	NAME	MAKE/REVISION	AE TITLE	IP ADDRESSES	PORT
Store 1					
Store 2					
Store 3D_1					
Store 3D_2					
Print					
Worklist					
Structured Reporting					
Storage Commit					
MPPS					

2.5.6 Wireless Network Settings

NOTE

Please ensure the wireless adapter is connected before you configure wireless network settings. Refer to [Connecting the Wireless Network Adapter](#) on page 115.

Table 2-7 Wireless Network Settings

Network Name (SSID):	
Network Autentication	
Open	☐
Shared Key	☐
WPA PSK	☐
WPA2 PSK	☐
Encryption Type:	
Data Encryption	
Network Key:	
Key Index:	
Available Channels:	

2.6 Electromagnetic Interference Survey

2.6.1 Electromagnetic Compatibility (EMC)

Electromagnetic compatibility describes a level of performance of a device within its electromagnetic environment. This environment consists of the device itself and its surroundings including other equipment, power sources and persons with which the device must interface. Inadequate compatibility results when a susceptible device fails to perform as intended due to interference from its environment or when the device produces unacceptable levels of emission to its environment. This interference is often referred to as radio-frequency or electromagnetic interference (RFI/EMI) and can be radiated through space or conducted over interconnecting power or signal cables. In addition to electromagnetic energy, EMC also includes possible effects from electrical fields, magnetic fields, electrostatic discharge and disturbances in the electrical power supply.

For product-specific information, refer to the user/service documentation for the Ultrasound system.

2.6.1.1 Compliance

Ultrasound system conforms to all applicable conducted and radiated emission limits and to immunity from electrostatic discharge, radiated and conducted RF fields, magnetic fields and power line transient requirements.

For applicable standards, refer to the Safety Chapter of the Ultrasound system's User's Manual.

NOTE

For CE compliance, it is critical that all covers, screws, shielding, gaskets, mesh, clamps, are in good condition, installed tightly without skew or stress. Proper installation following all comments noted in this service manual is required in order to achieve full EMC performance.

2.6.1.2 Electrostatic discharge (ESD) prevention



WARNING



DO NOT touch any boards with integrated circuits prior to taking the necessary ESD precautions.

Always connect yourself, with an arm-wrist strap, to the advised ESD connection point. Location of the ESD connection points vary by product. Refer to the product-specific service documentation for specific locations and for further guidance.

Follow general guidelines for handling of electrostatic sensitive equipment.



WARNING



Risk of electrical shock, Ultrasound system must be turned off. Avoid all contact with electrical contacts, conductors and components. Always use non-conductive handles designed for the removal and replacement of ESD sensitive parts. All parts that have the potential for storing energy must be discharged or isolated before making contact.



WARNING

If the covers are removed from an operating Ultrasound system, some metal surfaces may be warm enough to pose a potential heat hazard if touched, even while in shutdown mode.

2.6.2 Interference caution

Intended healthcare environments: Professional healthcare facility environment



CAUTION

Use of devices that transmit radio waves near the system could cause it to malfunction.

Before installing or using the system, be aware of the proximity of known radio frequency (RF) sources, such as:

- Radio and TV stations
- Portable and mobile RF communication devices (cell phones including 5G communication, two-way radios)
- Fixed RF communication devices including RFID (Radio-frequency identification), EAS (electronic article surveillance), and Wireless power transfer (WPT)
- High-frequency surgical units, such as diathermy, electrocautery, argon beam coagulators, etc.
- X-ray, CT, or MRI devices

These devices are also possible sources of interference as they may emit higher levels of electromagnetic radiation.

Devices which intrinsically transmit radio waves, such as cellular phones, radio transceivers, mobile radio transmitters, radio-controlled toys, etc., should preferably not be operated near the system.

Medical staff in charge of the system are required to instruct technicians, patients, and other people who may be around the system to fully comply with the above recommendations.

Any electrical device can unintentionally emit electromagnetic waves. However, minimum device separation distances cannot be calculated for such unspecified radiation. When the ultrasound system is used adjacent to or in close proximity to other equipment the user should be attentive to unexpected device behavior which may be caused by such radiation.

The ultrasound system is intended for use in the electromagnetic environment specified in the tables below.

The user of the ultrasound system should ensure that the device is used in such an environment.



WARNING

Use of accessories, transducers and cables other than those specified or provided by the manufacturer of this equipment could result in increased electromagnetic emissions or decreased electromagnetic immunity of this equipment and result in improper operation.



WARNING

Use of this equipment adjacent to or stacked with other equipment should be avoided because it could result in improper operation. If such use is necessary, this equipment and the other equipment should be observed to verify that they are operating normally.



WARNING

Portable RF communications equipment (including peripherals such as antenna cables and external antennas) should be used no closer than 30 cm (12 inches) to any part of the Vivid Explorer/Vivid Advanced/Vivid Focus series system, including cables specified by the manufacturer. Otherwise, degradation of the performance of this equipment could result.

2.6.3 Electromagnetic emissions

The system is suitable for use in the specified electromagnetic environment and it has meets the following standard's emission requirements.

Guidance and manufacturer's declaration – electromagnetic emissions.		
Emissions test	Compliance	Electromagnetic environment - guidance
RF emission CISPR 11 EN55011	Group 1	The ultrasound system uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
RF emission CISPR 11 EN55011	Class A	This system is suitable for use in all establishments, other than domestic establishments and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes, provided the following warning is heeded: Warning: This system is intended for use by healthcare professionals only. This system may cause radio interference or disrupt the operation of nearby equipment. It may be necessary to take mitigating measures, such as re-orienting or relocating the system or shielding the location.
Harmonic emission IEC 61000-3-2	Class A	
Voltage fluctuations/flicker emissions IEC 61000-3-3	Complies	

2.6.4 Electromagnetic interferences

The ultrasound system is approved for use in hospitals, clinics, and other environmentally qualified facilities, in terms of the prevention of radio wave interference. Operation of the system in an inappropriate environment can cause electronic interference to radios and television sets situated near the equipment.

Ensure that the system is protected from electromagnetic interferences as follows:

- Operate the system at least 4.5 meters (15 feet) away from equipment that emits strong electromagnetic radiation.
- Shield the system when operating it in the vicinity of radio broadcasting equipment, if necessary.

2.6.5 EMI limitations

Ultrasound systems are susceptible to Electromagnetic Interference (EMI) from radio frequencies, magnetic fields, and transients in the air or wiring. They also generate EMI. The Ultrasound system complies with limits as stated on the EMC label. However, there is no guarantee that interference will not occur in an installation.

Possible EMI sources should be identified before the Ultrasound system is installed.

Electrical and electronic equipment may produce EMI unintentionally as the result of a defect. Some of these sources include:

- medical lasers
- scanners
- cauterizing guns
- computers
- monitors
- fans
- gel warmers
- microwave ovens
- plasma ovens
- light dimmers
- mobile phones
- in-house wireless phones (DECT phones)
- wireless computer keyboard and mouse
- air conditioning system
- High Frequency (HF) surgery equipment
- general AC/DC adapters
- RFID access
- ID card readers
- Ultrasonic cleaners

The presence of a broadcast station or broadcast van may also cause interference.

See [EMI prevention/abatement](#) on page 75 for EMI prevention tips.

2.6.6 EMI prevention/abatement

Prevention/abatement details are provided below:

Table 2-8 EMI Prevention/Abatement

EMI Rule	Details
Be aware of Radio Frequency sources	- Keep the ultrasound system at least 5 meters (15 feet) away from other EMI sources. - Special shielding may be required to eliminate interference problems caused by high frequency, high powered radio or video broadcast signals.
Ground the Ultrasound system	Poor grounding is the most likely reason an ultrasound system will have noisy images. Check grounding of the power cord and power outlet.
Replace all screws, Radio Frequency gaskets, covers, cores	- After you finish repairing or updating the ultrasound system, replace all covers and tighten all screws. - Any cable with an external connection requires a magnet wrap at each end. - Install all covers. Loose or missing covers or Radio Frequency gaskets allow radio frequencies to interfere with the ultrasound signals.
Replace broken Radio Frequency gaskets	If more than 20% or a pair of the fingers on a Radio Frequency gasket are broken, replace the gasket. Do not turn on the ultrasound system until any loose metallic part is removed.
Do not place labels where Radio Frequency gaskets touch metal	Where applicable, never place a label where Radio Frequency gaskets meet the ultrasound system. Otherwise, the gap created will permit Radio Frequency leakage. Or, if a label has been found in such a position, move the label.
Use GE HealthCare specified harnesses and peripherals	The interconnect cables are grounded and require ferrite beads and other shielding. Also, cable length, material, and routing are all important; do not change from what is specified.
Take care with cellular phones	Cellular phones may transmit a 5 V/m signal; that could cause image artifacts.
Properly route peripheral cables	Where applicable, do not allow cables to lie across the top of the Card Rack or hang out of the peripheral bays. Loop the excess length for peripheral cables inside the peripheral bays. Attach the monitor cables to the frame.

2.7 Environmental Requirements for Probes

Probes can be used in clinical environment.

Ensure that the probe face temperature does not exceed the normal operation temperature range.

Probes must be operated, stored, or transported within the parameters outlined below.

	Operational	Storage	Transport
Temperature	+18° to +30° C (+64.4°F to +86°F)	-10° to +50° C (+14°F to +122°F)	-10° to +50° C (+14°F to +122°F)
Humidity	30% to 75% RH non-condensing	10% to 85% RH non-condensing	10% to 85% RH non-condensing
Pressure	700hPa (3000m) to 1060hPa	700hPa (3000m) to 1060hPa	700hPa (3000m) to 1060hPa

2.8 Time and Manpower Requirements

Site preparation takes time. Begin site preparation checks as soon as possible. If possible, six weeks before delivery, to allow enough time to make any changes.



WARNING

Have two people available to deliver and unpack the ultrasound system.

Attempts to move the ultrasound system considerable distances or on an incline by one person could result in personal injury or property damage or both.

2.9 Environmental protection

2.9.1 System and battery disposal

The USER/SERVICE staff should dispose of all the waste properly, per federal, state, and local waste disposal regulations.

Regulations vary for different countries. Dispose of a used battery in accordance with local regulations.

NOTE

Used batteries should not be placed with common household waste products. Contact local authorities for the location of a chemical waste collection program nearest you.

Please contact the manufacturer or other authorized disposal company to decommission your equipment.

2.10 Appendix

2.10.1 EMC Compliance

The ultrasound system conforms to all applicable conducted and radiated emission limits and to immunity from electrostatic discharge, radiated and conducted RF fields, magnetic fields and power line transient requirements as mentioned in IEC60601-1-2.

NOTE

For CE Compliance, it is critical that all covers, screws, shielding, gaskets, mesh, clamps, are in good condition, installed tightly without skew or stress. Proper installation following all comments noted in this service manual is required in order to achieve full EMC performance.

2.10.2 How to Access Supplier Manuals for Product Options

- The documentation for options manufactured by GE HealthCare are either described in the User Manual and/or the Service Manual.
- The documentation for options manufactured by a third party vendor, are delivered with the option. For updated or lost documentation, contact the vendors of the respective product options.

2.10.3 Option Catalog Numbers

Contact your GE HealthCare representative for information about available options.

Chapter 3

Tools

In this chapter

Tools Available via the Service Desktop, InSite etc..	82
---	----

3.1 Tools Available via the Service Desktop, InSite etc.

Please refer to the descriptions in

- [Service desktop](#) on page 259
- [Request for Service \(RFS\)](#) on page 213

Chapter 4

System Setup

In this chapter

System Setup.	84
Upgrades.	105
Installation Instruction for Vivid Explorer Update Kit.	106
Options.	110
Accessories.	112
Product Locator Installation Card.	126

4.1 System Setup

4.1.1 Average Setup Time

Table 4-1 Average Setup Time

DESCRIPTION	AVERAGE SETUP TIME	COMMENTS
Unpack the Vivid Explorer/Vivid Advanced/Vivid Focus	0.5 hour	
Set up the Vivid Explorer/Vivid Advanced/Vivid Focus (without options)	4 hours	Depends on the configuration
Configure DICOM Network	2 hours or more	Depends on the configuration
Install InSite	0.5 hour	

4.1.2 Installation Warnings



CAUTION

Two people are needed to unpack the ultrasound system because of its weight. Attempts to move the ultrasound system considerable distances or on an incline by one person could result in personal injury or property damage or both.

Two people are required whenever a part weighing 16 KG (35 LBS) or more must be lifted.

4.1.2.1 Acclimation time

Following transport, allow the system to acclimate before switching it on. Acclimation will take one hour for each 2.5 °C increment when the system's temperature is below 10 °C or above 35 °C.

°C	-20	-15	-10	-5	0	5	10	35	40	45	50	55	60
°F	-4	5	14	23	32	41	50	95	104	113	122	131	140
Hours	12	10	8	6	4	2	0	0	2	4	6	8	10

4.1.2.2 Control Console Positioning

The control console can be rotated and adjusted in height. There is a Release Handle located in the middle of the Control Panel.



Figure 4-1 Release Handle

- Pull and hold the release handle to rotate the Control panel horizontally, up to 45 degrees on each side. When reaching the desired swivel angle, release the handle.
- Pull and hold the release handle to move the Control panel on a vertical axis. When reaching the desired height, release the handle.



CAUTION

The ultrasound system should not be moved with the control console (UI) extended.
Do not put your hand between the control console and the main unit when moving the control console to its centered and locked position:
Danger of injuries!



CAUTION

Make sure that nothing would be jammed while moving!

NOTE

Shipping the Vivid Explorer/Vivid Advanced/Vivid Focus ultrasound system in its original packaging is only possible when the control console is centered and locked in position, the system is lowered to its minimum height and the monitor is flapped down and locked.

4.1.2.3 Brake Pedal Operation

The ultrasound system is driven by four free-swivel wheels, controlled by black brake pedals situated above the wheel. Step on the brake pedals to check whether the wheel brakes are functioning properly. For detailed information of Wheels, please see [Wheels](#) on page 34.

4.1.3 Safety Reminders

**DANGER**

When using any test instrument that is capable of opening the AC ground line (i.e., electrical safety analyzer's ground switch is open), don't touch the ultrasound system!

**CAUTION**

To prevent electrical shock, connect the ultrasound system to a properly grounded power outlet. Do not use a three to two prong adapter. This defeats safety grounding.

**CAUTION**

Connectors and plugs of any kind (AC Mains, DC-voltage supply, signal input/signal output (SIP/SOP) connectors, internal power/signal connectors) must always be firmly connected, i.e. the connector/plug must be completely pushed into the socket, to ensure good electrical contact.

When available, devices for connector/plug securing must be engaged.

**CAUTION**

The ultrasound system requires all covers!

Do not operate this system unless all board covers and frame panels are securely in place, to ensure safe operation, optimal system performance and cooling. When covers are removed, EMC and EMI compliance might be compromised.

For service procedures where the operation of the ultrasound system is necessary with board covers or frame panels removed, ensure that there are no EMI sensitive devices in the vicinity of the system. If in doubt, relocate the ultrasound system to a different environment where no EMI sensitive devices are in the vicinity.

**CAUTION**

Two people are needed to unpack the ultrasound system because of its weight. Attempts to move the ultrasound system considerable distances or on an incline by one person could result in personal injury or property damage or both.

Two people are required whenever a part weighing 16 KG (35 LBS) or more must be lifted.

**CAUTION**

Do not wear the ESD wrist strap when you work on live circuits and more than 30 V peak is present.

**CAUTION**

If the ultrasound system is very cold or hot, do not turn on its power until it has had a chance to acclimate to its operating environment.

**CAUTION**

Instructions for Use

The User Manual(s) should be fully read and understood before operating the ultrasound system. Keep manuals near the system for reference.

**CAUTION**

Acoustic Output Hazard

Although the ultrasound energy transmitted from the ultrasound probe is within AIUM/NEMA standards, avoid unnecessary exposure. Ultrasound energy can produce heat and mechanical damage.

4.1.3.1 Power ON Installation (Completing the Setup)

4.1.3.1.1 Power ON Warnings

**DANGER**

Always connect the ultrasound system to a fixed power socket which has the protective grounding connector.

**DANGER**

Never use a three-to-two prong adapter. Using a three-to-two prong adapter defeats the safety ground.

**DANGER**

Ensure that the power cord and plug are intact and that the power plug is the proper hospital-grade type (where required).

**CAUTION**

Ultrasound system requires all covers.

Operate this ultrasound system only when all board covers and frame panels are securely in place. The covers are required for safe operation, good ultrasound system performance, and cooling purposes.

**CAUTION**

Use only power supply cords, cables, and plugs provided or designated by GE HealthCare.

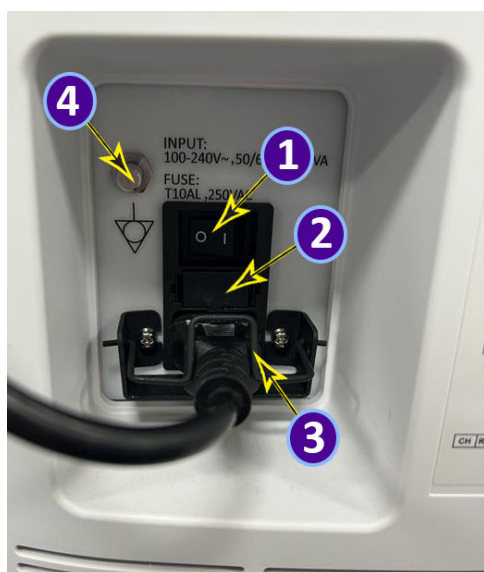
NOTE

Do not cycle the Circuit Breaker ON-OFF-ON in less than five (5) seconds. When turning OFF the Circuit Breaker, the ultrasound system should de-energize completely before turning the circuit breaker ON.

4.1.3.1.2 Connecting the Ultrasound System to a Power Source

Connecting the ultrasound system to the mains, involves preliminary checks of the power cord, voltage level, and compliance with electrical safety requirements.

1. Ensure that the wall outlet is of appropriate type.
2. Uncoil the power cable, allowing sufficient slack so that the ultrasound system can be moved slightly.
3. Verify that the power cable is without any visible scratches or any sign of damage.
4. Verify that the on-site mains voltage is within the limits indicated on the rating label near the **Circuit Breaker** at the rear of the ultrasound system.
5. Verify that the **Mains Power Circuit Breaker** is in OFF position. If not, switch it OFF.
6. If not already done, connect the mains power cable's female plug to the power inlet at the rear of the ultrasound system, then pull down the pull-out protection for power cable.



1. circuit breaker
2. fuses (T10AH,250P, size(mm):5*20)
3. mains power cable incl. pull-out protection
4. equipotential pin

Figure 4-2 Circuit breaker, fuses and mains power cord at rear of system

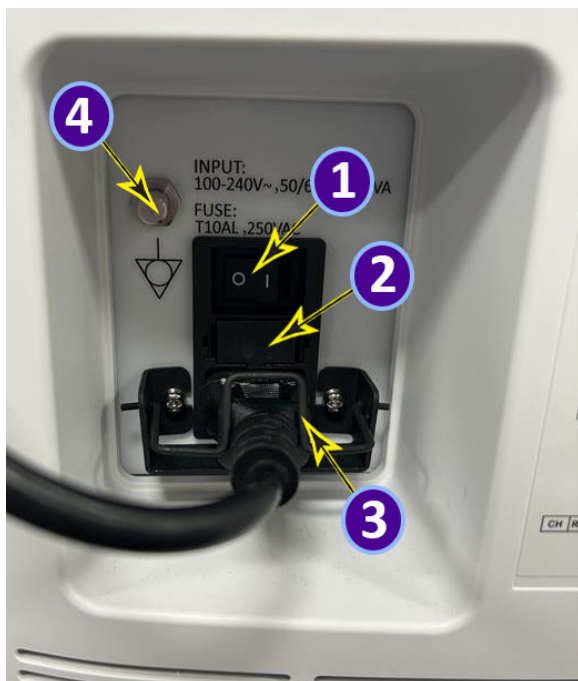
7. Connect the power cable's other end (male plug) to a hospital grade mains power outlet with the proper rated voltage.

The ultrasound system is ready for Power ON/ Boot Up.

4.1.3.1.3 Power ON/Boot Up

If not already done, connect the mains power cable to the back of the ultrasound system, then pull down the pull-out protection for power cable.

1. Connect the mains power cable to a hospital grade power outlet with the proper rated voltage. Never use an adapter that would defeat the safety ground.
2. Switch ON the **Circuit Breaker** [1] at the rear of the system.



1. circuit breaker
2. fuses (T10AH,250P, size(mm):5*20)
3. mains power cable incl. pull-out protection
4. equipotential pin

Figure 4-3 Circuit breaker, fuses and mains power cord at rear of system

NOTE

When AC power is applied to the system, the ON/OFF standby button on the control console illuminates (white light), indicating that the system (including the Back-End Processor) is in standby mode.

3. Press the button on the control console for ~3 seconds.



Figure 4-4 Press the On-Off switch

NOTE

The power switch of any attached printer(s) needs to be in ON position before starting the system. However, be aware some auxiliary equipment may switch itself to standby mode (e.g., Color video printer) and must therefore be switched on separately.

When the ON/OFF standby button on the control console is pressed, the system (including the back-end processor) starts and the operating system is loaded which leads to activation of the application software.

The system automatically performs an initialization sequence which includes the following:

- loading the operating system
- running a quick diagnostic check of the system
- detecting connected probes

As soon as the software has been loaded, the system enters 2D-Mode with the probe and application that were used before the system shutdown. Total time used for start-up is about 2 minutes.

4.1.3.1.4 Connect the probe

Probes can be connected at any time, whether the system is on or off.

Before connecting the probe:

- Inspect the probe and the probe cable for any damage.
- Do a visual check of the probe pins and system sockets. Remove any dust or foam remains from the probe pins.



CAUTION

Check the probe cable and lens to confirm they are not broken before scanning.

**DANGER**

Do NOT touch the patient and any of the connectors on the ultrasound system simultaneously, including ultrasound probe connectors.

**CAUTION**

Do not allow the probe head to hang freely. Impact to the probe head may result in irreparable damage.

**CAUTION**

Take the following precautions with the probe cables:

- Keep free from the wheels.
- Do not bend.
- Do not cross cables between probes.

NOTE

Further information on handling of ICE and TEE probes is located in the particular user manuals supplied with these probes.

DLP Probe ports

The Vivid Explorer and Vivid Advanced system has four DLP probe ports ([Figure 4-5 Connecting the probe to DLP Probe ports](#) on page 90).



1. Connecting the probe to DLP Probe ports
2. 1 doppler probe port to support the P2D and P6D probes

Figure 4-5 Connecting the probe to DLP Probe ports

To connect the probe to the DLP probe port:

1. Hold the probe connector vertically with the cable pointing upward.
2. Turn the connector locking handle counterclockwise.
3. Align the connector with the probe port and carefully push into place.
4. Turn the locking handle clockwise to the full vertical position to lock in place.
5. Position the probe cable so that it is not resting on the floor.

RS Probe ports

To connect the probe to the RS probe port:

1. Hold the probe connector vertically with the cable pointing upward.
2. Slide the connector lock to the left (unlocked position).
3. Align the connector with the probe port and carefully push into place.
4. Slide the connector lock to the right position to lock the probe connector.
5. Carefully position the probe cord so it is free to move and is not resting on the floor.

4.1.3.1.5 Power Shut Down

When the ultrasound system is switched off, it performs an automatic shutdown sequence.

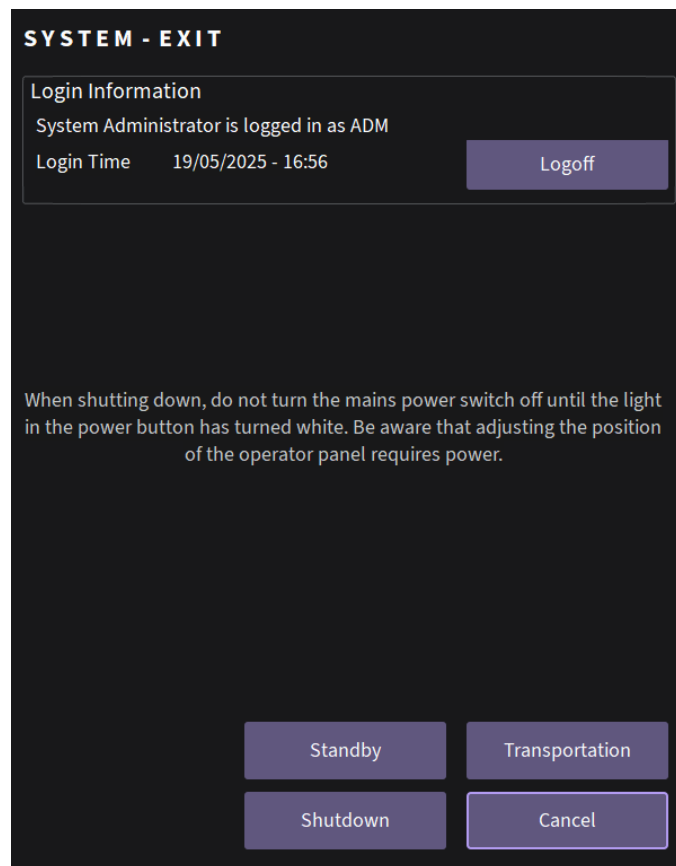


Figure 4-6 The Exit dialog window

The **SYSTEM - EXIT** menu, used when switching off the ultrasound system, gives you these choices:

- **Logoff**
Use this button to log off the current user.
The system remains ON and ready for a new user to log on.
If the **Logoff** button is dimmed, it indicates that no user is logged on to the ultrasound system at the moment.
- **Shutdown**
Use this button to shut down the system. The entire system will shut down. It is recommended to perform a full shutdown at least once a week.
If the **Shutdown** button is dimmed, use the key-combination **Ctrl+Alt+R** to shut down the unit.
- **Transportation mode**

Parts of the system will be powered down. This mode allows a shorter wakeup time when the system moved from one place to another. Also, when AC power is interrupted while in the Transportation mode the main screen is dimmed and the operator panel will not accept any user input.

The touch panel and the rest of the system is still powered by an internal battery.

When AC power is resumed, the system will turn ON in less than 5 seconds.

- **Standby**

With battery, most parts of the system will be powered down except RAM, the system keeps all currently running data to memory (RAM). With AC power without battery, all parts of the system will be powered down, the system saves all currently running data to the hard drive. By pressing on/off button can quickly wake up the system.

-  **CAUTION**
Press and hold the power on button to force shutdown system for a reset if console fails to wake up to operation mode from standby mode.

 **CAUTION**
After waking up from standby, the system should be restarted if abnormal image is noticed.

NOTE

The system can complete patient data saving before entering Standby mode.

When no battery is installed, the system can only be awakened by pressing the on/off button. When a battery is installed, the system can be awakened by pressing the on/off button, any key on the control panel, or any key on the keyboard.

Booting up from this mode takes less than 40 seconds.

- **Cancel**

Use this button to exit from the **System-Exit** menu and return to the previous operation.

4.1.3.1.6 Complete Power Down

1. Before power down, secure the Top Console in its lower, locked position. This is required when moving or transporting the Vivid Explorer/Vivid Advanced/Vivid Focus to ensure maximum stability.

NOTE

For service purposes, you may want to move the Top Console after Power Down. If you leave the Top Console in unlocked position, it will be sensed during Power Down, and the brakes that usually stabilize the position of the Top Console, will be released so you can move the Top Console in the XY directions.

2. Press once on the **On/Off** key on the Operating Panel to display the **System - Exit** menu.



Figure 4-7 Press once on the On/Off key

3. Select **Shutdown** to do a complete power down of the ultrasound system.

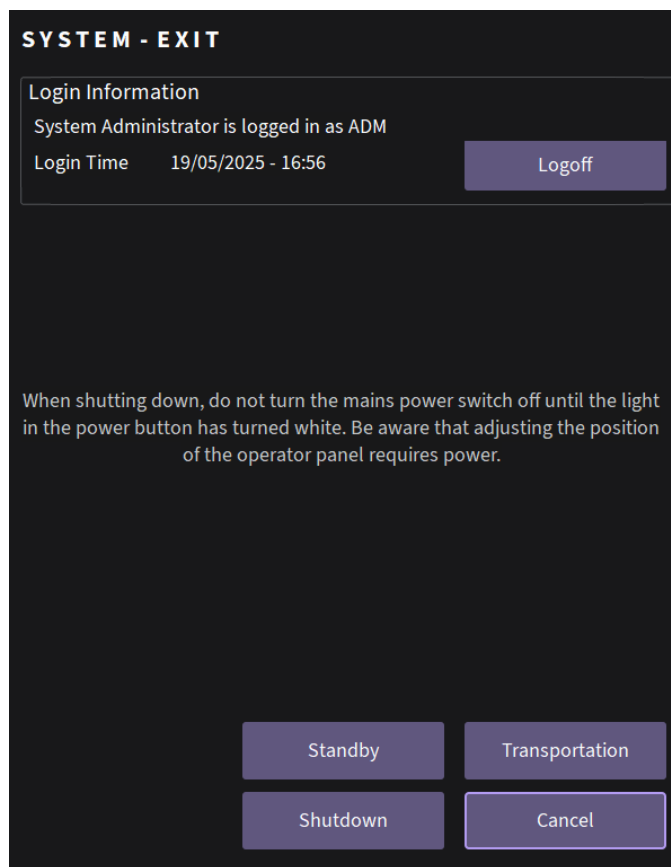


Figure 4-8 Select Shutdown

The Back End Processor will first turn off the scanner activity and display the message “Please wait - Shutdown in progress” on the Touch Panel.

Next, the ultrasound system will begin to shut down. The time it takes to shut down the system, including the Back End Processor, may vary from 10 seconds to approximately 1 minute.

The last thing that shuts down, is the light on the Operating Panel, indicating that you can continue with the next step.

4. Switch off the **Mains Power Circuit Breaker**, located at the rear of the ultrasound system.
This will cut power distribution within the Vivid Explorer/Vivid Advanced/Vivid Focus.
5. Disconnect the mains power cable.

4.1.3.1.7 Disconnect the Power Cable

The ultrasound system must be switched off before you disconnect the power cable.

This procedure is used as part of the LOTO for the power cord for the ultrasound system

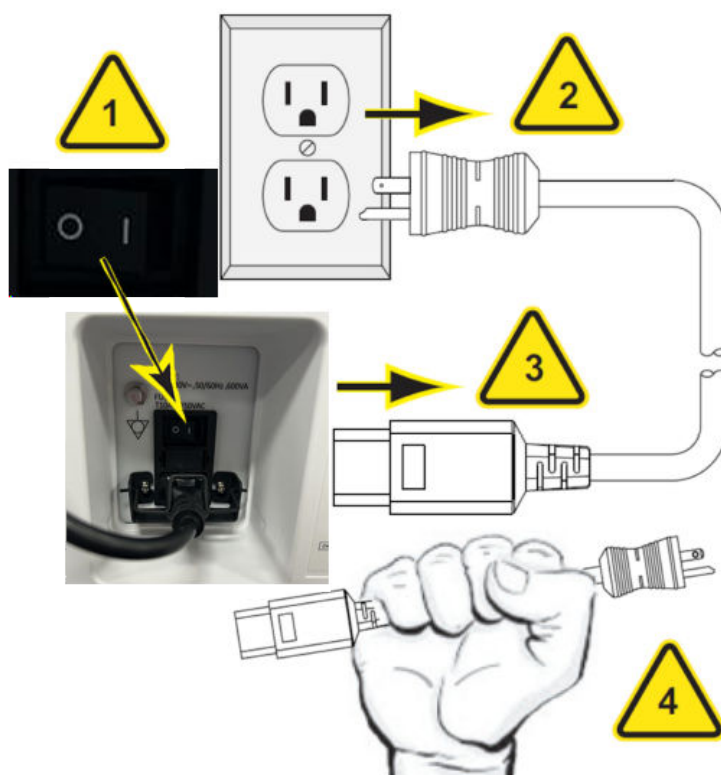


Figure 4-9 Disconnect Power Cord



CAUTION

Do not turn off the breaker before the power button on the Operator Panel is amber. Data may be lost or system software damaged if the breaker is turned off before the power button is amber.

1. Turn off the breaker [1] on the back of the system.
2. Disconnect the power cord [2] from the wall outlet.

NOTE

Do not disconnect the mains power cable [3] from the systems plug unless it is absolutely required for specific replacement procedures or if it is for any reason not possible to disconnect it from the AC wall outlet socket.

3. Move the power cord latch out of the way.
4. Disconnect the power cord from the ultrasound console [3].
5. Maintain exclusive control of the removed power cord [4].
6. Confirm that the ON/OFF button on the Operator Panel turns OFF.

4.1.3.1.8 Preparing for Transportation(for flexible arm)

To ensure that no part of the monitor can be damaged when transporting or moving the system, the monitor must be in a secure position.



Figure 4-10 Secure the flexible monitor

1. Move the upper monitor arm parallel to the lower arm and push it down.
2. Rotate the monitor to the center position and turn knob counterclockwise until arm rotation and height adjustment is locked.
3. Incline the monitor to the horizontal position.
4. Press the Release Handle in the middle of the Operator Panel and lower the system to its minimum height.

The ultrasound system can now be safely transported. Nevertheless, be careful when transporting or moving the device.

4.1.3.1.9 Preparing for Transportation(for fix arm)

To ensure that no part of the monitor can be damaged when transporting or moving the system, the monitor must be in a secure position.



Figure 4-11 Secure the fix monitor

1. Incline the monitor to the horizontal position.
2. Press the Release Handle in the middle of the Operator Panel and lower the system to its minimum height.

The ultrasound system can now be safely transported. Nevertheless, be careful when transporting or moving the device.

4.1.4 Service Form

Please refer to the [Site Log](#) on page 51.

4.1.5 Mechanical Installation

4.1.5.1 Receiving and Unpacking the Equipment

4.1.5.1.1 Warnings for Receiving and Unpacking



CAUTION

Two people are needed to unpack the ultrasound system because of its weight. Attempts to move the ultrasound system considerable distances or on an incline by one person could result in personal injury or property damage or both.

Two people are required whenever a part weighing 16 KG (35 LBS) or more must be lifted.



CAUTION

Remember to use relevant personal protecting equipment (PPE) during packing and unpacking. Check with your local EHS representative.

4.1.5.2 Before unpacking the equipment

4.1.5.2.1 Tilt Watch Indicators

Improper handling during transportation may harm the equipment inside the package even if the package itself is undamaged.

To make it easier to detect improper handling during transportation, a set of Tilt indicators have been attached to the transportation box.

The purpose of the tilt indicator label is to alert package handlers that the package content is sensitive to tipping and that it must remain upright at all times. It functions as an active “Up Arrow” that will change color if the package is tilted 89 degrees or more from its upright position. On rare occasions these labels have been falsely activated if the package was tipped less than 89 degrees, but shocked or vibrated at the same time. This is considered uncommon.



OK (Not Triggered)



Triggered

Figure 4-12 Tilt Indicator

Related reference

[If Tilt Indicator has Triggered or is Missing](#) on page 98

[Position of the Tilt Indicators](#) on page 98

4.1.5.2.2 If Tilt Indicator has Triggered or is Missing

An activated tilt indicator label does not indicate specifically whether the package was simply “Tipped” (laid down with no impact shock) or “Tipped Over” (free fall, with an impact shock).

If a package is received with an activated tilt indicator label, there is high degree of certainty it has been tilted 89 degrees or more from its upright position during shipment.

Table 4-2 Tilt Indicator has triggered or is missing

Step	Task
1.	If the Tilt Indicator is missing: Note on the shipping papers at the time of receipt that the Tilt Indicator label is missing. If the Tilt Indicator has triggered: Note on the shipping papers at the time of receipt that the Tilt Indicator label was activated.
2.	Inspect the product for possible concealed damage.

4.1.5.2.3 Position of the Tilt Indicators

The Tilt indicators have been attached to the transportation box as illustrated in the figure below.



Figure 4-13 Tilt indicators (position)

NOTE

Before cutting the straps, check the Tilt indicators to make sure they have not been triggered. If damaged (or missing), refer to: [If Tilt Indicator has Triggered or is Missing](#) on page 98.

4.1.5.2.4 Receiving the Ultrasound System

Examine all Packages

Examine package closely at time of delivery, as described in the procedure below.

1. Is damage apparent?
 - If YES; continue with the instructions in: [Damage in Transportation](#) on page xvi.
 - If NO; continue with the next step.
2. Is the Tilt Indicator red?

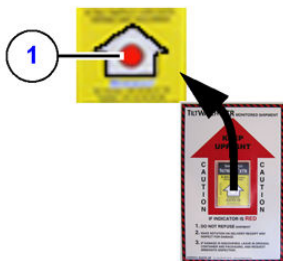


Figure 4-14 Tilt indicator has triggered (1)

- If **YES**: The Tilt Indicator has been activated. Continue with the instructions in: [Damage in Transportation](#) on page xvi before you continue with the next step.
 - If **NO**: continue with the next step.
3. Continue with the instructions in [Unpacking the Equipment](#) on page 99.

Damage in Transportation

Follow this procedure if damage is apparent:

1. Write "Damage in Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE HealthCare representative or hospital receiving agent.
2. Report the damage to the carrier.
 - Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier.
 - A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

4.1.5.3 Transportation Box Label for Vivid Explorer/Vivid Advanced/Vivid Focus

The transportation box label for Vivid Explorer/Vivid Advanced/Vivid Focus is located at the front of the transportation box.

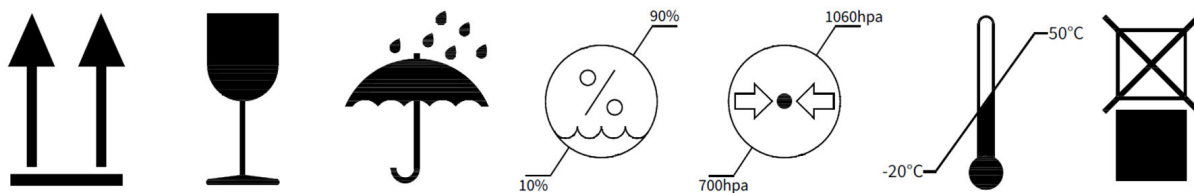


Figure 4-15 Transportation box label

4.1.5.4 Unpacking the Equipment

This procedure describes how to unpack the Vivid Explorer/Vivid Advanced/Vivid Focus from the cardboard box.

Before cutting the straps, check the Tilt Tags to make sure they have not been triggered. If damaged, report it to the carrier. If not, continue with the procedure below.

Table 4-3 Uncrating the Vivid Explorer/Vivid Advanced/Vivid Focus

Step	Task	Illustration
1.	Cut the straps around the crate. NOTE To avoid injury, hold the strap clasp with one hand when cutting the strap.	
2.	Remove the Top Cover	
3.	Remove the three Plastic Locks. NOTE Press the inside plastic lock gently to release it, then remove the outside lock.	 
4.	Open the sidewall cardboard and remove it.	

Step	Task	Illustration
5	Remove the dust bag from the system.	
6	Cut the wrapping bag by scissors and remove the PE bag from the system.	
7	Unlock the strap. NOTE Press the metal part shown in the diagram with your thumb, and use your other hand to pull out the packing strap.	

Step	Task	Illustration
8	Remove the front stopper and position it on the ground to create a ramp.	
9	Then, guide the machine down along the ramp.	

4.1.5.4.1 Physical Inspection

Verify that the Vivid Explorer/Vivid Advanced/Vivid Focus arrived intact (visual inspection).

If the Vivid Explorer/Vivid Advanced/Vivid Focus has been damaged, please refer to [Damage in Transportation](#) on page xvi in the beginning of this manual.

4.1.5.5 Preparing for Setup

4.1.5.5.1 Verify Customer Order

1. Compare items received by the customer to those that are listed on the delivery order.
2. Report any items that are missing, back ordered, or damaged.

4.1.5.5.2 Physical Inspection

Verify that the Vivid Explorer/Vivid Advanced/Vivid Focus arrived intact (visual inspection).

If the Vivid Explorer/Vivid Advanced/Vivid Focus has been damaged, please refer to [Damage in Transportation](#) on page xvi in the beginning of this manual.

4.1.5.5.3 EMI Protection

The Vivid Explorer/Vivid Advanced/Vivid Focus has been designed to minimize the effects of Electro-Magnetic Interference (EMI). Many of the covers, shields, and screws are provided primarily to protect the Vivid Explorer/Vivid Advanced/Vivid Focus from image artifacts caused by this interference. For this reason, it is imperative that all covers and hardware are installed and secured before the Vivid Explorer/Vivid Advanced/Vivid Focus is put into operation.

See [EMI limitations](#) on page 75 for more information about EMI protection.

4.1.5.6 Packing the Console

Follow the unpacking procedure in reverse order to pack the equipment.

Related reference

[Unpacking the Equipment](#) on page 99

4.1.5.7 Packaging Information

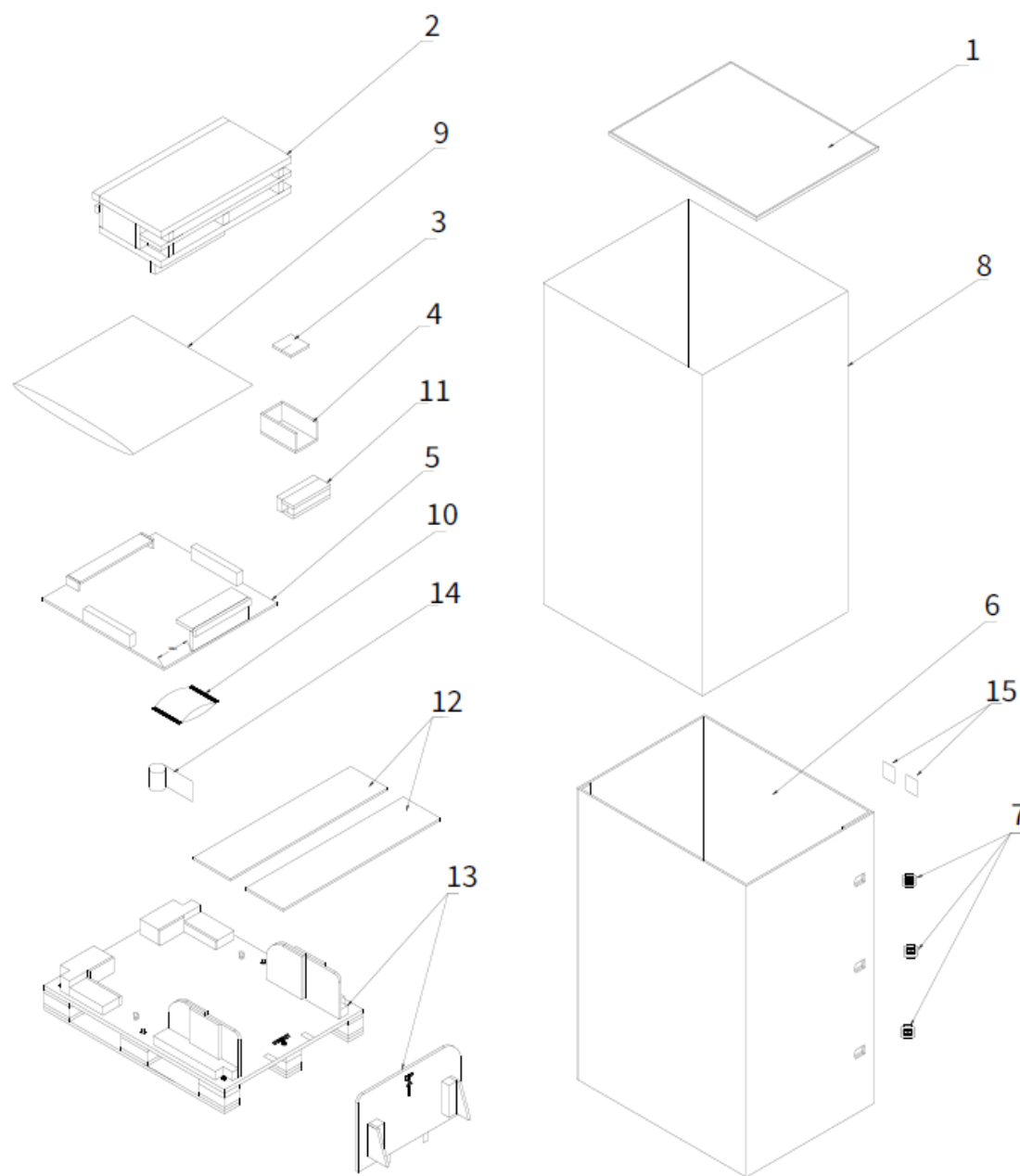


Figure 4-16 Packaging structure

Table 4-4 Packaging details

#	Description	Qty	Material	Details
1	Lid	1	Plywood	NA
2	Monitor cushion	1	EPE	NA
3	Monitor arm cushion-1	1	EPE	NA

#	Description	Qty	Material	Details
4	Monitor arm cushion-2	1	EPE	NA
5	Middle monitor and keyboard cushion	1	EPE	NA
6	Sleeve	1	US337g/200g/US337g/ 200g/US337g	NA
7	Plastic lock	3	PP	NA
8	ESD LDPE bag	1	LDPE	NA
9	LCD bag	1	LDPE	NA
10	Desiccant	2 or 1	Bag: none woven fabric Content: silica gel	2 for Ocean ship and 1 for air ship or other ship
11	Back handle cushion	1	EPE	For Vivid AdvancedVivid Explorer and Vivid Advanced
12	Strap cushion	2	EPE	NA
13	Pallet & ramp	1	Plywood	NA
14	Fastening rope with frap	1	NA	Size:25*0.5*4000mm Weight:~225.5g
15	Tilt watch	2	NA	Size:74*60mm, weight:~12.5g

NOTE

Always dispose of all the waste properly, per federal, state, and local waste disposal regulations.

4.2 Upgrades

There are no upgrades available for the v211 ultrasound system.

4.3 Installation Instruction for Vivid Explorer Update Kit

We provide optional update kit for Vivid Explorer ultrasound system. This update kit is applicable for the following HCAT:

Table 4-5 HCAT for Vivid Explorer Update Kit

HCAT	Description	Remarks
H48822DD	2D to 4D Update Kit	Updating from Vivid Explorer 2D to Vivid Explorer 4D does not require a software update. Only a new option key needs to be regenerated and re-entered.
H48822DE	4D to Dimension Update kit	Dimension configuration is supported only for international regions. Please consult your regional marketing team to confirm availability. Don't forget to attach dimension sticker to the bottom-right corner of the Monitor frame.

4.3.1 Tools

None.

4.3.2 Manpower

15 minutes.

4.3.3 Preparations

Prepare the option key of the configuration that you intend to update to.

4.3.4 Install the Optional Update Kit

This section uses the update from 'Vivid Explorer 4D' to 'Vivid Explorer Dimension' as an example to illustrate the complete update process

1. Press **Utility** → **Config** on the Touch Panel and log on as ADM.
2. Select the **Admin** category and **Licenses** subgroup.
The **Licenses** subgroup is displayed.

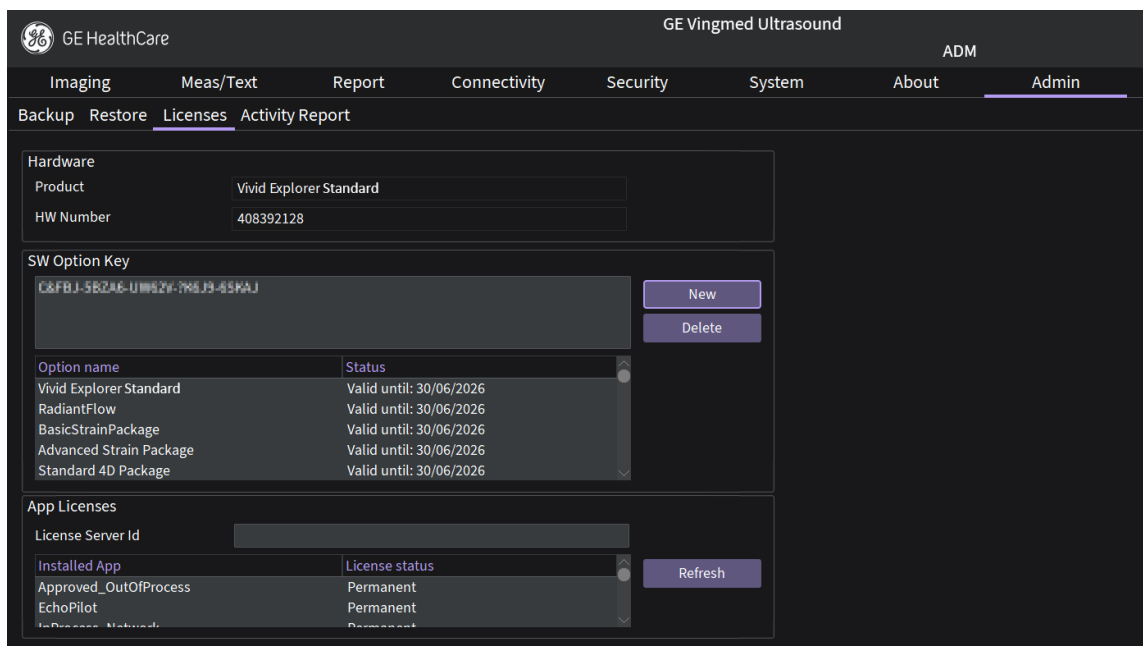


Figure 4-17 The License Subgroup

- Press the New button to enter the option key for Vivid Explorer Dimension and then press **Save**.

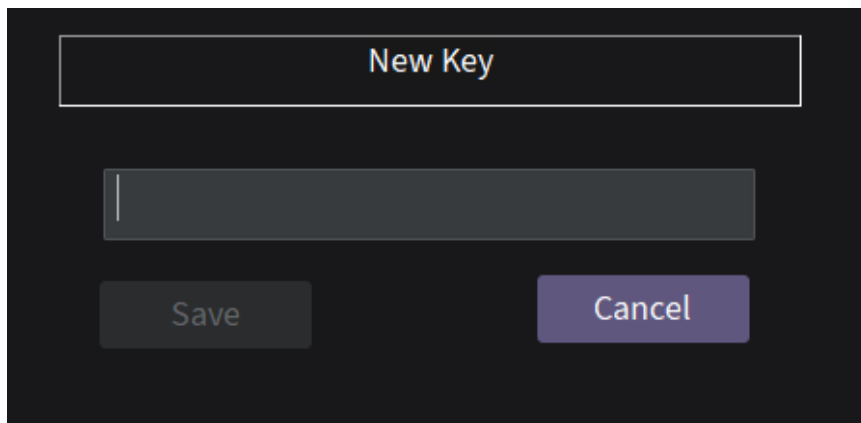


Figure 4-18 Add Option Key

- Ensure that the option key for Vivid Explorer Dimension is listed.

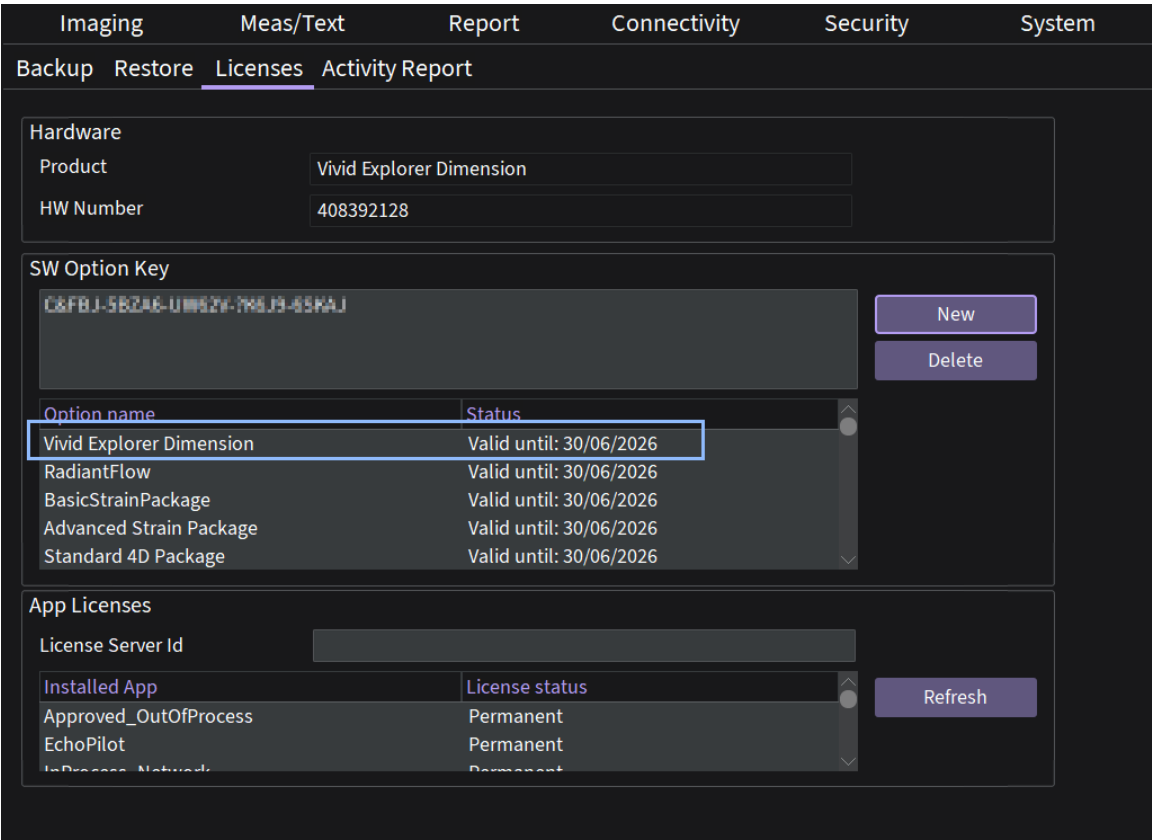


Figure 4-19 Vivid Explorer Dimension is Successfully Installed

5. Reboot the ultrasound system to ensure the name 'Vivid Explorer Dimension' is correctly displayed on the boot-up screen.

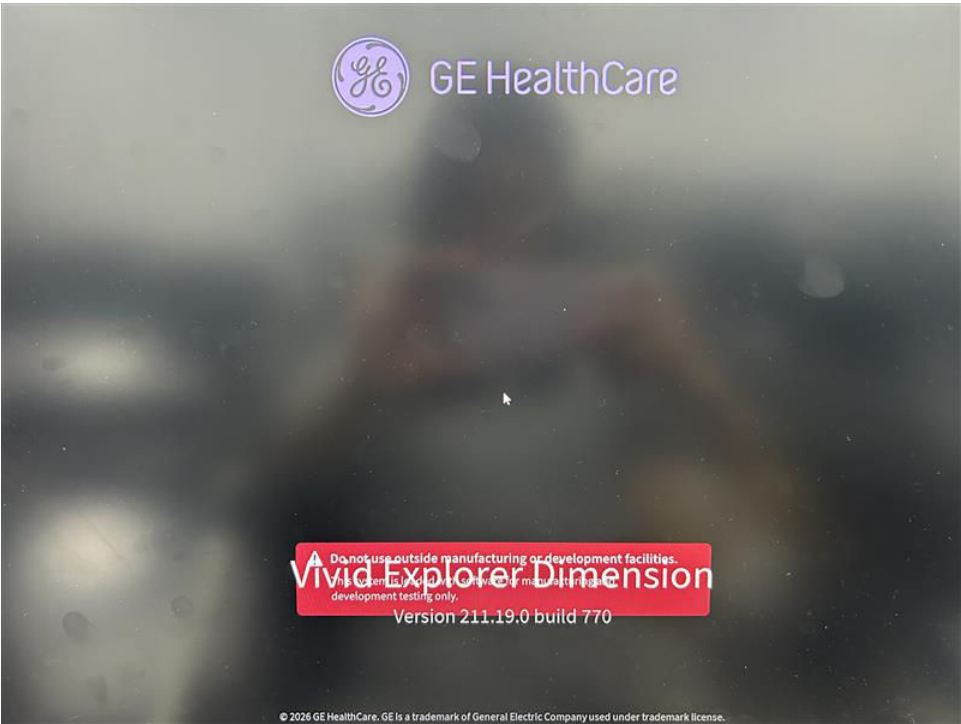


Figure 4-20 'Vivid Explorer Dimension' is Correctly Displayed on Boot-up Screen

6. For H48822DE(4D to Dimension Update), remember to apply the Dimension monitor sticker to the lower-right corner of the monitor. For H48822DD(2D to 4D Update). please ignore this step.



Figure 4-21 Apply 'Dimension' Sticker

4.4 Options

The following options are available for Vivid Explorer/Vivid Advanced/Vivid Focus:

- Ensure option keys are installed when you intend to use the following two features about battery:
 - Transportation mode
 - Battery scan mode
- Contact your region representative if you want to know more detailed information about software options for your system.

4.4.1 Software Options Configuration

Software Option Introduction

The needed software for the software options are preinstalled on the ultrasound system. A Software Option Key is required to activate the (purchasable) software options.

A Software Option Key is an alphanumeric text string that enables a software option or a combination of software options.

The Software Option Key is specific to each ultrasound system.

NOTE

More than one Software Option Keys may be in use, depending on the installed options.

Installing a Software Option Key

Follow these steps to install a Software Option Key:

1. Select **Config (F2)** and log on as ADM.
2. Select **Admin** (lower part of window).
3. Select the **System Admin** tab.
4. Select **New** to open the **New Key** dialog where you type the Software Option Key.

Remote Check and Configurations

Contact the Online Center for InSite checkout.

4.4.2 View-X

View-X is an option that enables streaming from an external video signal in a dedicated window on the system. The video signal is received by means of multimedia streaming over the IP network and without audio. For instructions to connect and set up View-X, consult the documentation for the device itself and the Service Manual.

Video streaming setup is described in the user manual and in the View-X with Epiphan Pearl Nano Broadcaster Installation Manual, Direction Number: GD294142-1EN..

View-X is available from the **Imaging** screen and toggles the streaming window On/Off. When toggled On a dedicated window showing the streamed video appears in the lower right corner by default.

NOTE

Network connection quality may impact streaming quality. Under ideal networking conditions a delay of approx. 1 sec will still persist.



Figure 4-22 Scanning window with View-X on

The size and position of the window can be adjusted by using the rotary control below the View-X button.

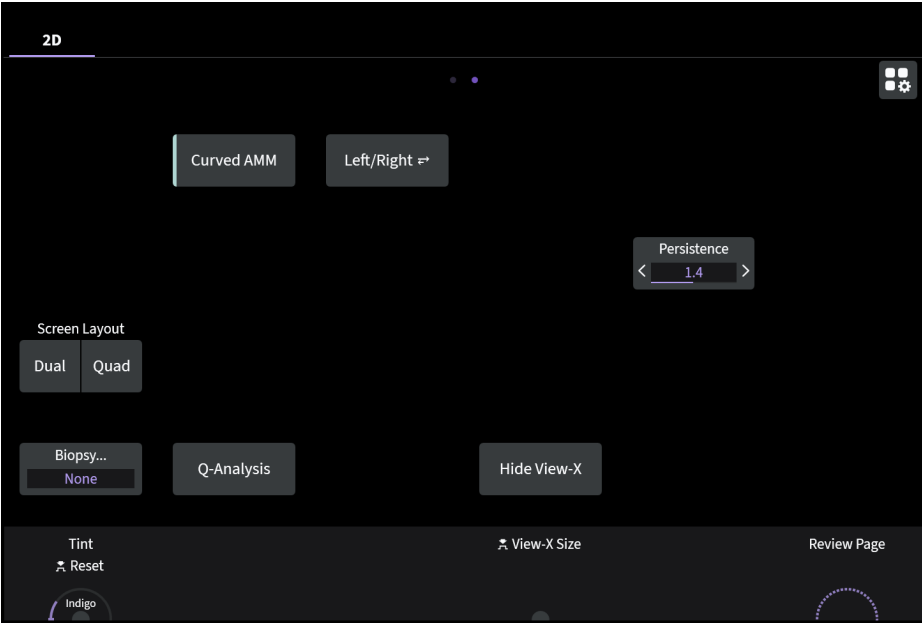


Figure 4-23 Rotary control for View-X size and position

4.5 Accessories

- [Accessory List](#) on page 112
- [Connection of Accessories](#) on page 112

4.5.1 Accessory List

The following external accessories are available for the Vivid Explorer/Vivid Advanced/Vivid Focus:

- Printer
- Probes
- ECG cables
- Wired Footswitch
- Transend DVD Writer kit
- External USB hard disk or USB memory stick
- Wireless USB Adapter
- ICE RS-DLP Adapter
- Elastic Cable Hook
- Network Breakway Adapter
- Gel Warmer
- Basket and Trays
- Charge Box
- Battery Pack
- Printer Shelf with Cable Kit
- Ergonomic P2D Probe Holder
- Crystal ICECord-RS
- EP Lab Accessory Kit
- Ferrite cores
- TEE Probe Storage Rack
- View-X

View-X is an option that enables streaming from an external video signal in a dedicated window on the system. The video signal is received by means of multimedia streaming over the IP network and without audio.

For instructions to connect and set up View-X, see Direction Number GD294142-1EN, View-X with Epiphan Pearl Nano Broadcaster Installation Manual.

4.5.2 Connection of Accessories

4.5.2.1 Connecting the Printer

An optional internal printer can be delivered with the ultrasound system.

4.5.2.2 Connecting the ECG

Connect the ECG cable to the ECG connector on the front of the system trolley.



4.5.2.3 Connecting the Footswitch

The Footswitch should be directly connected to any accessible USB-port on the Vivid Explorer/Vivid Advanced/Vivid Focus (e.g., on rear of the system or under speakers on both sides).

To configure functionality navigate from **Config > imaging > Appliation**.



Figure 4-24 Footswitch functional assignment

The graphic on the bottom of the main screen can be configured to show the functional assignment of the different pedals.

4.5.2.4 Connecting the DVD Writer Kit

The DVD Writer Kit should be directly connected to any accessible USB-port on the Vivid Explorer/Vivid Advanced/Vivid Focus.

1. Take out DVD and the cable and attach one end of he supplied cable to the DVD drive.



Figure 4-25 DVD and Cable

2. Connect the other end of the DVD drive to the USB port located under the speaker and put the DVD drive on the platform behind the Operator Panel.

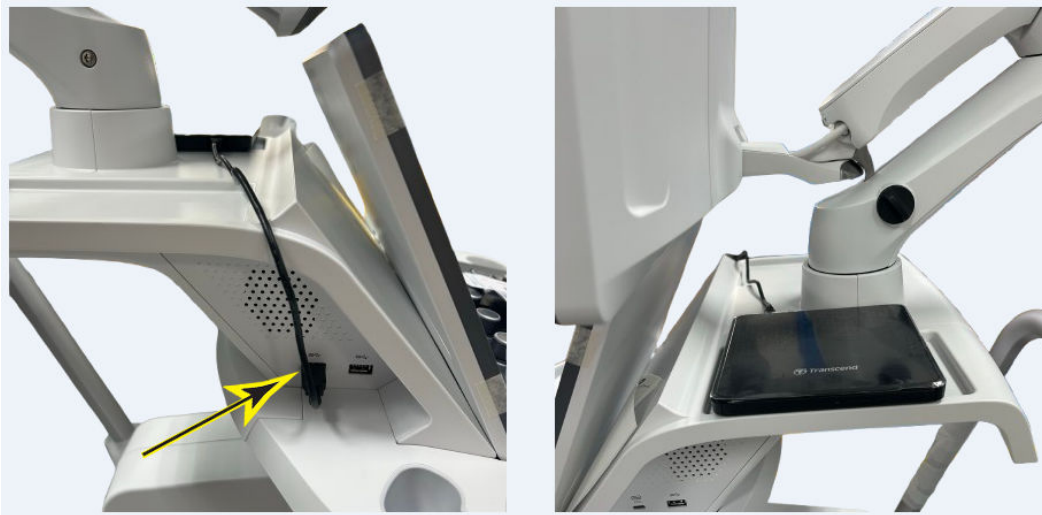


Figure 4-26 Connect DVD to Vivid Explorer/Vivid Advanced/Vivid Focus

NOTE

To prevent the DVD drive from falling, please connect the USB to the opposite side of where you place the drive. For example, if you put it on the left shelf behind the touch panel, plug the USB into the right USB port under the speaker and vice versa.

NOTE

Remove the DVD from the place behind the Operator Panel before moving the system. Failure to do so may result in the DVD falling and breaking during transport.

4.5.2.5 Connecting an External USB Hard Disk or USB Memory Stick

NOTE

Before connecting an USB device, please read [General Remarks when using external USB-Devices](#) on page 115. An external HDD or USB stick may be connected to any accessible USB port of the Vivid Explorer/Vivid Advanced/Vivid Focus system.

An external USB Hard Disk Drive or USB stick can be connected once the system is powered ON, or after shutdown. The Vivid Explorer/Vivid Advanced/Vivid Focus, Windows detects the device and automatically installs a driver. During this process, the message „New Physical Drive Has Been Attached” will be displayed at the bottom of the main screen.

NOTE

Memory drives or sticks may be sensitive to EMC interference. This may affect system performance and/or image quality.

4.5.2.5.1 General Remarks when using external USB-Devices**CAUTION**

Do not connect or disconnect any external USB-devices to or from the system while scanning a patient! The appearing dialogs could distract you from the scan!

When an external USB-storage device (such as an USB-memory stick or an external hard disk) is connected to the Vivid Explorer/Vivid Advanced/Vivid Focus, Windows detects the device and automatically installs a driver. During this process, the message „New Physical Drive Has Been Attached” will be displayed at the bottom of the main screen.

The device is then accessible using the drive letter the system assigned to it.

NOTE

If an external drive was not recognized automatically after connecting it, click **Rescan Drive**.

NOTE

When connecting external USB devices, be sure to execute Safety Directions found in the user manual.

4.5.2.5.2 Ejecting removable media

1. Press **Utility/Eject** on the Touch Panel.

NOTE

Do not eject CD/DVD using the button on the CD/DVD drive.

The **Eject device** menu is displayed ([Figure 4-27 The Eject device menu](#) on page 115).

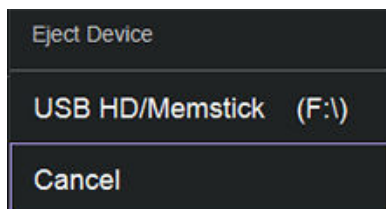


Figure 4-27 The Eject device menu

2. Select the relevant media.

Wait for the display of the Information window indicating that the device can be safely removed before removing the media.

4.5.2.6 Connecting the Wireless Network Adapter

It is recommended to plug the optional Wireless Network Adapter into the intended USB A-port, which can be seen by removing the Operator Panel Top Cover of the ultrasound system.

1. Turn the system power ON and wait until the system has booted.
2. Remove the OPIO Top Cover rear as described in [Operator Panel Top Cover Rear Replacement](#) on page 344.
3. Plug the Wireless Network adapter into the intended USB-A port of the ultrasound system as shown in [Figure 4-28 Designated USB A port for Wireless Network Adapter](#) on page 116. All software drivers are pre-installed for the designated Wireless Network adapter only.

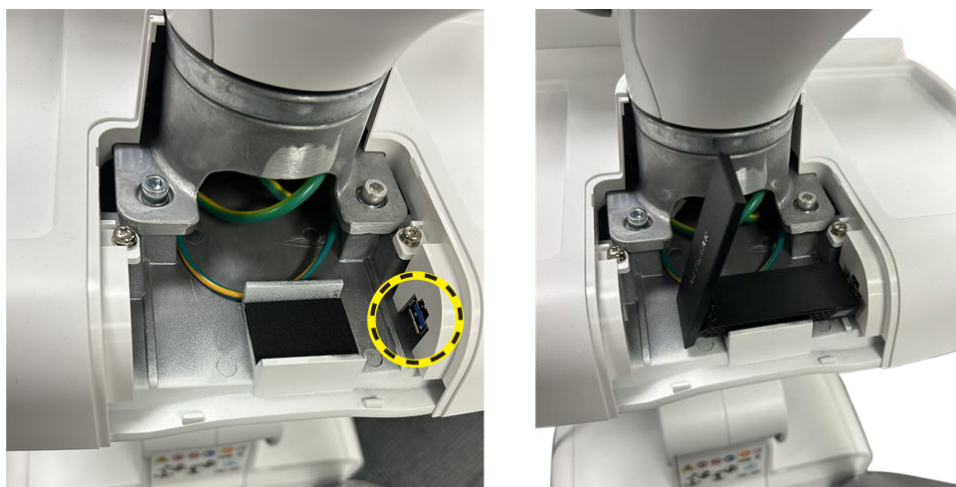


Figure 4-28 Designated USB A port for Wireless Network Adapter

4.5.2.7 Install the ICE RS-DLP Adapter

The ICE RS-DLP Adapter is designed to provide a connection between the DLP connector to the RS connector, connecting the Vivid Explorer and Vivid Advanced to ICE Catheters with RS connector.

Follow these steps to correctly use the ICE RS-DLP Adapter.

NOTE

Before connection, please ensure the 'DLP' knob is in the unlocked position; otherwise, it cannot be installed.



Figure 4-29 'DLP' Knob in Unlock Position before Connection

1. Connect the ICE RS-DLP Adapter to the DLP port and turn the knob to lock the adapter.

NOTE

There is an arrow on the adapter indicating the correct installation orientation. Please ensure the arrow is facing upwards when connecting to the system.



Figure 4-30 ICE RS-DLP Adapter



Figure 4-31 Connect the ICE RS-DLP Adapter to the DLP port and turn the Knob to Lock Position

2. Connect the ICE Catheters with RS connector to the Adapter and flip the connector locking lever to the right.



Figure 4-32 Connect the ICE Catheters with RS connector

4.5.2.8 Install the Elastic Cable Hook

The optional Elastic Cable Hook can be used to manage the cables.

1. Align the Elastic Cable Hook with the protruding part behind either of the two front cable hooks and insert it.

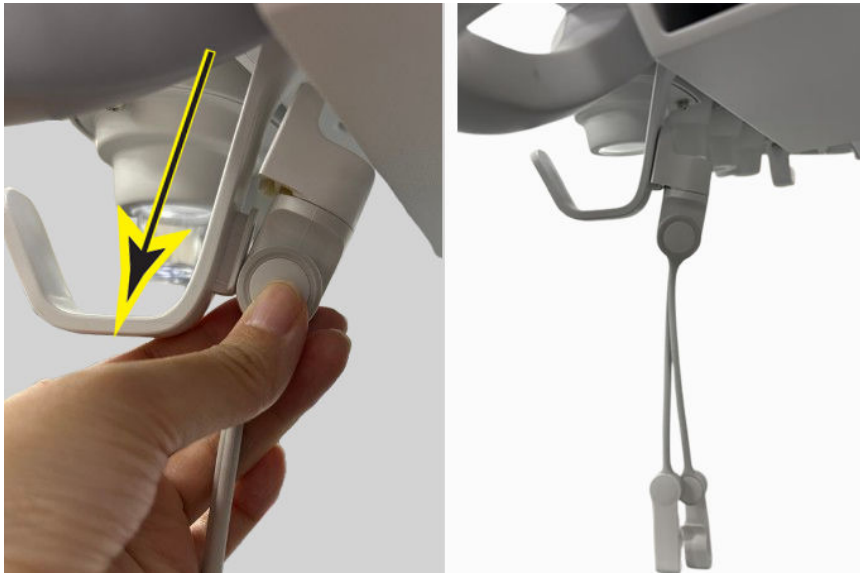


Figure 4-33 Install the Elastic Cable Hook

2. Hang the cable on the other end of the cable hook.

NOTE

Since the cable hook's wire is elastic, you can also hang the probe cable on the hook during the scanning of the probe.



Figure 4-34 Hang the Cable on the Elastic Cable Hook

4.5.2.9 Install the Network Breakway Adapter

Directly insert the network cable into the WLAN port at the rear of the system, the network cable tends to come loose easily. An optional Network Breakaway Adapter can be used to prevent the network cable from falling out.

1. Connect one end of the Network Breakway Adapter to the WLAN port at the rear of the system.
2. Insert the network cable into the other end of the Network Breakway Adapter.



Figure 4-35 Install the Network Breakway Adapter

4.5.2.10 Install Gel Warmer

An optional Gel Warmer can be delivered with the ultrasound system.

Follow the following instruction to install the Gel Warmer.

1. Unscrew 3 screws securing the Gel Holder to take out the Gel Holder.
2. Install the Gel Warmer in the original location of the Gel Holder from underneath, and secure it with 3 screws.

NOTE

The Gel Warmer can only be installed on the left-side gel holder position.

3. Plug the Gel Warmer's connection cable into the designated pre-installed socket under the Operator Panel.

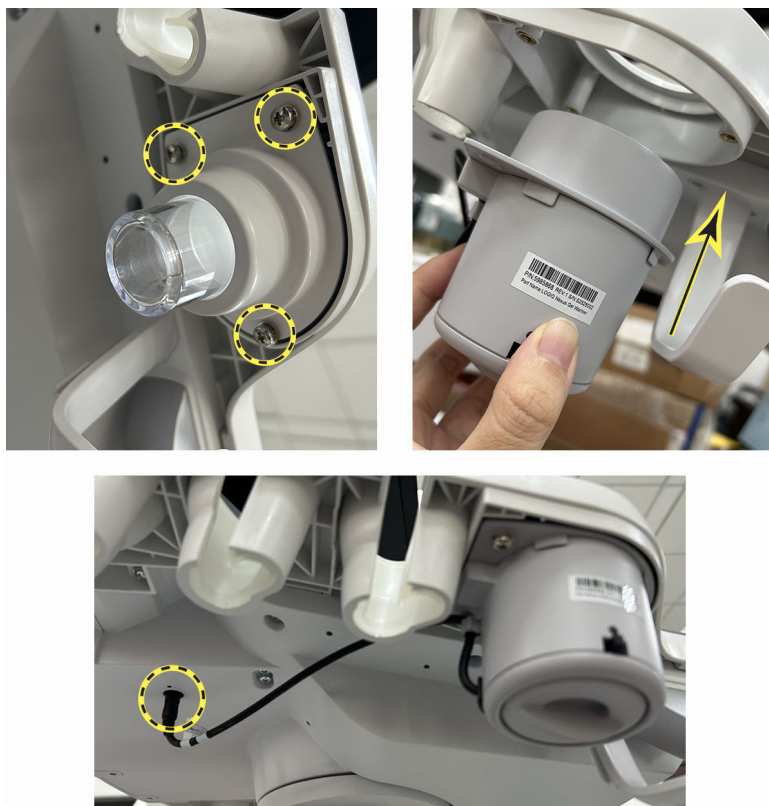


Figure 4-36 Install Gel Warmer

4.5.2.10.1 Functional Check

When the Gel Warmer switch is in the middle position, it is set to '0'. In this state, the indicator light is off. Turn the Gel Warmer switch to Level I and II, and check whether the indicator light is illuminating. Then place a coupling gel to verify if the heating function is working properly.



Figure 4-37 Gel Warmer Level Switch

4.5.2.11 Install the Baskets and Trays

Optional baskets and trays can be delivered with the ultrasound system.

For installation of a replacement unit, see the following information:

- [Trolley Rear Basket Replacement](#) on page 356
- [Trolley Mounted Dual Storage Bin Replacement](#) on page 355
- [OPIO Upper Deck Storage Tray](#) on page 357
- [Trolley Lower Front Tray Replacement](#) on page 359

4.5.2.12 Install Charge Box

An optional Charge Box can be delivered with the ultrasound system.

For installation of a replacement unit, see [Charge Box Module Replacement](#) on page 352.

4.5.2.13 Install Battery Pack

An optional Battery Pack can be delivered with the ultrasound system.

For instruction of a replacement unit, see [Battery Replacement](#) on page 348.

4.5.2.14 Install Crystal ICECord-RS

An optional Crystal ICECord-RS can be delivered with the ultrasound system.

For instruction of a replacement unit, see 6007620-1EN, Vivid™ Explorer/Vivid™ Advanced Addendum for 2D ICE option.

4.5.2.15 Install Ergonomic P2D Probe Holder

An optional Ergonomic P2D Probe Holder can be delivered with the ultrasound system.

1. Place the P2D Probe Holder onto the probe holder of the ultrasound system, ensuring that the opening direction aligns with the opening direction of the system's probe holder. And then securely install it.

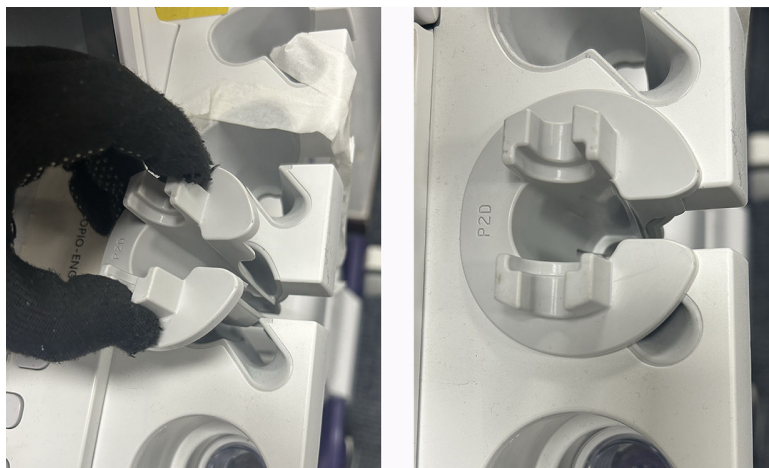


Figure 4-38 Install Ergonomic P2D Probe Holder

2. To uninstall the P2D Probe Holder, first push it upward from the bottom by hand, then remove it.



Figure 4-39 Uninstall Ergonomic P2D Probe Holder

4.5.2.16 Install Ferriter Core

The Ferrite Core (5834536) is used with the Crystal ICE Cord-RS (5877427-S).

1. Use the key to open the Ferrite Core.



Figure 4-40 Install Ergonomic P2D Probe Holder

2. Then wind the Crystal ICE Cord around the Ferrite Core as shown below and close it.



Figure 4-41 Wind the Crystal ICE Cord

4.5.2.17 TEE Probe Storage Rack

4.5.2.17.1 Introduction

The Storage rack is an accessory developed to provide secure storage of the TEE probes. It can be used as a probe stand in the operation room or echo-lab, or as an extension to the TEE Cleaning and Storing System in the cleaning room.

4.5.2.17.2 Overview

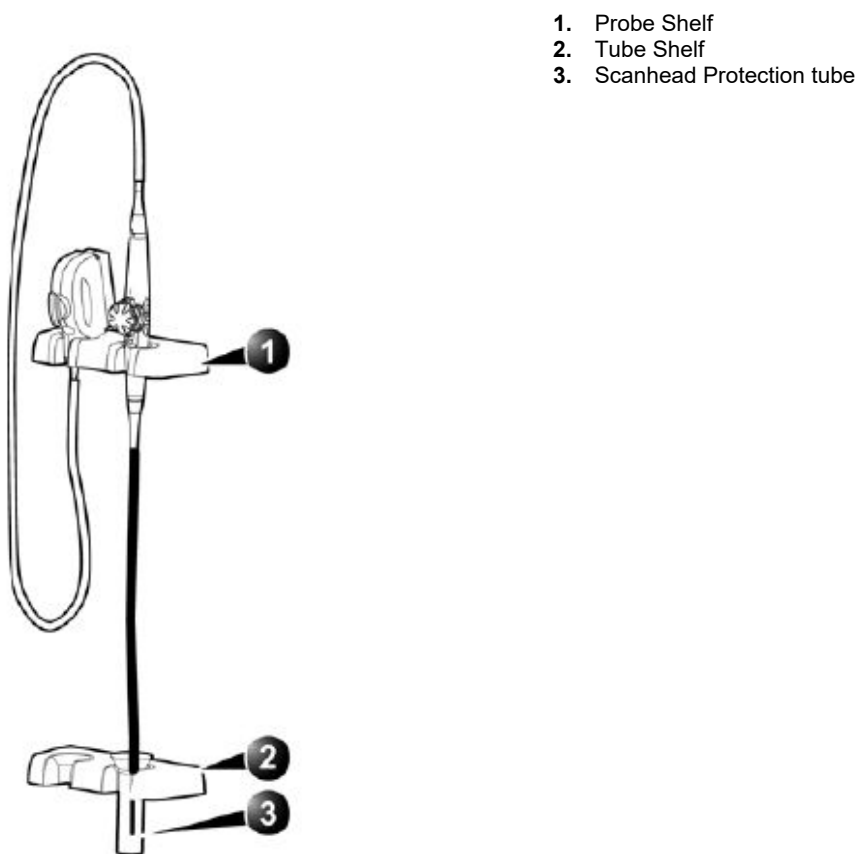


Figure 4-42 Storage Rack

4.5.2.17.3 Installation

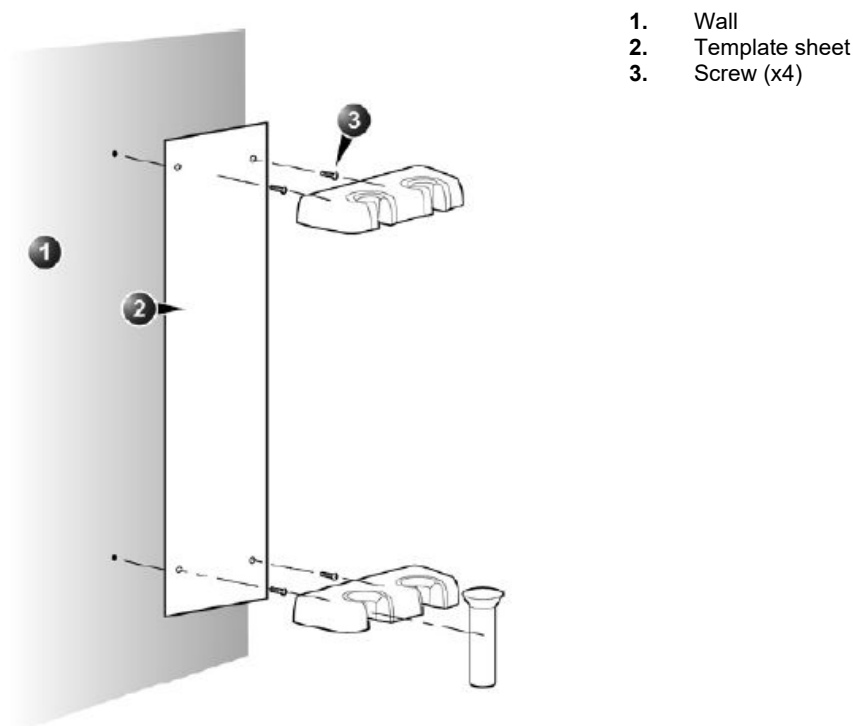


Figure 4-43 Installation Diagram

The unit is mounted on the wall using four screws. Two types of screws are provided; for wooden/gypsum or concrete wall.

The unit must be mounted on a solid wall (not a free-standing wall).

Table 4-6

Wall Type	Minimum Thickness
Wooden	10 mm
Gypsum board	13 mm

- Unpack the Storage rack and read the installation manual.
- Using tape, mount the template sheet vertically onto the wall. Make sure that there is enough clearance from the floor for the scanhead protection tube.
- Mark the position of the four mounting holes on the wall and remove the template sheet.
- Drill the holes with an appropriate gauge.
- Enter the screws into the wall.
- Mount the Storage rack elements onto the protruding screws.
- Mount the tube into the tube shelf.

4.5.2.18 Connect external monitor



CAUTION

Read the external monitor's user manual thoroughly before attempting to use the monitor.

1. Connect the external monitor to the system using an HDMI cable.

2. Connect the external monitor to mains as described in the monitor's user manual.



Figure 4-44 External monitor connection

NOTE

It might take some seconds before the external monitor is detected and the flexible video out connection stabilizes.

4.5.2.19 View-X

View-X is an option that enables streaming from an external video signal in a dedicated window on the system. The video signal is received by means of multimedia streaming over the IP network and without audio. For instructions to connect and set up View-X, consult the documentation for the device itself and the User Manual.

Video streaming setup is described in the user manual and in the View-X with Epiphan Pearl Nano Broadcaster Installation Manual, Direction Number: GD294142-1EN..

4.6 Product Locator Installation Card

NOTE

The provided Product Locator Installation Card may not be the same as shown in the Figure below.

 GE Medical Systems Mailing Address Product Locator File P.O. Box 414 Milwaukee, WI 53201-0414		GEMS-E Product Locator Administration BP 34 78533 Buc Cedex, FRANCE		Yokogawa Medical Systems Ltd. GEMSA Service Administration 4-7-127 Asahigaoka Hino-shi Tokyo 191, JAPAN	
DESCRIPTION	FDA	MODEL	REV	SERIAL	
SHIPMENT		OCP	BS	ORD	
		DISTRICT	CUSTOMER NO.		DATE (MO-DA-YR)
		DESTINATION NAME AND ADDRESS ZIP CODE			
 GE Medical Systems Mailing Address Product Locator File P.O. Box 414 Milwaukee, WI 53201-0414		GEMS-E Product Locator Administration BP 34 78533 Buc Cedex, FRANCE		Yokogawa Medical Systems Ltd. GEMSA Service Administration 4-7-127 Asahigaoka Hino-shi Tokyo 191, JAPAN	
DESCRIPTION	FDA	MODEL	REV	SERIAL	
SYSTEM I.D.		OCP	BS	ORD	EMPLOYEE NO.
INSTALLATION		DISTRICT	ROOM		DATE (MO-DA-YR)
		CUSTOMER NO.			
		DESTINATION NAME AND ADDRESS ZIP CODE			

Figure 4-45 Product Locator Installation Card

4.6.1 Paperwork

NOTE

During and after setup, the Vivid Explorer/Vivid Advanced/Vivid Focus and peripherals documentation, i.e. User Manuals, Installation Manuals, removable documentation media, et cetera, must be kept as part of the original system documentation package. This ensures that all relevant safety and user information is readily available during the operation and service of the ultrasound system.

4.6.2 Instructions for Use

1. Check that the correct manual(s) for the system and software revision is included with the installation.

NOTE

Specific language versions of the instructions may also be available. Check with your Sales Representative for availability.

2. Check that the preferred manual language is available on the ultrasound system.

Chapter 5

Alignment, Setup, Calibration

In this chapter

Mechanical Alignment Procedures.	128
Central Listing.	130
Configuration.	133
Calibration.	209
InSite.	213

5.1 Mechanical Alignment Procedures

5.1.1 Height Adjustment of the Operator Panel

Refer to [Control Console Positioning](#) on page 84 to get the detailed information about the height adjustment of the Operator Panel.

5.1.2 Position Controls for the Main Monitor

5.1.2.1 Lock the main display

1. Turn the release knob counter-clockwise to raise the lock.
2. Move the main display sideways to lock the arm's joints.
3. Push the display down into parked position.



1. Unlocked main display
2. Locked main display
3. Release knob

Figure 5-1 Main display locking mechanism

5.1.2.2 Unlock and move the main display

1. Turn the release knob clockwise to unlock the main display ([Figure 5-1 Main display locking mechanism](#) on page 128).

The main display can be moved freely in all directions.

2. Grab the bottom of the main display to adjust the position of the display ([Figure 5-2 Positioning the main display](#) on page 129).



Figure 5-2 Positioning the main display

NOTE

When flipping up the display from the flip down state, you can grab the upper corner of the display.

5.1.3 Castor Controls

5.2 Central Listing

5.2.1 LED Descriptions

- LEDs on the Operator Panel:
 - Backlight On/Off (Standby) switch (states: green= system on / white = system standby / blue = running on batteries / off = power is off)
 - Backlight buttons on the Operator Panel (states: green = active / white = available / off = not available)
 - Decoration lights in front of Operator Panel - White
- LEDs on the left side of E-cage (TBE Power on Diagnostic LEDs):
Remove the Trolley Left Cover (refer to [Trolley Left Cover Replacement](#) on page 345) and there is a label for LEDs on the left side of E-cage.



Figure 5-3 Labels for LEDs on the left side of E-cage

- **LED1: ACBOX Power OK**
Indicate that AC Power BOX 12V DC output power to TMBE is OK.
 - Connect AC power and switch AC breaker to on position
 - If LED1 light on or PWR_BTN's white LED backlight on OPIO panel light on , mostly ACBOX works fine
 - If LED1 does not light, consider ACBOX failure. Try to switch AC break to OFF position and unplug power cord for at least 1 minute, then plug back and switch on again. If ACBOX recover and output can work normally, nothing else need. Otherwise, replace ACBOX.
- **LED2: CPU PWR OK**
Indicate DC/DC buck converter circuit for CPU is ok on TMBE board.
 - Press power button to power on system.
 - If LED2 does not light, mostly DC/DC buck converter on TMBE do not work correctly. Try to switch AC break to OFF position and unplug power cord for at least 1 minute, then plug back and

switch on again. If TMBE recover and power on, nothing else need. Otherwise, replace TMBE board. replace TMBE board.

- **LED3: CPU POST OK**
Indicate CPU BIOS post ok on TMBE board.
 - If LED3 does not light on after LED2 is lighted (normally several seconds), mostly CPU part of TMBE do not work, consider replacing TMBE board first.
 - Also considering LED4 together, sometimes TMX board defect(such as PCIE hung) may also cause CPU cannot boot up normally.
- **LED4: MST PWR OK**
Indicate MSP MCU, DSP, FPGA power up is ok on TMX board.
 - If LED4 does not light on after LED2 is lighted (normally several seconds), mostly TMX board need replacement.
 - On some situations, try install newest software (with VividFirmwareUpdater) first before replacing TMX, some failures may be caused by firmware and software miss match, this can be fixed by software reloading, do not need hardware replace.

LED4 is only an indicator. It shows whether the FPGA and MCU on the TMX have completed the firmware loading and initialization process and have asserted the PWR_OK signal.

If LED4 does not light as expected, the possible causes include:

- Hardware issues, such as a faulty TMX board that may require replacement.
- Firmware issues, which may be resolved by reloading or reinstalling the software using the VividFirmwareUpdater built into the SW.
- Software issues, or an unexpected power loss during a firmware upgrade, the firmware can be repaired by reloading or reinstalling the software using the VividFirmwareUpdater built into the SW.

NOTE

All four diagnostic LEDs are intended for use only when the system cannot boot to the Windows level. If the system is able to boot into Windows and reach EchoLoader, there is no need to check the LED status.

*TMBE : Tandem Backend board with CPU and IO

*PWR : Power

*POST: Power-On Self-Test

5.2.2 Adjustable Components Description

The components below can be adjusted by the user of the ultrasound system:

- **Main Monitor**
 - sideways
 - up/down
 - tilt
- **Monitor Arm**
 - sideways
 - up/down
- **Operator Panel (Control Console)**
 - sideways
 - up/down
- **Casters (wheels)**
 - engaging and releasing brakes

There are no components on the ultrasound system that require service adjustment.

5.3 Configuration

Overview of the Configuration section:

- [System Configuration](#) on page 133
- [Connectivity tab](#) on page 135
- [Security tab](#) on page 181
- [System tab](#) on page 194
- [About tab](#) on page 196
- [Admin tab](#) on page 197
- [Service tab](#) on page 203

5.3.1 System Configuration

5.3.1.1 About system Configuration

The configuration package enables customization of the global configuration of the system, application-specific settings, system connectivity, and data management settings.

The configuration management package is divided into different setup categories with subgroups.

Only users with administrative rights have access to all the subgroups.

Setup category	Description	Subgroup	Access
Imaging	Global imaging, shortcuts for touch panel buttons, scan information, application settings, TEE probe button assignment, probe test intervals, FlexiViews configuration, 4D Markers setup and Scan Assist configuration	Global	All
		Shortcuts	All
		Scan Info	All
		Application	All
		Probe Buttons	All
		Probe Check	All
		FlexiViews	All
		Scan Assist Pro	Admin
		4D Marker	All
		Scan Assist Pro	All
Meas/Text	Measurements and Annotation configuration	Measurement menu	All
		Advanced	All
		Modify Calcs	All
		OB Tables	All
		DICOM Mapping	Admin
		Text	All
		Bodymark	All
		Options	All
		Advanced Quantification	All

Setup category	Description	Subgroup	Access
Report	Report configuration	Template	All
		Diagnostic Codes	All
		Comment Texts	All
		Structured Findings	All
		Options	All
Connectivity	Connectivity configuration	Network	Admin
		DICOM	Admin
		Dataflow	Admin
		Transfer Media	Admin
		Buttons	All
		Tools	All
		Remote Path	Admin
		Disk management	Admin
		Printer	Admin
		Interventional	Admin
		Other	All
Security	Security settings and user management	Users	Admin
		User Policies	Admin
		LDAP	Admin
		System Password	Admin
		Disk Encryption	Admin
		Login banner	Admin
		Audit	Admin
System	General system settings and system test	Settings	Admin
		Battery	All
		Test	Admin
		Power Saving	All
About	System information	System Version	All
		SW version	All
		Firmware Version	All
		HW Version	All
		Probes	All
		Battery	All

Setup category	Description	Subgroup	Access
Admin	Data management and licenses	Backup	Admin
		Restore	Admin
		Licenses	Admin
		Activity report	Admin
Service	Service	Service	Admin

5.3.1.2 Imaging, Meas/Text, Report

The **Imaging** tab, the **Meas/Text** tab and the **Report** tab are described in the User Manual.

5.3.2 Connectivity tab

5.3.2.1 Connectivity

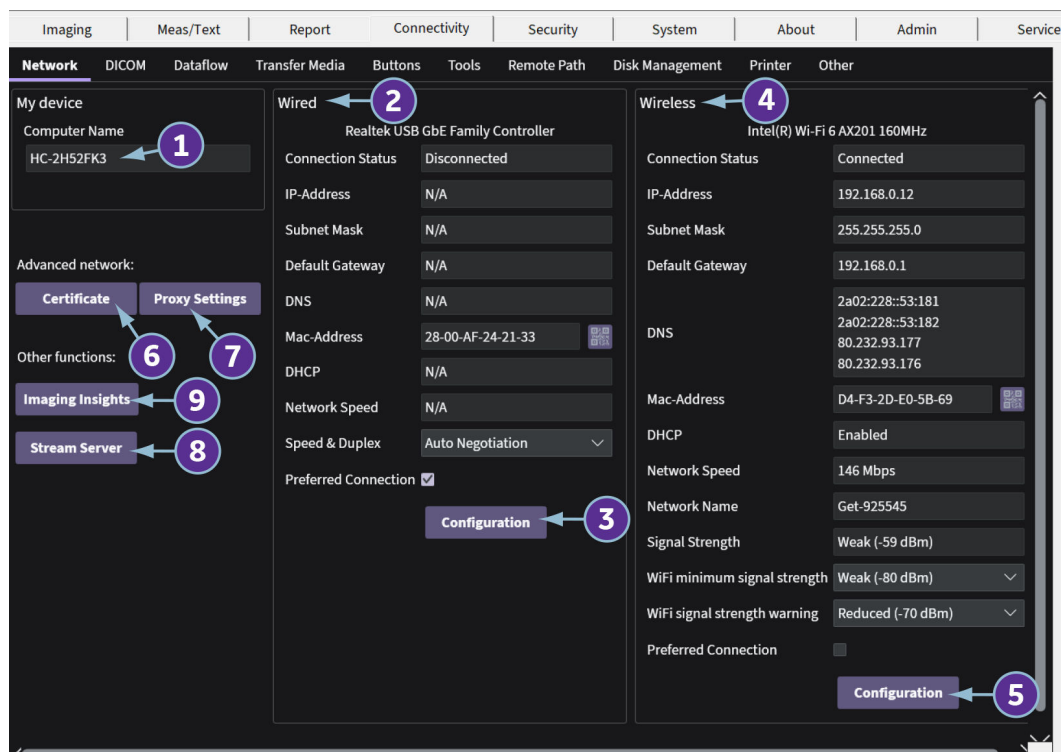
This section describes the communication and connection options for the Vivid Explorer/Vivid Advanced/Vivid Focus ultrasound system with other devices in the hospital information system. This section covers the procedures for configuration and optimal data management from a Vivid Explorer/Vivid Advanced/Vivid Focus in the following scenarios:

- A stand-alone Vivid Explorer/Vivid Advanced/Vivid Focus
- A Vivid Explorer/Vivid Advanced/Vivid Focus and workstation in a network
- A Vivid Explorer/Vivid Advanced/Vivid Focus and a DICOM server in a network.
- A Vivid Explorer/Vivid Advanced/Vivid Focus and an EchoPAC Software Only in a direct connect environment.

5.3.2.2 Network sheet

5.3.2.2.1 System's Network settings

1. Press **Utility > Config** on the Touch panel and log on as administrator.
2. Select the **Connectivity** category and **Network** subgroup.
The **Network** subgroup is displayed.



1. My Device.

Computer name: is shown but not editable.

2. Wired network adapter. Fields are automatically populated from the network adapter settings. In order to change them, press the **Configuration** button (3) in Windows. Speed and Duplex can be selected on this page.

3. Configuration: configure the Wired connection.

4. Wireless network adapter. Fields are automatically populated from the network adapter settings. To change them, press the **Configuration** button (5), (see [Wireless Setup](#) on page 139).

Note that both Wi-Fi minimal strength and strength warning can be selected on this page.

Note that the **Preferred Connection** method (wired/wireless) can be selected by checking a box.

5. Configuration: configure the Wireless connection.

6. Certificate: Click this button to enter certificates.

7. Proxy Settings: Configure network proxy settings (see [Network Proxy Settings](#) on page 138).

8. Stream Server Settings: Configure streaming of live ultrasound image data.

9. Imaging Insights: Configure connection to the Imaging Insights server by providing device usage information.

Figure 5-4 Network subgroup

3. Select **Configuration** (in the Wired network section) to configure:

- IP address for the system
- Subnet mask for the system
- IP address for the Default Gateway

This is done at the Windows level.

5.3.2.2.2 Connectivity Setup

NOTE

If connected to a stand-alone network (Peer-to-Peer network with a Vivid Explorer/Vivid Advanced/Vivid Focus ultrasound system, an EchoPAC workstation and an optional network printer), use default delivery settings.

5.3.2.2.3 Imaging Insights

Imaging Insights automatically collects DICOM data from GE HealthCare and other vendors' ultrasound equipment or equipment fleet and displays system utilization and operator usage insights in a plotted dashboard. Operational insights include exam volumes, first and last exam time, probes utilization, exam type; operator usage data includes length of exam, scan mode, probes, and exam type. It helps optimize system and probe fleet investment plans, identify staff assignment and training needs, and monitor variability of staff usage patterns.

1. Log on as ADM, press **Utility > Config** on the Touch Panel.
2. Select the **Connectivity** category and the **Network** subgroup.

The **Network** sheet is displayed.

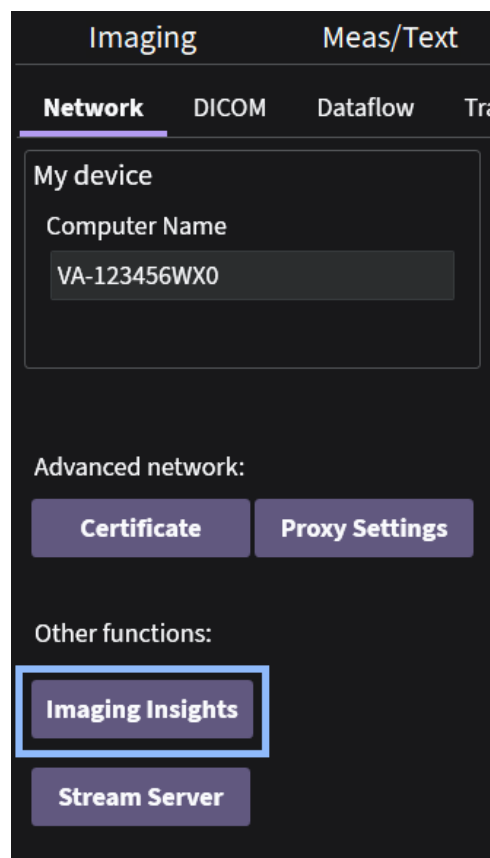


Figure 5-5 Imaging Insights button

3. Press **Imaging Insights** button and a window will pop up.
4. Configure the server information from the pop-up window and then click **Check**.

Imaging Insights

☒ Send data to Imaging Insights **View statistics**

Server

IP-Address: 10.191.63.132 **Check** ✓

AE Title: JS

Port: 5104

Save Settings **Close**

Figure 5-6 Server Configuration

5. If the network connection to the server is OK, it will be illustrated by the “Pass” sign, a green check mark .

The “Fail” sign  indicates that the connection has failed.

Typical causes for the issue:

- Network cable is not connected
- Configuration error(s)

6. Choose **Send data to Imaging Insights**, then the data from this scanner will be sent to the Imaging Insights server.

NOTE

If an exam is done with patient information created, the data will be sent to server immediately when selecting **End Exam**. If an emergency scan is done without creating patient information, the data will be sent to the server every two hours until system shutdown.

NOTE

Press the **View statistics** button to check whether the data is sent successfully or not.

5.3.2.2.4 Network Proxy Settings

How to setup Network Proxy Settings

Press the **Proxy Settings** button on the **Connectivity > Network configuration** screen to show Network Proxy Settings dialog.

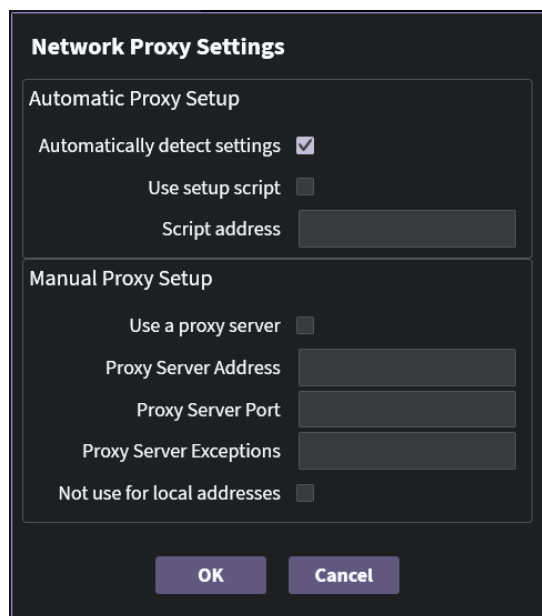


Figure 5-7 Network Proxy Settings window

Automatic Proxy Setup

Enable the **Automatically detect settings** option to use Web Proxy Auto-Discovery Protocol (WPAD) for detecting proxy settings. If the connected network requires a proxy and it provides that proxy via WPAD, the proxy will be automatically configured and used.

Enable the **Use setup script** option and insert the network address of the script into the Script address box to setup script for your proxy configuration. This script may also be referred to as a .PAC file.

Manual proxy setup

Enable the **Use a proxy server** option to manually configure the proxy server. The manual configuration of a proxy requires you to have a specific IP address and port for the server that you want to use. This information should be entered in the **Proxy Server Address** and **Proxy Server Port** fields.

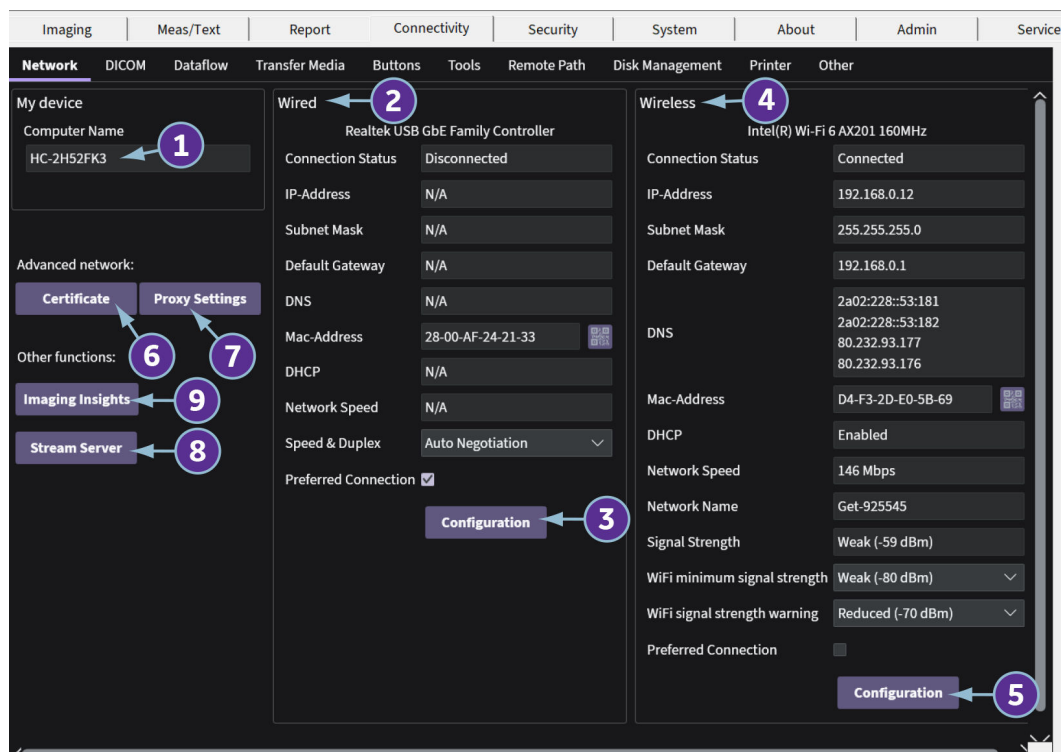
For addresses you do not want the system to use the proxy server on, enter web or IP addresses separated with a semicolon (;) in the **Proxy Server Exceptions** field.

Enable the **Not use for local addresses** option to bypass the proxy server when you connect to resources on your local network or intranet.

Click **OK** to save and apply network proxy settings or click **Cancel** to ignore the recent changes.

5.3.2.2.5 Wireless Setup

In order to configure the Wireless adapter (if used), click the Configuration box (5) in the Wireless section.



This is done at Windows level, but both **Wi-Fi Minimum Signal Strength** and **Wi-Fi Signal Strength Warning** can be configured directly on the **Connectivity > Network** page.

NOTE

Refer to [System's Network settings](#) on page 135 for detailed introduction of **Network** tab.

Wireless Network Setup

The following procedure is used to configure the ultrasound system for a wireless network environment. This procedure is required for every new wireless network.

NOTE

Do not use any type of wireless network adapter other than a GE HealthCare-approved Wireless Adapter.

NOTE

If you have questions regarding the configuration, please contact your local IT expert.

Configuring the Wireless Network adapter

1. From the default ultrasound system screen, press **Config** and log on as ADM.
2. Select **Connectivity** (lower part of window).
3. Select the **Network** tab.

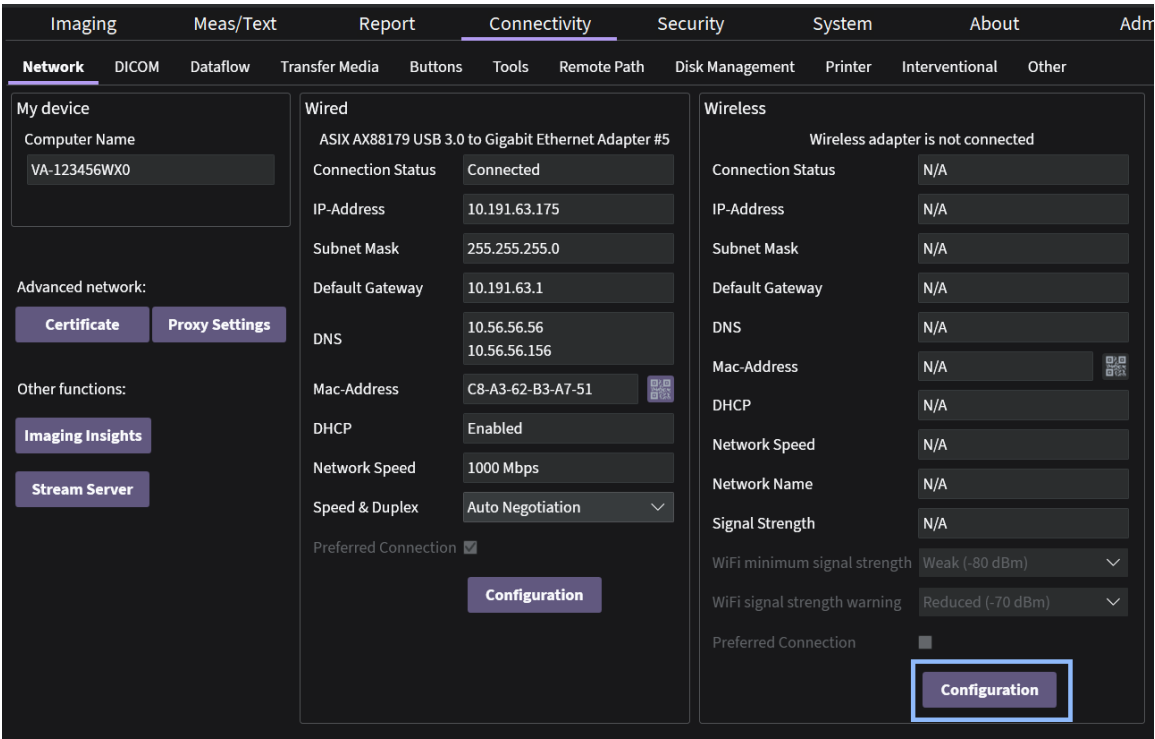


Figure 5-8 Connectivity - Network tab

4. Click the **Configuration** button in the lower right corner.

NOTE

If a wireless network adapter is not connected or if the connected wireless network adapter is not the correct model, no wireless networks will be listed in the **Wireless Settings** page.

5. Either double-click or select the network you want to connect to, and press the **Connect** button.
If this is the first time you attempt to connect to the network, a dialog will pop up on screen asking you to configure the network setting.

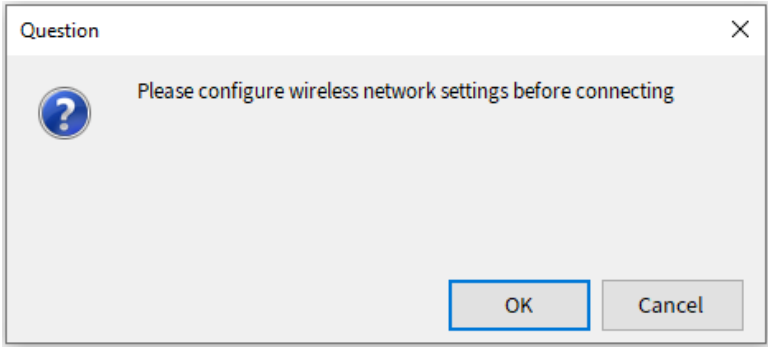


Figure 5-9 First Time Connection Dialog

6. Press **OK** in the dialog window. A new window for setting up your connection will open.

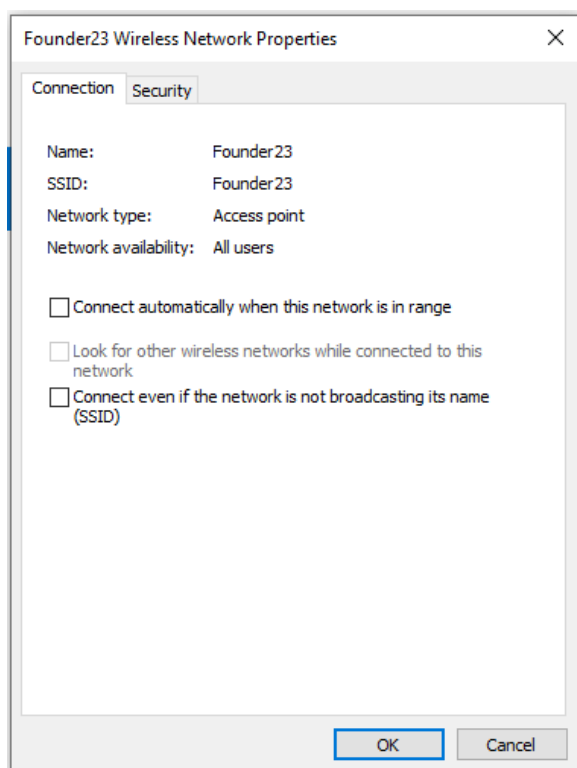


Figure 5-10 Network Settings Window - Connection

7. Select check-boxes according to preference.

If the **Connect automatically when this network is in range** setting is selected, then this network will auto-connect when available, without needing to enter the **Wireless Settings** page.

8. Set up the security options in the **Security** tab of connection setup dialog.

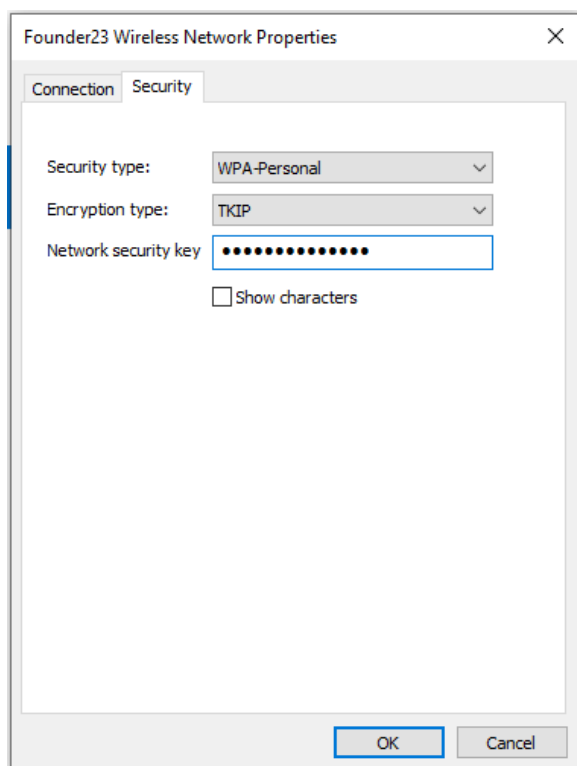


Figure 5-11 Network Settings Window - Security

9. If the wireless network is to be configured for WPA/SWPA2 Enterprise and a customer-specific certificate is to be installed on the system, this can be done by using the certificate dialog available from the **LDAP** config page in the **Connectivity — Security** section or from **Connectivity — Network** section.
 - a. First, go to the **LDAP** config.

Imaging Meas/Text Report Connectivity **Security** System

Users User policies **LDAP** System password Disk encryption Login banner Audit

☒ Enable LDAP authentication

Connection configuration

Directory server URI Port ☒ SSL

Domain

DN for users

User caching

☒ Enable caching Remember user days

Attribute mapping

Attribute	LDAP attribute
firstname	givenName
lastname	sn
prefix	personalTitle
phone	telephoneNumber
email	mail
address	postalAddress

Group mapping

Group	LDAP group(s)
Cardiologist	
ConsultingPhys	
DiagPhys	
Fellow	
GEAdmin	
HospAdmin	
Operator	
Physician	
RefDoc	

Figure 5-12 LDAP config screen

Imaging Meas/Text Report **Connectivity**

Network DICOM Dataflow Transfer Media Buttons Tools Remote Path

My device

Computer Name

Advanced network:

Other functions:

Wired

ASIX AX88179 USB 3.0 to Gigabit Ethernet Adapter

Connection Status

IP-Address

Subnet Mask

Default Gateway

DNS

Mac-Address

DHCP

Network Speed

Speed & Duplex

Preferred Connection ☒

Figure 5-13 Network config screen

- b. Then click on the **Certificates...** button to open the **Certificate** dialog.
10. After you have finished setting up your connection, press **OK**. The ultrasound system will then try to establish a connection to your network. A dialog will be shown on screen while this is in progress.

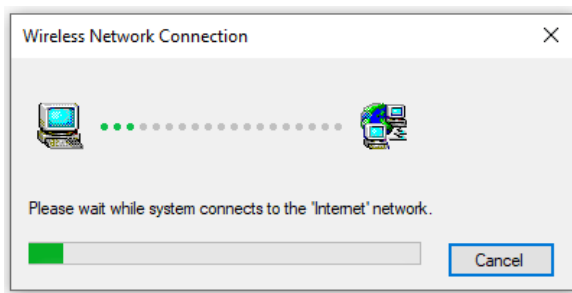


Figure 5-14 Connecting to Network Dialog

11. Once a connection has been established you will see the status **Connected** in the **Wireless Settings** page, next to the network you have connected to.

All network connections that are configured will also be displayed with a star icon.

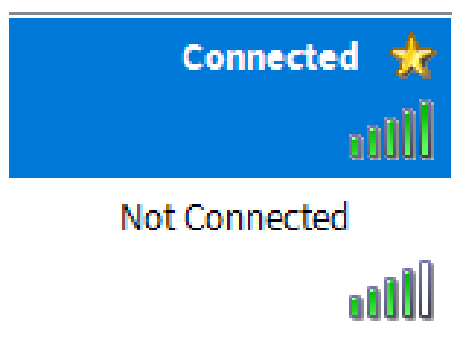


Figure 5-15 Connection Established

NOTE

Whenever connection to a new/different wireless network is required, it will be necessary to repeat all procedure steps above.

5.3.2.3 DICOM sheet

5.3.2.3.1 System in a network environment with a DICOM server

The system's Network settings must be configured as described in [System's Network settings](#) on page 135.

In addition, to work against the DICOM server, the following information has to be entered in the system:

- The DICOM server IP address
- The DICOM server port number
- The DICOM server AE title (the server's name)

NOTE

Changing the DICOM server AE title requires a reboot of the system to take effect.

Typically, source for this information is the network administrator.

5.3.2.3.2 Create or modify a DICOM server IP address

Follow the steps below if the IP address settings for the DICOM server need to be modified or created:

1. Press **Utility/Config** on the Touch panel and log on as administrator.
2. Select the **Connectivity** category and **Dataflow** subgroup.
The **Dataflow** sheet is displayed.
3. Click on the **Servers and Diagnostics** button in the lower right corner.

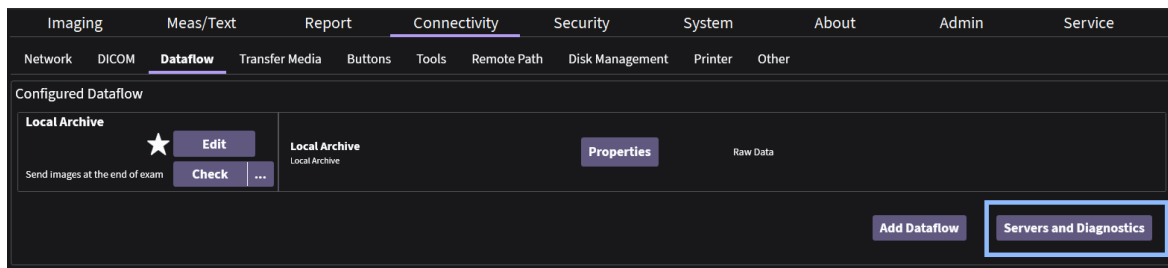


Figure 5-16 Dataflow-Servers location

- Click **Add** to add a new server or **Modify** to change an existing one. To remove one, highlight the server and click **Remove**.

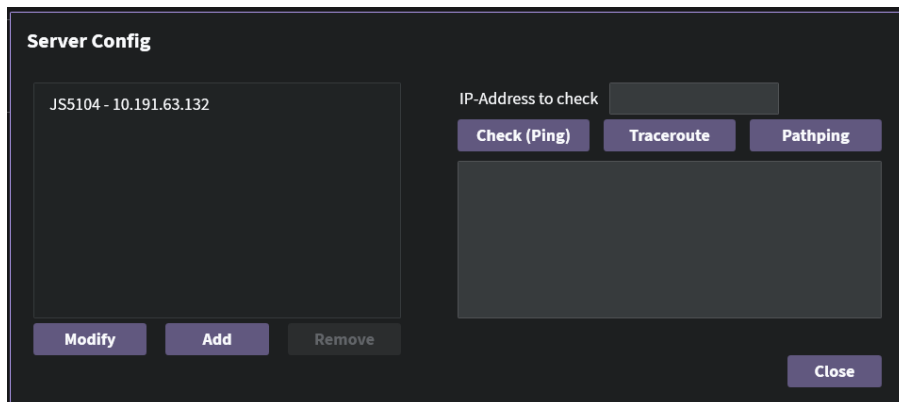


Figure 5-17 Server configuration window

- Enter the name of the server, and typically deselect **Automatic** and enter the IP address manually.

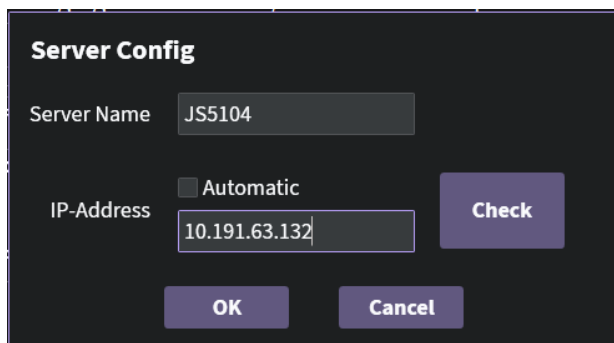


Figure 5-18 Entering the server name

- Verify connection to the server by pressing the **Check** button.
- Press **OK** to complete the setup of the server Name and IP address.

5.3.2.3.3 DICOM devices configuration

Depending on the DICOM dataflow selected, one or several DICOM devices may have to be configured.

- Press **Utility > Config** on the Touch panel and log on as administrator.
- Select the **Connectivity** category and **Dataflow** subgroup.
The **Dataflow** sheet is displayed.
- Select **Add Dataflow** (bottom right) to enter a new dataflow and choose from a factory template or create a Custom one from scratch.

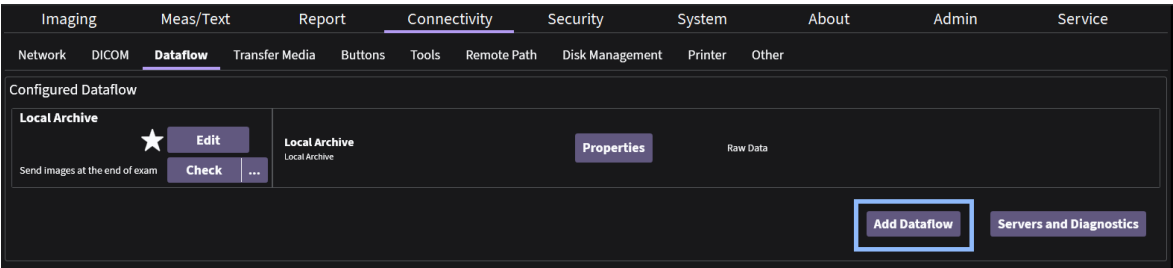


Figure 5-19 Dataflow selection window



Figure 5-20 New Dataflow selection

4. In this example, the factory **Worklist / Local Archive / DICOM Storage** is used. The screen shows a list of what type of services are used and various preference settings for the dataflow.

Custom Dataflow

Dataflow Name:

☐ Default dataflow Default dataflow is on the top of the list

☒ Show in Patient search / Transfer screen

☒ Search as you type

☐ Show today's exams by default

Send images: ☒ Immediately ☐ At the end of exam

☐ Use for transfer purpose only

Available Devices

- ☒ Dicom Worklist Get list of scheduled examinations
- ☐ Remote Archive Use the database of an EchoPAC server
- ☒ Local Archive Use the internal patient database
- ☐ Dicom QR Read patients from a PACS
- ☒ Dicom Storage Store to PACS
- ☐ Dicom Storage Store to PACS

OK Cancel

Figure 5-21 Custom Dataflow Settings

- Click **OK** when completed.
- Note that the different service need to be set up next. Each service must be configured by clicking on the respective **Properties** buttons.

Imaging Meas/Text Report **Connectivity** Security System About Admin Service

Network DICOM **Dataflow** Transfer Media Buttons Tools Remote Path Disk Management Printer Other

Configured Dataflow

Local Archive Edit Send images at the end of exam Check	Local Archive Local Archive Properties	Raw Data
Worklist / Local Archive / DICOM Storage Edit Send images immediately Check	Worklist Dicom Worklist IP: AE Title: Properties	Query result limit: 70 Raw Data
	Local Archive Local Archive Properties	
	DICOM Storage Dicom Storage IP: AE Title: Properties	MF: SR private data

Add Dataflow Servers and Diagnostics

Figure 5-22 Configured dataflow window

- DICOM Storage Properties:

In the Properties for DICOM Storage:

- Select the server previously defined from the pull-down menu in the **IP address** field.
- Enter the DICOM server **AE title**. This entry is case-sensitive and must match exactly.
- Enter the DICOM server port number.

For some DICOM servers, the default **Timeout** setting may be too low.

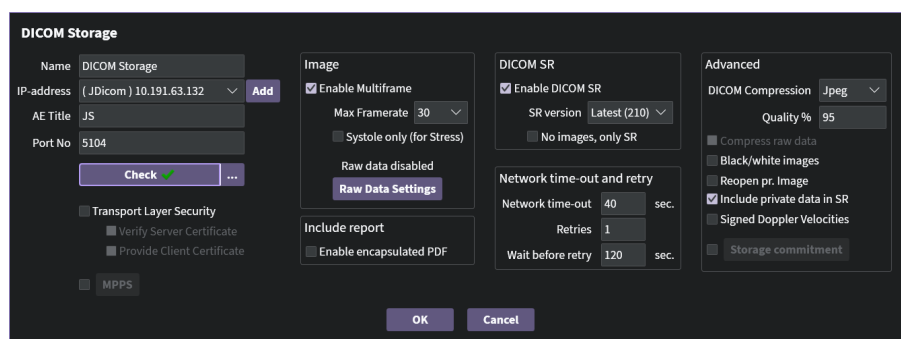


Figure 5-23 DICOM storage properties

8. When configuring the DICOM storage device, the following image settings are recommended to enter in the Properties.

- Check **Enable DICOM SR** if required (see [DICOM SR](#) on page 158).
- Keep **Reopen pr. image** unchecked.
- Keep **Raw data** disabled.
- Set **Max Frame rate** to 30.
- Keep **Black/White images** unchecked.
- Set **DICOM Compression** to JPEG.
- Set **Quality** to 95.
- Check **Enable Multiframe**.

Raw Data Settings gives you the ability to select which images to transfer in the raw data format by mode:

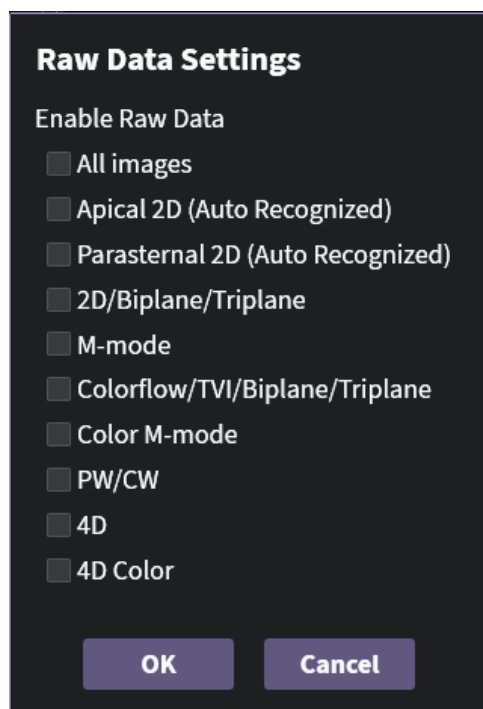


Figure 5-24 Raw Data settings

Systole only (for Stress). When this setting is active, all stress images will be sent to the DICOM server showing only systole. The setting is selectable when transferring only multiframe (not raw data).

NOTE

Setting **Compression** to **None** may result in long transfer time and cineloop with more than 500 frames will be truncated. If **Compression** is set to **None**, set **Max frame rate** to either 25 or 30 frames per second to reduce the risk of truncating loops.

5.3.2.3.4 DICOM / TLS

To use TLS protocol for secure communication with the DICOM server, the “Transport Layer Security” option must be checked off on the DICOM dataflow (the default setting).

NOTE

DICOM / TLS must be supported by the DICOM server.

1. Open the **Properties** view for the DICOM dataflow, as described in [DICOM devices configuration](#) on page 146.
2. Ensure the **Transport Layer Security** box is checked as shown below:

Figure 5-25 Check the ‘Transport Layer Security’

3. (Optional) Select **Verify Server Certificate** to ensure the connected DICOM server is trusted by the system (the default setting).
4. (Optional) To add server or client certificate, open the Certificate utility accessible from **LDAP config** view, see [Digital certificates](#) on page 150.
5. (Optional) To enable use of client certificate, check off for **Provide Client Certificate**. See [Select Client Certificate](#) on page 155.

NOTE

The client certificate must be selected before the **Provide Client Certificate** check box can be checked off.

5.3.2.3.5 Digital certificates

The digital certificates can be managed from the **LDAP** configuration.

Public certificates can be added to trust connected devices. While client certificates with a private key can be added for authenticating the Vivid system towards a connected device.

Import/delete Certificate

1. Press **Utility/Config** on the Touch Panel as administrator.
2. Select the **Admin** category and the **LDAP** subgroup.
The LDAP sub group is displayed.

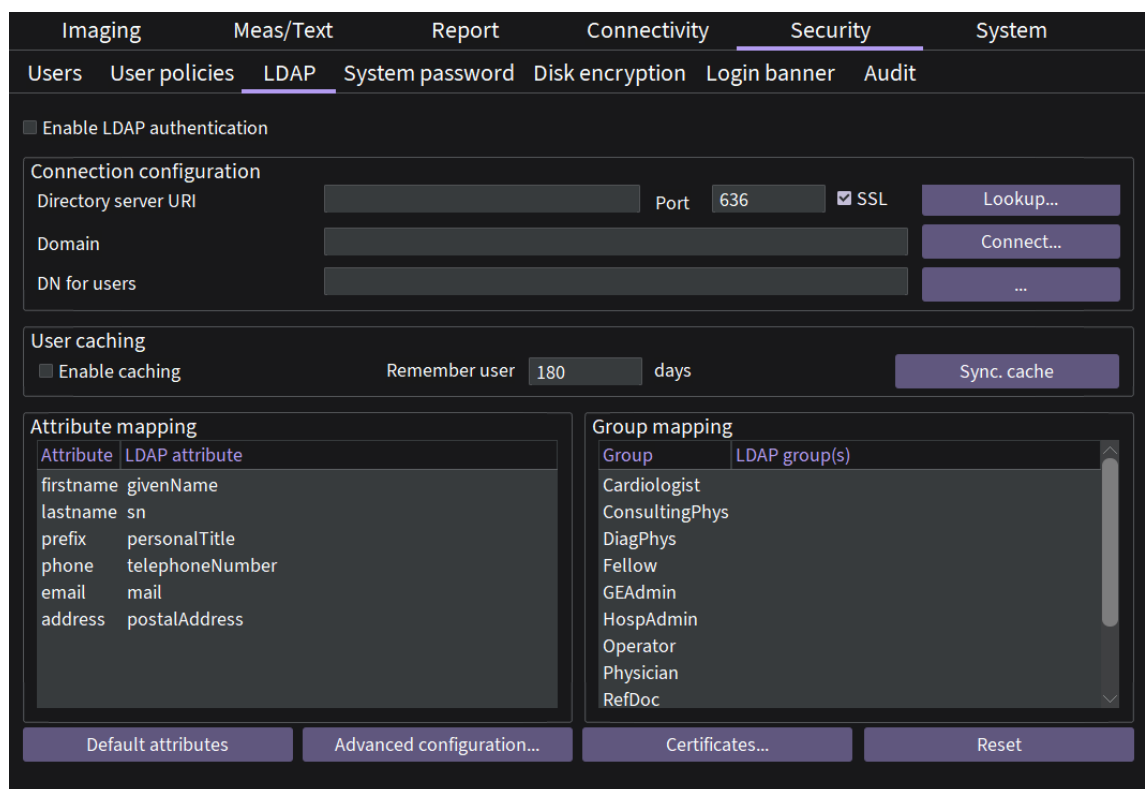


Figure 5-26 LDAP window

- Click on the **Certificates...** button.

The **Certificate Manager** windows is displayed.

- In the **Certificate Manager** window, select the folder where the certificate is to be imported to.
- Click **Action, All Tasks** and then **Import...** from the drop-down menu.

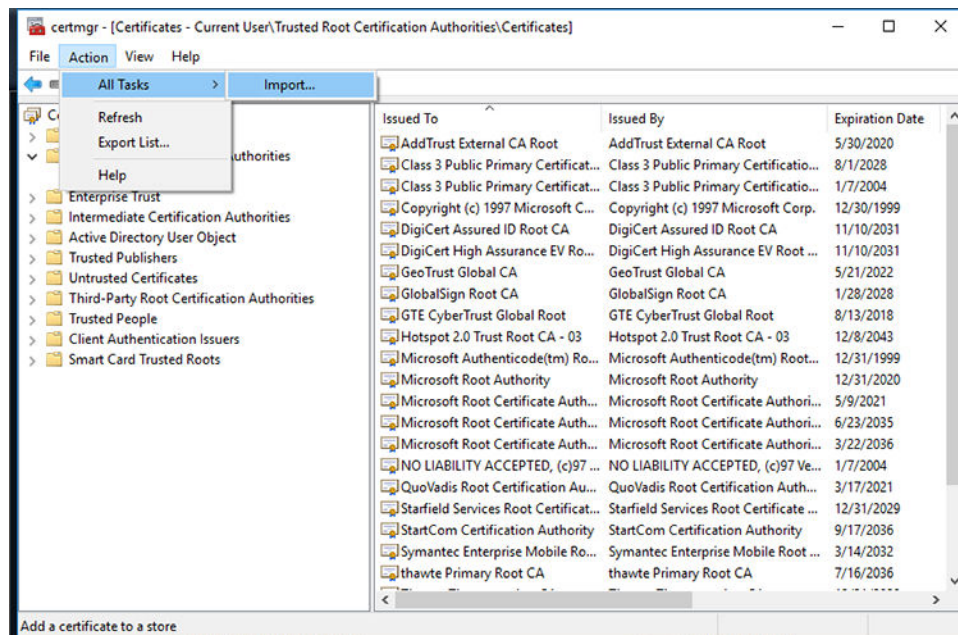


Figure 5-27 Import certificate

- Press the **Next** button in the pop-up welcome window.

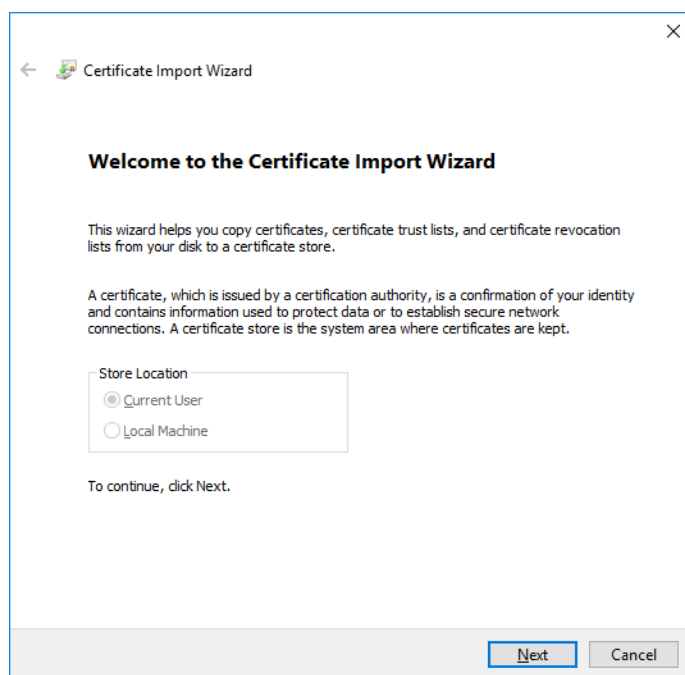


Figure 5-28 Welcome window

7. Click **Browse** to find the certificate file to import.

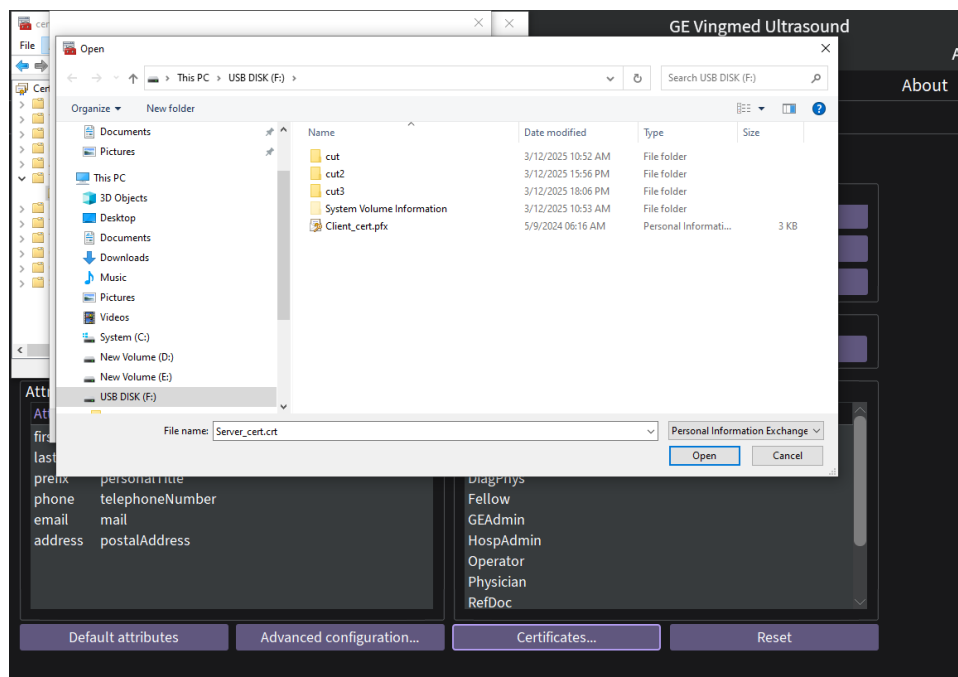


Figure 5-29 Find the certificate file to import

NOTE

If importing a Client Certificate including a private key (e.g. in case of importing a certificate used for authenticating towards the DICOM server, the "Provide Client Certificate" option) choose the certificate type from the list on the right of the "File Name" box. Only the ".pfx" certificate is supported for Client certificates.

8. Browse to the certificate file to import, select the file and press **Next** to confirm.
9. (Optional) Enter the password and make sure that the second checkbox is selected.

Press the **Next** button to confirm the certificate import.

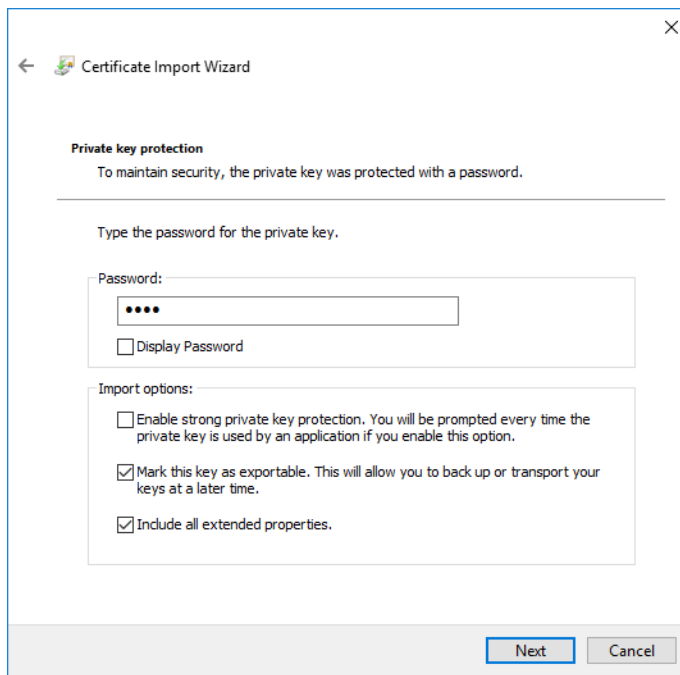


Figure 5-30 Enter Password

10. Confirm the certificate store by pressing **Next**.

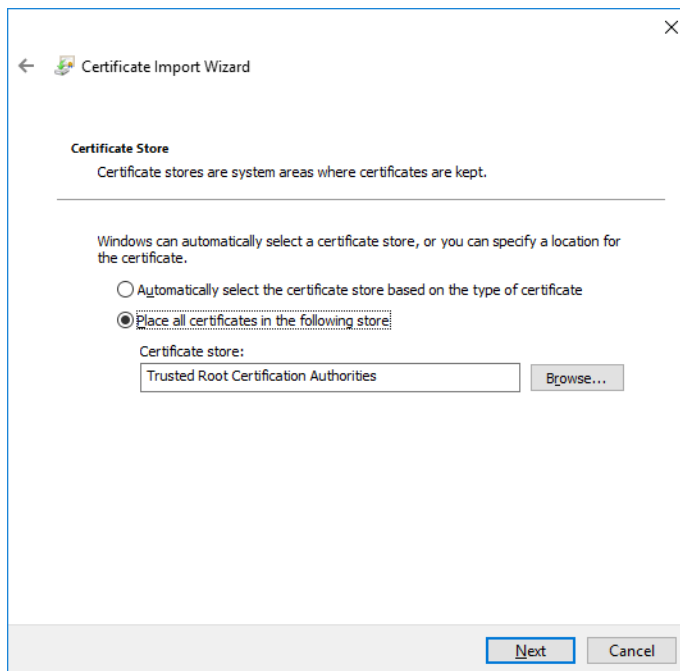


Figure 5-31 Certificate Store

11. Press the **Finish** button.

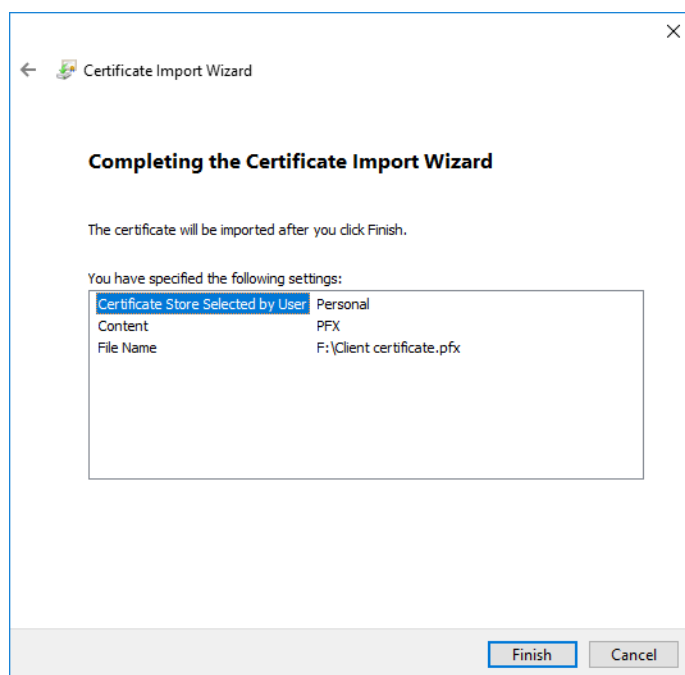


Figure 5-32 Completing the Certificate Import Wizard

12. A window is displayed to indicate the import was successful. Press **OK** to close the window.
13. Close the **Certificate Manager** window when finished.

Delete Certificate

1. Press **Utility/Config** as administrator.
2. Select the **Admin** category and the **LDAP** subgroup.
The LDAP sub group is displayed.

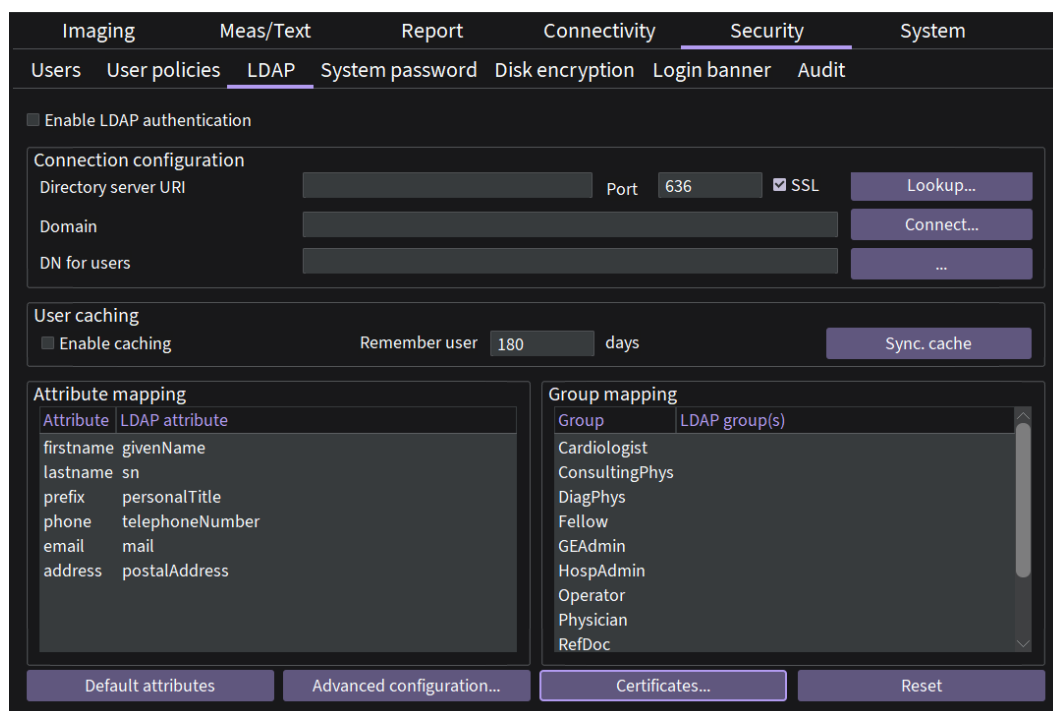


Figure 5-33 LDAP window

- Click on the **Certificates..** button.

The **Certificate Manager** window is displayed.

- In the **Certificate Manager** window, select the certificate to delete.
- Select **Delete** from the Action drop-down menu to delete the certificate.
- Press **Yes** to finally delete the certificate or **No** to keep it.

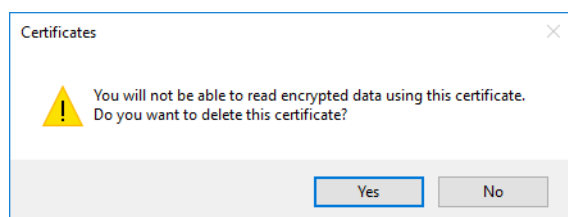


Figure 5-34 Delete Certificate confirmation window

Select Client Certificate

- Press **Utility/Config** on the Touch Panel as administrator.
- Select the **Connectivity** category and the **DICOM** subgroup.
The **DICOM** sub groups are displayed.

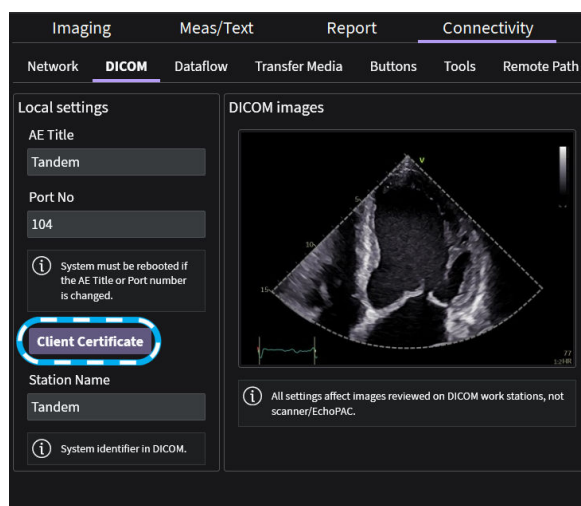


Figure 5-35 Connectivity - Network subgroup

- Click on the **Client Certificates** button.

The **Select Client Certificate** window is displayed.

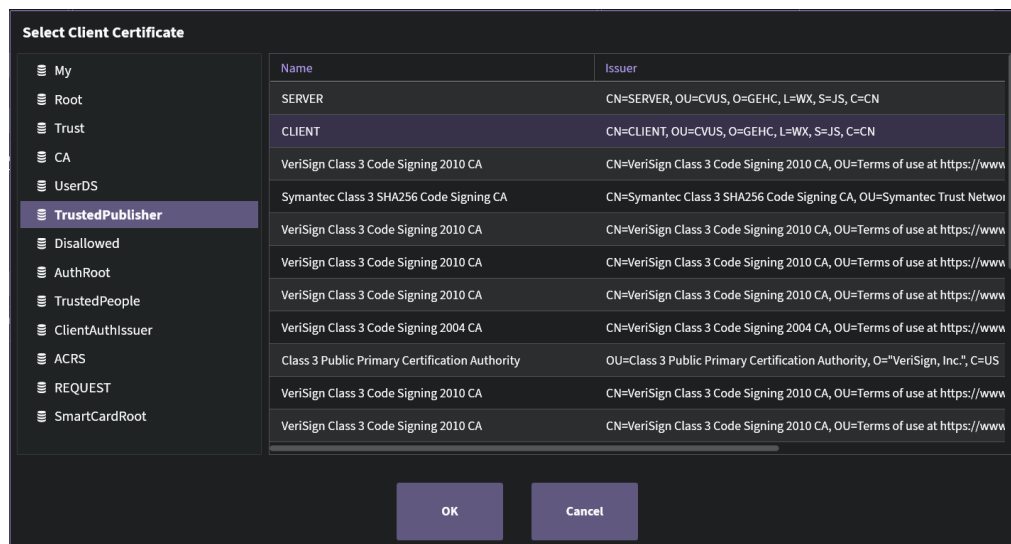


Figure 5-36 Select Client Certificate window

NOTE

Only the certificate with a private key can be selected.

- Select a folder and a certificate.
- Press **OK** and the certificate is selected as the client certificate to use.

5.3.2.3.6 Adjusting the Search Criteria

When selecting a DICOM Worklist dataflow or Query/Retrieve, search criteria can be set for the system to use when searching the database.

- Press **Utility > Config** on the Touch panel and log on as administrator.
- Select the **Connectivity** category and **Dataflow** subgroup.
The **Dataflow** sheet is displayed.
- Select a DICOM Worklist dataflow or the Query/Retrieve dataflow.
- Select the Worklist or Query/Retrieve device and press **Properties**.

The **Properties** window for the selected DICOM device is displayed.

Figure 5-37 DICOM Worklist properties window

NOTE

By reducing **Query result limit** the performance might be improved, however all results might then not be shown.

5. Press **Search criteria**.
The **Search criteria** window is displayed.
6. Select a Search criteria from the **Select tag** pull-down menu.
7. Enter a value if required or leave blank if not to be used. This entry is case sensitive and must match exactly.
8. Press **Update list**.
9. Press **OK** to close the **Search criteria** window.

Search Criteria

Select Tag: 00080060 Modality

Value: US ☐ Don't Use

Update list **Remove**

Name	Value	Don't Use
00080060 Modality	US	
00100024 Issuer of Patient ID Qualifiers Sequence		✓
00100021 Issuer Of Patient Id		✓

OK **Cancel**

Figure 5-38 The Search Criteria window

5.3.2.3.7 DICOM SR

DICOM Structured Reporting (SR) is a standardized format for medical results. Vivid Explorer/Vivid Advanced/Vivid Focus supports the specialized form for Echo Ultrasound ("TID 5200 Echocardiography Procedure Report"), Pediatric ("TID 5220 Pediatric Cardiac Ultrasound Report") and Vascular Ultrasound ("TID 5100 Vascular Ultrasound Procedure Report") for M&A results.

With the DICOM SR support, M&A for an exam can be sent at the end of the exam or when exported from local archive. The destination can be either a server on the network (Storage SCP) or a removable media (DICOM Media) depending on the DICOM dataflow selected.

NOTE

In order for the patient's blood pressure to be populated in the DICOM SR file, the user needs to enter the patient's weight and height as well.

"TID 5220 Pediatric Cardiac Ultrasound Procedure Report" is sent if the exam contains M&A from category Pediatric (Heart).

"TID 5200 Echocardiography Procedure Report" is sent if the exam contains M&A from category Cardiac and does not contain M&A from category Pediatric (Heart).

In addition, "TID 5200 Echocardiography Procedure Report" is sent per third-party application if the exam contains any measurements stored from third-party software.

"TID 5100 Vascular Ultrasound Procedure Report" is sent if the exam contains M&A from category Vascular/Abdominal.

"TID 5200 Echocardiography Procedure Report" is sent per third-party application if the exam contains any measurements stored from third-party software. These measurements are now fully supported by all of the available dataflows. The measurements can be transferred and exported for DICOM Storage dataflow and they will be recorded to separate SR files, called TID 5200 "Echocardiography Procedure Report". The workflow is similar to the Worksheet functionality where a single file stores all measurements for an application.

"TID 5200 Echocardiography Procedure Report", "TID 5220 Pediatric Cardiac Ultrasound Report" and "TID 5100 Vascular Ultrasound Procedure Report" do not support all M&A results from Vivid Explorer/Vivid Advanced/Vivid Focus. They are limited to the following:

- No unassigned measurement.

Refer to the Vivid Explorer/Vivid Advanced/Vivid Focus Reference manual for a complete list of supported parameters.

- The following modes: 2D, M-mode, Color Flow, PW Doppler, CW Doppler, 3D and TDI.
- Not Modified Simpson method or Bullet methods.

Refer to the Vivid Explorer/Vivid Advanced/Vivid Focus Reference manual for a complete list of supported methods.

- Basic derivations (Average, Last, Min and Max), no references between the derived measurements and the ones they were made from.
- Wall Motion Scoring: individual segment scores only according to 16-segment model, no graded Hypokinesis (only Hypokinesis is used).

DICOM SR must be activated for each DICOM device.

1. Press **Utility/Config** on the Touch panel and log on as administrator.
2. Select the **Connectivity** category and **Dataflow** subgroup.
The **Dataflow** sheet is displayed.
3. Select the DICOM dataflow to configure in the **Dataflow** pull-down menu.
4. Select a DICOM storage device and press **Properties**.
The **Properties** window for the selected DICOM storage device is displayed.



Figure 5-39 DICOM SR Settings

5. Check the option **Enable DICOM SR** to enable DICOM SR.
The following additional options are available:
 - **No images, only SR**: no images are sent, only M&A.
 - **Signed Doppler velocities**: send signed Doppler velocities.
 - **SR version**: select the version from the pull-down menu. The current exam data will be sent in the same format as the selected SR version. Details about format and content of the SR version can be found in the corresponding user manual of the selected version.

These settings apply to both "TID 5200 Echocardiography Procedure Report" and "TID 5100 Vascular Ultrasound Procedure Report".

6. Select **OK**.

5.3.2.3.8 DICOM spooler

DICOM spooler displays the current DICOM output jobs. The jobs may be Storage, Print, Modality Performed Procedure Step (MPPS) or Storage Commitment. The DICOM spooler is used for checking the current job's status when a job is saved or when the total spooler status on the right of the **Archive** window displays an error.

From the DICOM spooler the user can also:

- **Delete** non-active jobs.
- **Resend** a job that has failed or is in hold.
- **Send** a job that has failed or is in hold, to a new destination.

The job's status displayed in the **DICOM spooler** window can be:

- **Pending:** the job is complete, waiting to be active.
- **Hold:** the job is complete, but suspended, waiting for an action from the user.
- **Append:** the job is incomplete, waiting for more images (Direct store function).
- **Active:** the job is complete and connected to the destination device.
- **Failed:** the job is complete but one or more images failed to transmit to the destination device.
- **Done:** the job is saved to the destination device. The jobs that are done are removed from the spooler after a few minutes.

To start the DICOM spooler:

- Do one of the following:
 - Press **Utility** and **Spooler** on the Touch panel
 - Press **F4** on the alphanumeric keyboard
 - Press **Alt + S** on the alphanumeric keyboard
 - Press the **Spooler** icon () in the lower left corner of the status bar

The **DICOM spooler** window is displayed (see [Figure 5-40 The DICOM job spooler window](#) on page 160).

The **DICOM spooler** window is automatically updated. Press **Refresh** to update the information displayed at any time.

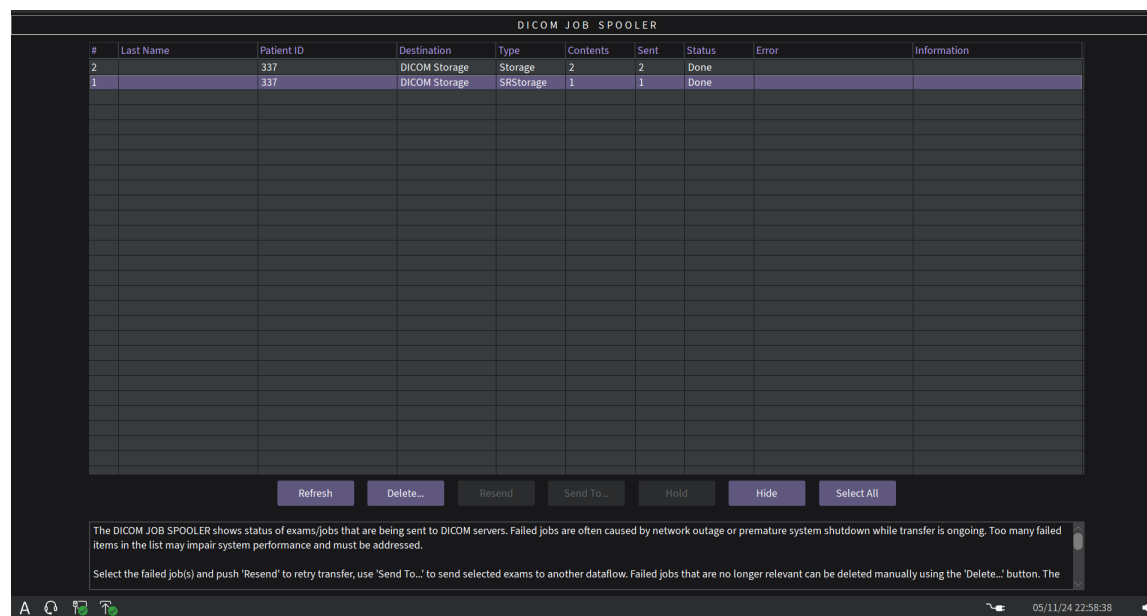


Figure 5-40 The DICOM job spooler window

The DICOM spooler window contains helpful troubleshooting information at the bottom of the screen.

To delete a job:

- Select the job(s) to delete in the **DICOM job spooler** window.

NOTE

Only non-active jobs can be deleted.

- Press **Delete**.

To resend a job:

- Select the job(s) to re-send in the **DICOM job spooler** window.

NOTE

Only jobs that failed or are in hold can be resent.

2. Press **Resend**.

To send a job to a new destination:

This is typically used if the configuration of the server is changed.

1. Select the job(s) to send in the **DICOM job spooler** window.

NOTE

Only jobs that failed or are in hold can be sent to a new destination.

2. Press **Send to....**
A dialogue window is displayed.
3. Select the new destination from the **Destination** pull-down menu.
4. Press **Send**.

To hold a job:

1. Select the job(s) to hold in the **DICOM job spooler** window.

NOTE

Only inactive jobs can be set on hold.

2. Press **Hold**.
3. To undo hold, press **Resend**.

Spooler status icon:

A Spooler icon is shown in the status bar (the right one below).



Figure 5-41 Spooler status bar

There are four different icons depending on the status of the data being transferred:

- | | |
|---|--|
|  | Transfer in progress. System connected to network. |
|  | Transfer paused or on hold. Transfer will resume automatically when network available. |
|  | Transfer completed successfully |
|  | Transfer failed. Please click spooler icon for more details. |

Figure 5-42 Data transfer statuses

When hovering over the spooler icon the text shown above will be shown in the status bar.

Shutdown when data transferring from the DICOM Spooler:

Whenever there are images and/or measurements currently in the process of being transferred from the DICOM Spooler at the time the power button is pressed, this is indicated in the **System Exit** dialogue as shown below. A progress bar with a count-down timer gives an estimate for time remaining until transfer is completed.

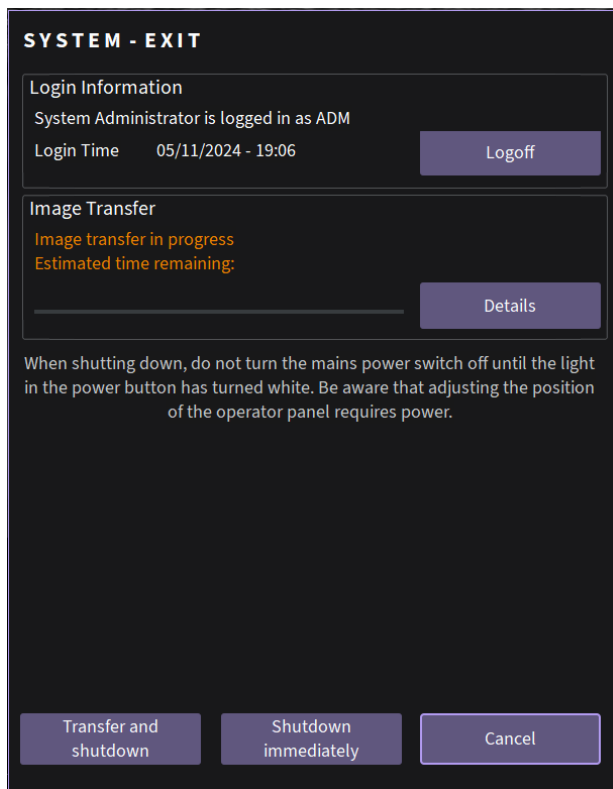


Figure 5-43 System Exit dialogue with data in DICOM Spooler

You can select the following options:

1. **Transfer and Shutdown**
The system will complete transfer of data before shutdown is performed.
2. **Shutdown immediately**
Transfer of data will be aborted, and the system will shut down immediately. Unsent data will remain in the spooler and will have to be resent from the DICOM Spooler at a later time when the system is connected to the network.
3. **Details**
Pressing the **Details** button launches the **DICOM Spooler** screen, where further details on the image transfer process can be seen.

NOTE
Pressing **Details** button aborts the shutdown process.
4. **Cancel**
Pressing **Cancel** will abort the Shutdown process, while data transfer will continue.

5.3.2.3.9 DICOM images

To configure DICOM images:

1. Press **Utility > Config** on the Touch panel and log on as administrator.
2. Select the **Connectivity** category and **DICOM** subgroup.
The **DICOM** sheet is displayed.

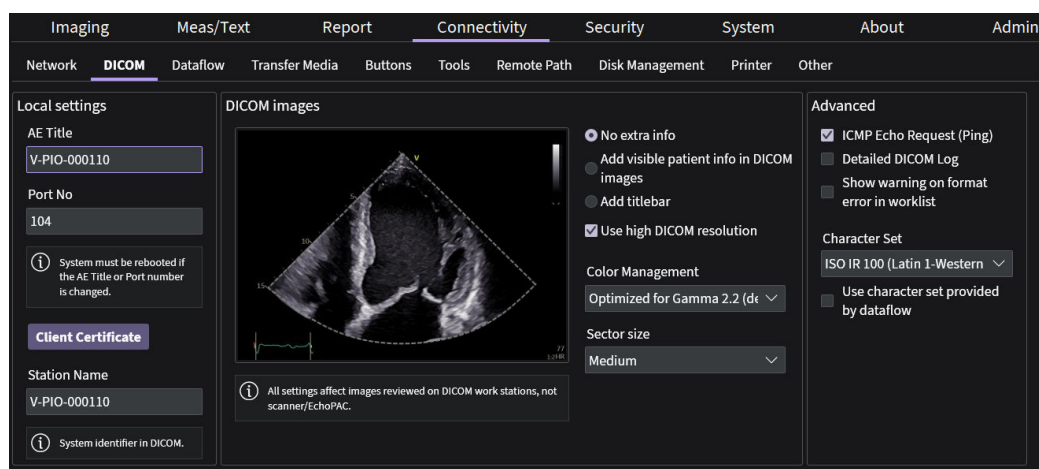


Figure 5-44 The DICOM sheet

From DICOM images the user can select between:

- **No extra info**
- **Add visible patient info in the DICOM images:** displays patient information (name, date of birth and ID) on DICOM images.
- **Add title bar:** adds the Title bar to the DICOM images. Must be enabled in order for images from the 4D Auto AVQ tool to contain traces, bullseye, 3D model and measurements as part of the stored DICOM Multiframe/Singleframe files when using a dataflow with RAW data disabled.

When **Use high DICOM resolution** is enabled, DICOM images are stored using a higher pixel density. Use this setting when exporting to systems accessed by high definition DICOM viewing stations.

NOTE

Using high DICOM resolution will double the file size of the DICOM data when using standard compression settings. Such files will consume more disk space and also slow down storage, recall and transfer of files.

Color management provides a selection of gamma settings for optimized representation of the Vivid images on DICOM workstations.

Sector size offers three different sizes of the 2D sector for viewing on DICOM work stations.

5.3.2.3.10 Configuring DICOM Encapsulated PDF reports

The user can configure the system to enable or disable storing DICOM Encapsulated PDF reports on the media.

1. Select **Config**.
2. Select the **Connectivity** category and the **Transfer Media** or **Dataflow** subgroup.
3. Select a dataflow (i.e DICOM USB, DICOM DVD, DICOM Local HD on **Transfer Media** sheet or **DICOM Storage on Dataflow** sheet).

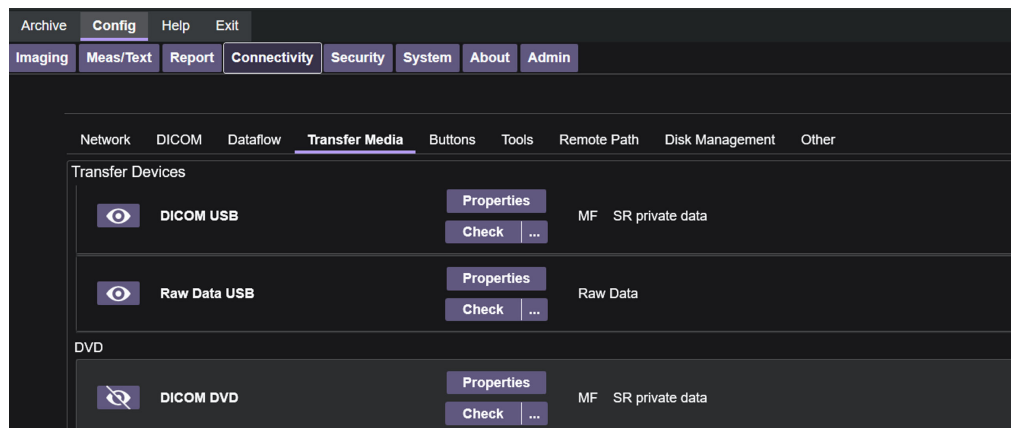


Figure 5-45 Transfer Media sheet

4. Select **Properties**.

The **DICOM Media Properties** window is displayed (see below).

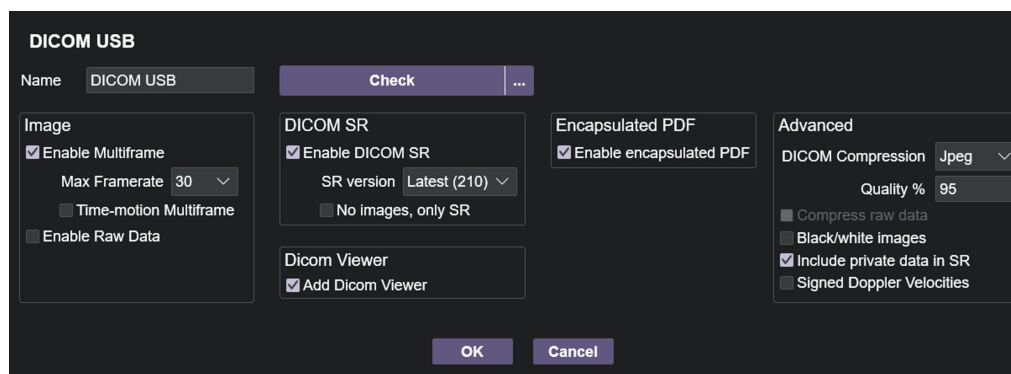


Figure 5-46 DICOM Media Properties window

5. Check the **Enable encapsulated PDF** option.

The PDF report will be exported when transferring examination containing report to the corresponding DICOM media.

NOTE

For Transfer Media dataflows (DICOM USB, DICOM DVD, and DICOM Local HD) a self-contained DICOM viewer can be used to display Encapsulated PDF report. To enable DICOM Viewer, check the **Add Dicom Viewer** option in the **DICOM Media Properties** window.

5.3.2.4 Dataflow Sheet

The Dataflow Concept

Communication between the Vivid Explorer/Vivid Advanced/Vivid Focus ultrasound system and other information providers on the network takes the form of dataflows. Each dataflow defines the transfer of patient information and images from an input source to the system, and from the system to one or several output sources.

A dataflow is a set of pre-configured settings. Selecting a dataflow will automatically customize the system to work according to the settings associated with this dataflow.

5.3.2.4.1 Dataflows available

A set of pre-defined dataflows is available on the system as listed in the table below.

NOTE

Not all dataflow listed below are visible by default.

NOTE

The list of dataflow available is configurable (see [Dataflow adjustments](#) on page 167).

Dataflow	Description
LocalArchive	The local database is used for patient archiving. Images are stored to internal harddrive.
LocalArchive / DICOM Server	The local archive is used for patient archiving. Images are stored to the internal hard drive and to a DICOM server. Some of the measurements are stored if DICOM SR is turned on (see DICOM SR on page 158 for more information about DICOM SR and supported measurements).
EchoPAC Archive / DICOM Storage Image Vault Archive / DICOM Storage	A remote database (either on ultrasound system or a server) is used for patient archiving. Images are stored to a network image volume (either internal HD on ultrasound system or a server).
EchoPAC Archive / DICOM Storage Image Vault Archive / DICOM Storage	A remote database is used for patient archiving. Images are stored to a network image volume and to a DICOM server. Some of the measurements are stored if DICOM SR is turned on (see DICOM SR on page 158 for more information about DICOM SR and supported measurements).
Worklist / Local Archive / DICOM Storage	Search in a DICOM Modality Worklist, the patient found is copied into local database. The patient information and the examination results are stored to the local database. Images are stored to a DICOM server and to an image volume on the local harddrive. Some of the measurements are stored if DICOM SR is turned on (see DICOM SR on page 158 for more information about DICOM SR and supported measurements).
Worklist / EchoPAC Archive / DICOM Storage Worklist / Image Vault Archive / DICOM Storage	Search in a DICOM Modality Worklist, the patient found is copied into a remote database. The patient information and the examination results are stored to a remote database. Images are stored to a DICOM server and to an image network volume. Some of the measurements are stored if DICOM SR is turned on (see DICOM SR on page 158 for more information about DICOM SR and supported measurements).
Worklist / Local Archive	Search in a DICOM Modality Worklist, the patient found is copied into the local database. The patient information and the examination results are stored to the local database. Images are stored to an image volume on the local harddrive.
Worklist / EchoPAC Archive Worklist / Image Vault Archive	Search in a DICOM Modality Worklist, the patient found is copied into a remote database. The patient information and the examination results are stored to a remote database. Images are stored to an image network volume.
DICOM DVD Read Only	Read DICOM images from the CD/DVD-drive. Read only dataflow, no data can be stored.

Dataflow	Description
DICOM Storage	Store pure DICOM images to a DICOM device. Raw data may also be saved depending on the dataflow configuration. Some of the measurements are stored if DICOM SR is turned on (see DICOM SR on page 158 for more information about DICOM SR and supported measurements).
DICOM Query Retrieve	Retrieve images from a DICOM server based on query parameters.
Worklist - DICOM Storage	Search in the DICOM Modality Worklist. Images are stored to a DICOM Server. Some of the measurements are stored if DICOM SR is turned on (see DICOM SR on page 158).
DICOM Query Retrieve / DICOM Storage	Retrieve images from a DICOM server based on query parameters. Images are stored to a DICOM server. Some of the measurements are stored if DICOM SR is turned on (see DICOM SR on page 158 for more information about DICOM SR and supported measurements).
Worklist / DICOM Query Retrieve - DICOM Storage	Search in a DICOM Modality Worklist, retrieve images from a DICOM server based on query parameters. Images are stored to a DICOM server. Some of the measurements are stored if DICOM SR is turned on (see DICOM SR on page 158 for more information about DICOM SR and supported measurements).
DICOM USB Read Only	Read DICOM data from an USB storage device. Read only dataflow, no data can be stored.
No Archive	Enables to perform an examination without storing the data to any archive.

5.3.2.4.2 Transfer sources and destinations available

NOTE

Not all items listed below may be visible by default.

Transfer source/destination	Description
DICOM USB	USB device defaulted to store DICOM data, but can also be configured to include raw data images as well as DICOM SR. No support for inclusion of other patient data.
DICOM DVD	DVD (or CD) device defaulted to store DICOM data, but can also be configured to include raw data images as well as DICOM SR. No support for inclusion of other patient data.
Export as XML	Media or remote path support from transfer of measurements and other parameters in XML format
DICOM Print	Support for printing of images on a DICOM printer
Raw Data USB	USB device defaulted to store raw data but can also be configured to include DICOM images as well as DICOM SR. Support for inclusion of other patient data included for database transfer.
DICOM Raw data DVD	DVD (or CD) device defaulted to store raw data but can also be configured to include DICOM images as well as DICOM SR. Support for inclusion of other patient data included for database transfer.

5.3.2.4.3 Dataflow adjustments

1. Press **Utility > Config** on the Touch panel and log on as administrator if required.
2. Select the **Connectivity** category and **Dataflow** subgroup.
The **Dataflow** sheet is displayed.
3. Select **Add Dataflow** (choose a custom or factory template).

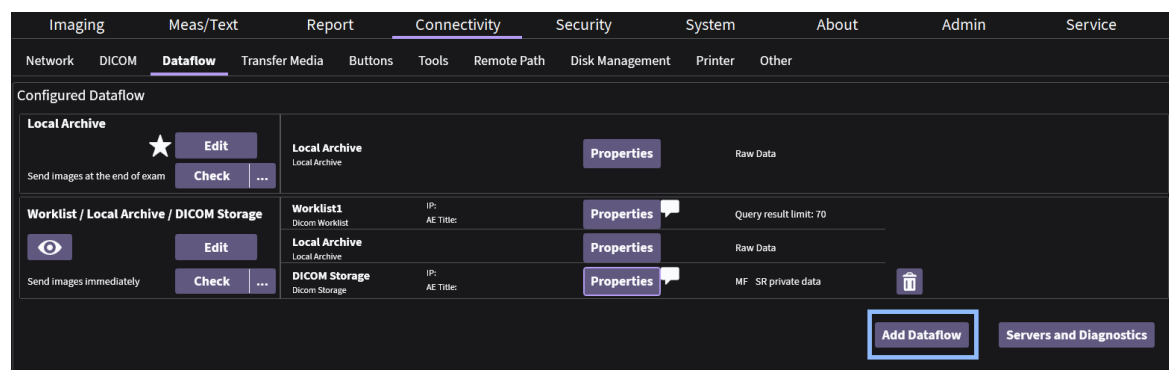
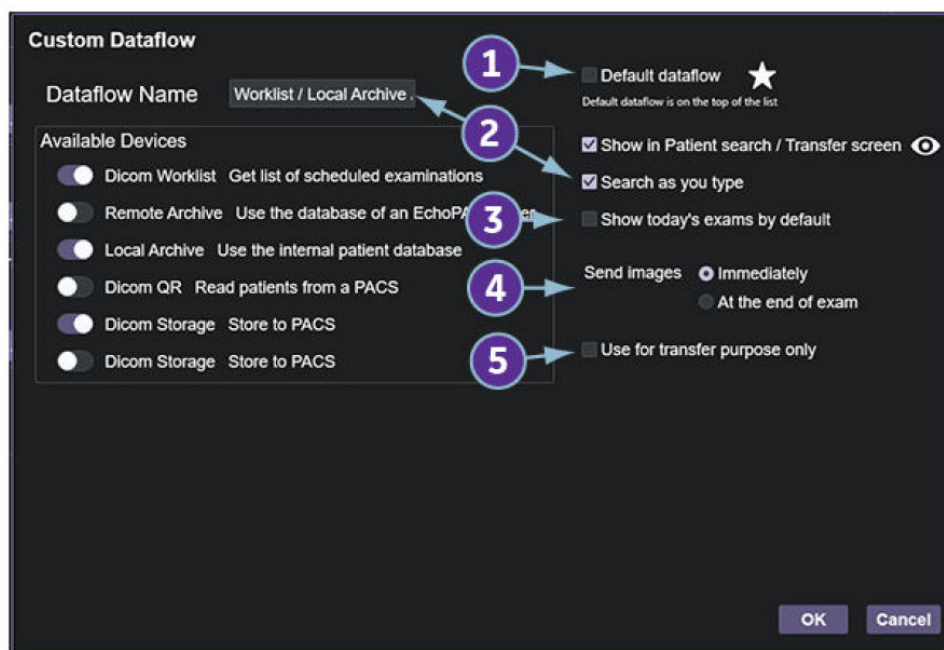


Figure 5-47 The Dataflow sheet

Then, click on the **Edit** button on the selected dataflow to open up the screen below:



1. Check **Default dataflow** to make it default. It will then show up at the top of the dataflow list — with the star icon next to it.
2. **Search as you type** will narrow the displayed patients in the archive the more letters/numbers are entered.
3. **Show today's exam by default** will limit the displayed patients when the box is checked.
4. Images can be sent immediately after Image Store (also called Direct Store) or at the end of the exam by configuring **Send images**.
5. If a dataflow is only to be used for transfer purposes and not as a dataflow used for acquiring exams, check the **Use for transfer purpose only** checkbox.

Figure 5-48 Custom Dataflow settings

5.3.2.4.4 Adjusting the assigned devices

1. Select the dataflow to configure.
2. Select the input or output device to configure.
3. Press **Properties**.
The **Properties** window is displayed.
4. Adjust the device specific parameters as desired (see table below). Not all the settings listed below apply to all devices.

General settings	Definition
Name	Free text: give a descriptive name for the device.
IP address	Select from drop-down menu (if available)
Database Location	Automatically selected according to the IP address
File destination	Automatically selected according to the IP address
MPPS	Modality Perform Procedure Step: send information (typically to a HIS) that a scheduled exam has been started, performed or interrupted.

Image settings	Definition
Raw Data Settings	Raw Data Settings gives you the ability to select which images to transfer in raw data format by mode: • All images • Apical 2D (auto classified) • Parasternal 2D (auto classified) • 2D/Biplane/Triplane • M-mode • Colorflow/TVI/Biplane/Triplane • Color M-mode • PW/CW • 4D • 4D Color
Raw Compression	Enables compression of raw data images upon storage and export. Raw compression is active only if the setting Allow raw data is checked.
Max Frame rate	Select 25, 30 or Full (original acquisition) from the pop-up menu. Applies to DICOM only.
Compression	Select compression type or no compression. Applies to DICOM only.
Quality	Set picture quality from 1 to 100%. A low picture quality level allows high data compression, while a high picture quality restrains the compression. Applies to DICOM only.
Allow Multiframe	☒ : Allow cineloop storage. Applies to DICOM only.
Systole only (for Stress)	When this setting is active, all stress images will be sent to the DICOM server showing only systole. The setting is selectable when transferring only multiframe (not raw data).

Connection settings	Definition
Retry	Set maximum number of connection retries, time interval between tentative and time-out.

DICOM settings	Definition
AE Title	The Application Entity Title is set during DICOM configuration. Refer to the network specifications. Please note that reboot of the system is required when changing the AE Title.
Port	The Port no. is allocated during DICOM configuration. Refer to your network specifications.
Verification	Verify the connection to another DICOM application.
Storage commitment	Send a request to a PACS, asking it to permanently archive image(s).
MPPS	Modality Perform Procedure Step: send information (typically to a HIS) that a scheduled exam has been started, performed or interrupted.

DICOM settings	Definition
DICOM SR settings	- Allow SR: enable DICOM SR. - Allow SR private data: include current exam data in a private format within DICOM SR to retain measurements and complete exam information when recalling an exam from a DICOM environment. This is especially important if EchoPAC Plug-in is used in the DICOM environment or if any exam at any later time is expected to be recalled to Vivid Explorer/Vivid Advanced/Vivid Focus or EchoPAC Software Only using DICOM Query/Retrieve. If the DICOM server does not allow the private data format within DICOM SR files, this feature should be disabled. - Signed Doppler velocities: send signed Doppler velocities. - Use older SR version: when checked, a Use older SR version pull-down menu is displayed. The current exam data will be sent in the same format as the selected SR version. Details about format and content of the SR version can be found in the corresponding user manual of the selected version. See also DICOM SR on page 158.

5.3.2.4.5 Dataflow selection

Select a dataflow from the **Archive** screen or configure the system with a default dataflow from the Configuration management package as described below.

Default dataflow selection

- Press **Utility > Config** on the Touch panel and log on as administrator if required.
- Select the **Connectivity** category and **Dataflow** subgroup.
The **Dataflow** sheet is displayed.
- Press **Add Dataflow** to select the desired dataflow from the pull-down menu.
The selected dataflow is displayed in the **Configured Dataflow** list.
- Press **Edit** button and check the option **Default dataflow**.

Figure 5-49 Dataflow Sheet

5.3.2.5 Transfer sheet

On this sheet you can configure source/destinations for media.

- hide/show by clicking the eye symbol
- configure by clicking **Properties**

5.3.2.5.1 XML Export configuration

The destination for transfer of patient records to XML format must be configured prior to use.

To configure the Transfer function:

1. Press **Utility > Config** on the Touch panel and log on as administrator.
2. Select the **Connectivity** category and **Transfer Media** subgroup.
The **Transfer Media** sheet is displayed.
3. Select the dataflow **Export as XML**.

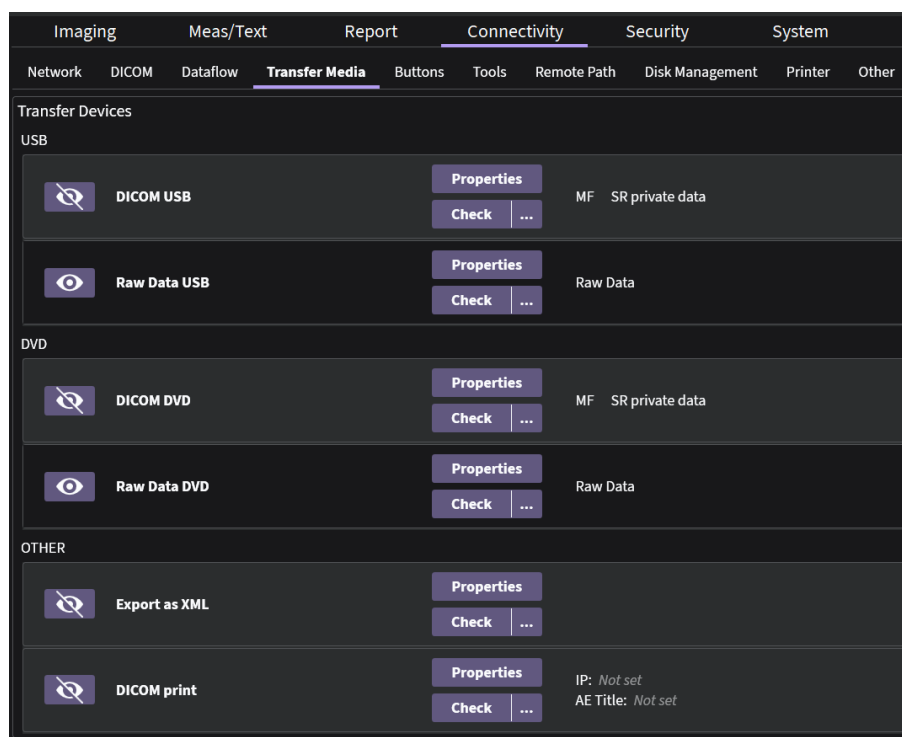


Figure 5-50 The Transfer Media sheet (Export to XML)

5.3.2.6 Buttons sheet

5.3.2.6.1 P1 / P2 button configuration

The **P1** and **P2** buttons on the Control panel can be configured to perform several actions (e.g. Video Print, Laser print, DICOM storage...etc.).

To configure the P1 and P2 buttons:

1. Press **Utility > Config** on the Touch panel.
2. Select the **Connectivity** category and the **Buttons** subgroup.
The **Buttons** sheet is displayed.
3. In the **Button** field, select **P1** or **P2**.

4. Select an output device in the **Available output** field and press the **Right arrow** button to assign the device to the selected button.

The **Properties** window for the selected device is displayed, if configurable.

5. Click the pull-down to change the printer.
6. Click **Properties** to configure the printer.

To remove a device, select the device in the **Selected devices** field and press the **Left arrow** button.

5.3.2.7 Tools sheet

5.3.2.7.1 Recommendation concerning CD and DVD handling

To avoid data loss, never touch the recordable surface of a disk. Handle the disk only by the outer edge. Do not place it face down on a hard surface. Fingerprints or scratches will make the disk unusable. Before usage, verify that the disk surface has no visible scratches. If there are any scratches, do NOT use the disk.

5.3.2.7.2 Formatting removable media

To format removable media:

1. Insert the media in the drive.
2. Press **Utility > Config** on the Touch Panel.
3. If required, log on to the system.
4. Select the category **Connectivity** and select the sheet **Tools** ([Figure 5-51 The Tools sheet](#) on page 172).

Figure 5-51 The Tools sheet

5. Select the removable media from the **Media** drop-down menu (CD-R, DVD-R or USB device).

NOTE

Select **Refresh** if the media does not appear on the list.

6. Enter a name for the removable media in the **Label** field.

NOTE

Only the following characters and signs can be used when labeling a media: A - Z, a - z, 0 - 9, “_” and “-”. Do not use more than 11 characters or signs. Do not use space.

7. Select **Format**.

A confirmation window is displayed.



CAUTION

The formatting process will erase any data present on the media.

8. Select **OK** to continue.
9. Wait for the display of the **Information** window indicating that the formatting process is completed.
10. Select **OK**.

11. Eject the media as described below.

NOTE

Removable media used during Disk space management, Backup, Export, or Save as do not need to be formatted in advance as the formatting process is part of these procedures if required.

Ejecting removable media

1. Press **Utility/Eject** on the Touch Panel.

NOTE

Do not eject CD/DVD using the button on the CD/DVD drive.

The **Eject device** menu is displayed ([Figure 5-52 The Eject device menu](#) on page 173).

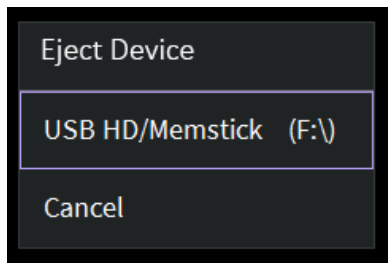


Figure 5-52 The Eject device menu

2. Select the relevant media.

Wait for the display of the Information window indicating that the device can be safely removed before removing the media.

5.3.2.7.3 Unlock patients

If for any reason an examination is not properly finished, the patient record is locked and cannot be opened again unless it is unlocked.

To unlock patient records:

1. Press **Utility > Config** on the Touch panel.
2. Select the **Connectivity** category and the **Tools** subgroup.
3. In the **Tools** sheet, select the patient record(s) to unlock.
You can search for a specific patient record or a group of patient record using the searching filters.
4. Select **Unlock** to unlock the selected patient record(s) or select **Unlock all** to unlock all patient records listed.
A **Confirmation** window is displayed.
5. Select **OK**.

5.3.2.8 Remote Path sheet

5.3.2.8.1 Default remote path setting

The user can define a default remote path for a network shared folder (\\server-name\share-name). The default remote path can then be selected as a destination archive for the following operations:

- Export traces function in Q-Analysis
- Export of system error log file
- Export of report templates
- Save as function for images
- Save as function for reports

To define a default remote path:

1. Press **Utility > Config** on the Touch panel and log on as administrator if required.
2. Select the **Connectivity** category and **Remote Path** subgroup.
The **Remote Path** sheet is displayed.

Figure 5-53 The Remote Path sheet

3. In the **Remote Path** section, enter a remote path of a shared folder on the network.
To check the connection, press **Check**.
4. In the **Configurable Remote Path User** section, enter the user name and password required to access the shared folder.

5.3.2.9 Disk Management sheet

5.3.2.9.1 Disk management

The Disk management function allows the user to manage hard disk space while maintaining the patient database on the system. The Disk management function can be used to move, copy or delete images and move or copy reports from the oldest patient records. The Disk management function has also an auto-purge feature that will automatically delete images and reports that have already been copied if the local hard disk is getting full.

Three different disk management scenarios are possible depending on the system configuration:

- Disk management is set to **move** files: the user runs the Disk management function on a regular basis to move images and reports from older patient records to removable media or to a network volume. Using this setting, moved images and reports are deleted from the local hard drive and copied to the specified destination. This scenario prevents the local disk to fill up and keeps images and reports from the most recent patient records on the local disk. Using this scenario, the user can control what should remain on the system while keeping the disk free space at an operational level.
- Disk management is set to **copy** files: the user runs the Disk management function on a regular basis to copy images and reports from older patient records to removable media or to a network volume. To prevent the local disk to fill up, the auto-purge function automatically deletes files that were previously copied when the disk free space has reached the minimum allowed limit (**15 GB**). This scenario lets the system automatically manage the disk space on the system.

NOTE

When using this setting, the original images will be retrieved from the local hard drive as long as they are available there. When the images are deleted from the local hard drive by the auto-purge function, the copied images will be retrieved.

- Disk management is set to **delete** files: the user runs the Disk management function on a regular basis to delete images from older patient records.

NOTE

Ensure that you have established a data management protocol for your office/institution. The user **MUST** manage the removable media used when running Disk management by keeping a log and by creating a media filing system.

A person should be in charge of performing the process. The Disk management system can be set up to display a reminder at regular intervals.

It is always highly recommended to take a backup of moved/copied files, which is the responsibility of the customer. The system does not offer functionality for taking backup of images and reports saved on long-term storage media.

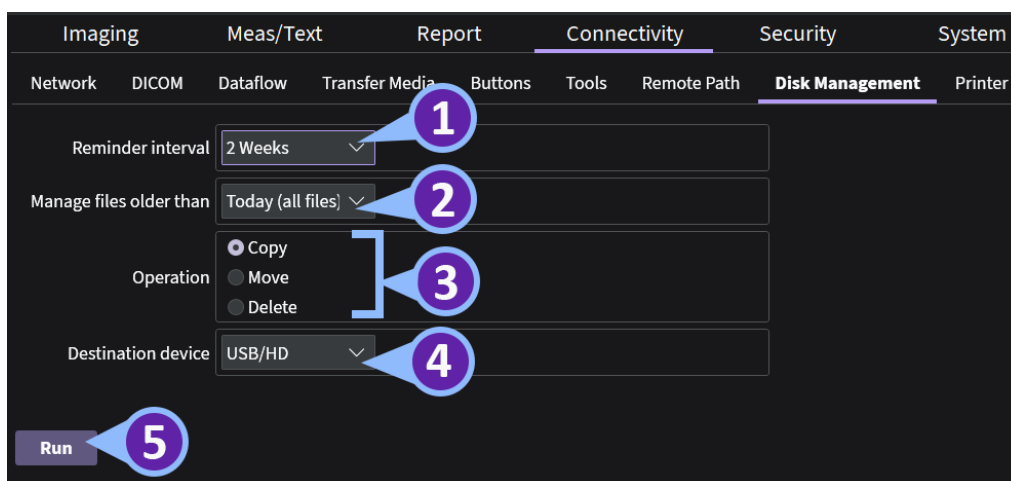
NOTE

It is recommended that users should do data management in a specific time interval.

5.3.2.9.2 Configuring the Disk management function

Configuration of the Disk management system can only be done by user with administration rights.

1. Press **Utility > Config** on the Touch Panel.
If required, log on as administrator.
2. Select the category **Connectivity**.
3. In the **Connectivity** category, select the sheet **Disk management**.



1. Sets the reminder time interval for running Disk management.
2. Sets the files to be managed based on the examination dates.
3. Sets the Disk management to copy, move or delete images.
4. Sets the destination device.
5. Starts Disk management.

Figure 5-54 The Disk management sheet

Disk management schedule setting

1. Next to **Reminder interval**, specify the number of days/weeks you want the system to prompt you to perform disk management.
This setting should be set based on the activity of your office/institution.

Data management settings

1. Select a number of days, weeks or months next to **Manage files older than**. Only files older than the specified setting will be copied, moved or deleted.
If **Today (all files)** is selected, all files will be copied or moved.
2. Next to **Operation check**:

- **Copy:** the images and reports from the examinations older than the specified setting defined in step [1](#) on page 175 are copied to the specified destination. Using this setting, the files will exist in two locations, the local hard drive and the destination.
- **Move:** the images and reports from the examinations older than the specified setting defined in step [1](#) on page 175 are copied to the specified destination, verified and then deleted from the local hard drive. Using this setting, the files will exist in one location, the destination media. They are removed from the local hard drive.
- **Delete:** the images from the examinations older than the specified setting defined in step [1](#) on page 175 are deleted from the hard drive.

Destination device setting

1. Next to **Destination device**, select a removable media or a network share folder.

NOTE

When a network share folder is selected, the path to the folder must be entered. Press **Check** to verify the connection.



CAUTION

If using removable media, it is recommended to use dedicated media to the Disk management process. Removable media used for data backup must not be used when performing Disk management.

Do not use the same removable media on several systems.

5.3.2.9.3 Running the Disk management function

The Disk management function can be run at any time. In addition, the user may be prompted to run Disk management if the time since the last Disk management operation performed has reached the setting for the Reminder interval (see [Disk management schedule setting](#) on page 175), or if the local hard drive is about to be full.

Disk management can be run from the **Archive** screen (see below) or from **Config/Connectivity/Disk management** ([Figure 5-54 The Disk management sheet](#) on page 175).

Manual start of Disk management

1. Press **Patient**.
The **Archive** screen is displayed.
2. Press **Disk management**.
The **Disk management** window is displayed ([Figure 5-54 The Disk management sheet](#) on page 175).

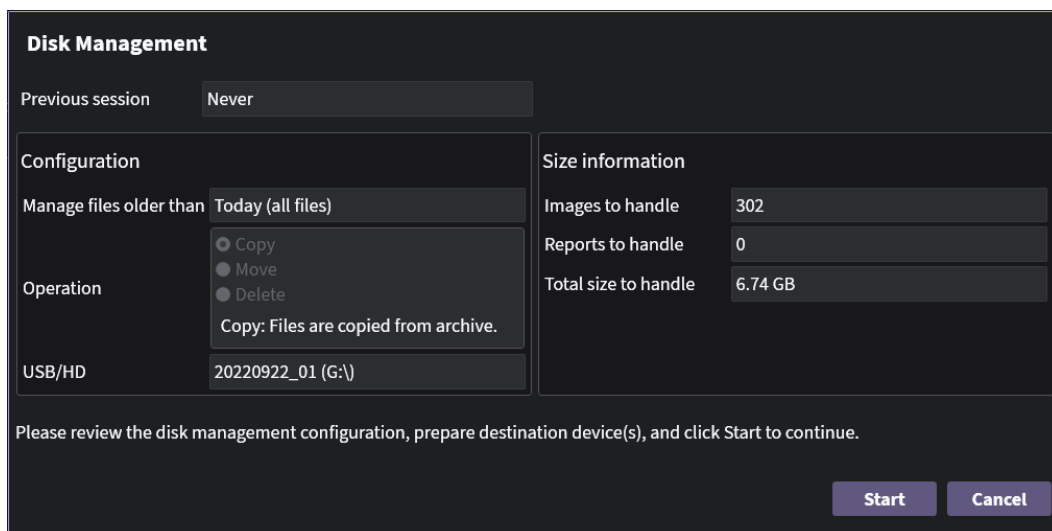


Figure 5-55 The Disk management window

The Disk management operation will either copy, remove or delete files from the local archives depending on the Disk management configuration (see [Configuring the Disk management function](#) on page 175). Make sure the correct configuration is set.

Prepare the destination device(s). If a connected USB device is used, make sure the correct device is selected.

If using CD/DVD, the operation may require several disks as specified in the Disk management window. Make sure the specified number of disks are available.

NOTE

CD/DVD do not need to be formatted.

3. Press **Start**.

The **Disk management processing files** window is displayed showing progression of the process ([Figure 5-56 The Copying files window](#) on page 177).

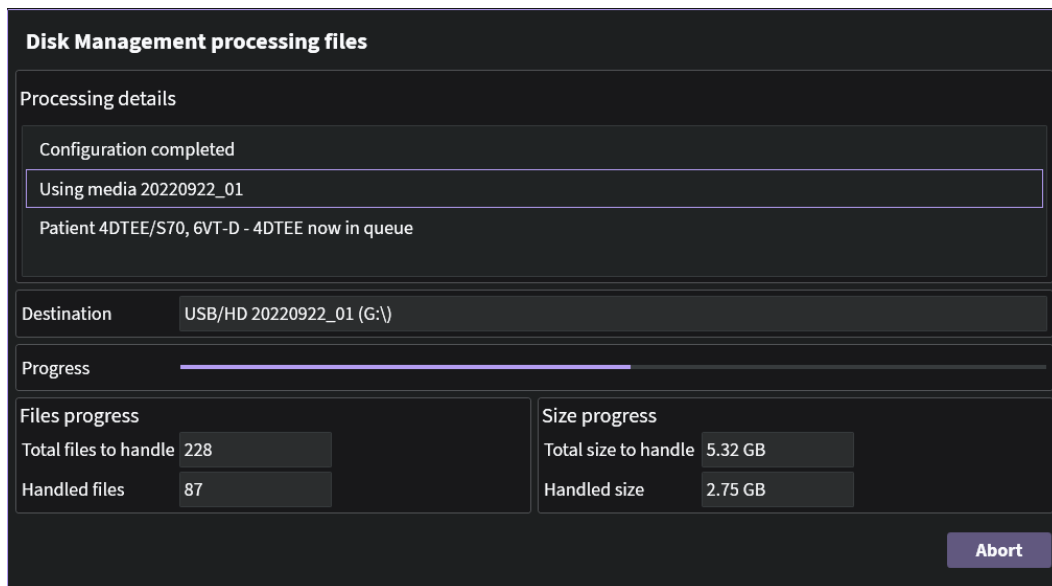


Figure 5-56 The Copying files window

If using CD/DVD as destination device, the system automatically formats the disks if required. If the media is formatted, the user will be asked to enter a label for the media.

NOTE

The media label should have an identification of the system the Disk management is run from.

NOTE

Disk management is aborted if the destination device contains a database backup or exported patient data.

The information displayed in the **Disk management processing files** window is updated while the files are being copied.

4. If more than one media is necessary, a dialogue window is displayed asking the user to insert a new media.

Press **OK** after the new media is inserted.

The operation is resumed.

5. When all the files are copied, the **Disk management completed** window is displayed ([Figure 5-57 The Disk management completed window](#) on page 178), showing the list of processed examinations, the media used and a detailed log.

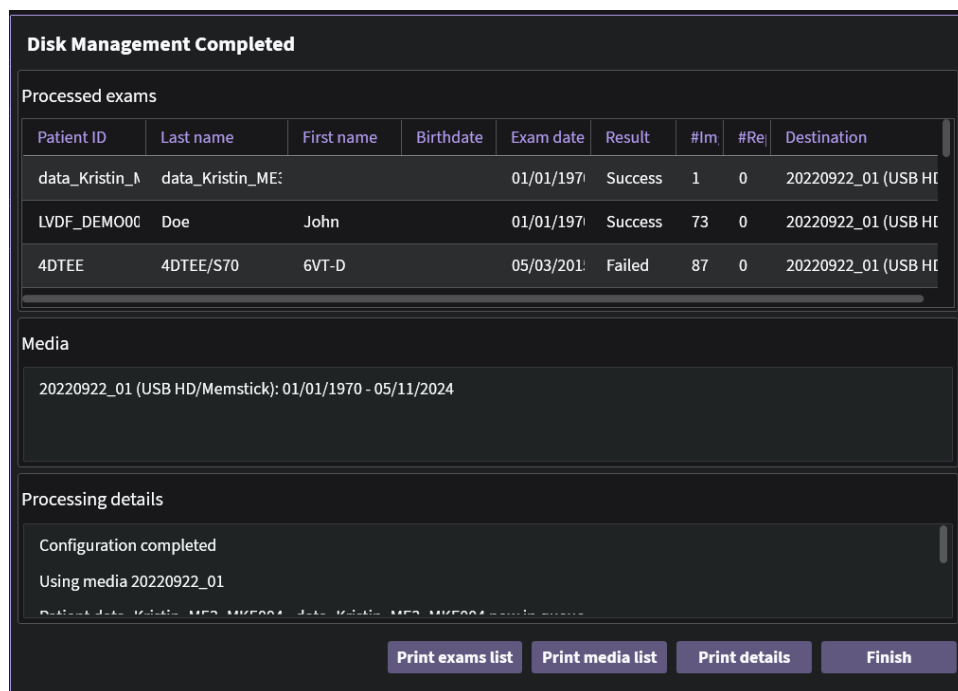


Figure 5-57 The Disk management completed window

- Select **Print exam list** to print the list of processed examinations.
 - Select **Print media list** to print the list of media.
 - Select **Print details** to print the detailed log.
6. Make sure all media are physically labeled according to the list displayed in the **Disk management completed** window. The media label should also have an identification of the system the Disk management was run from.
 7. Press **Finish** to complete the Disk management operation and file the media.
A backup reminder window is displayed.

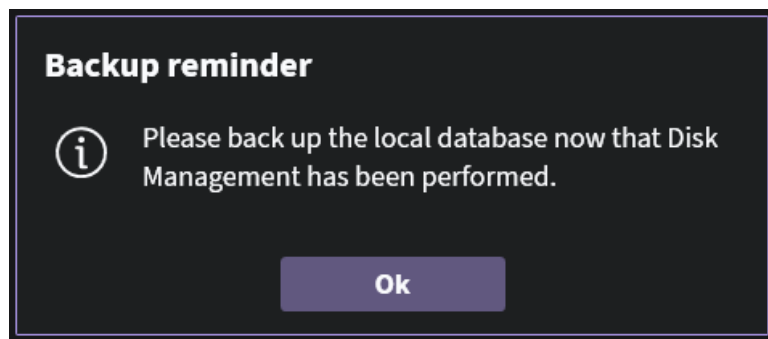


Figure 5-58 The Backup reminder window

8. Press **OK**.

See [Data Backup and restore](#) on page 197 to perform a database backup.

5.3.2.10 Printer

5.3.2.10.1 Printer setup

Printers are configured in the **Connectivity > Printer** tab.

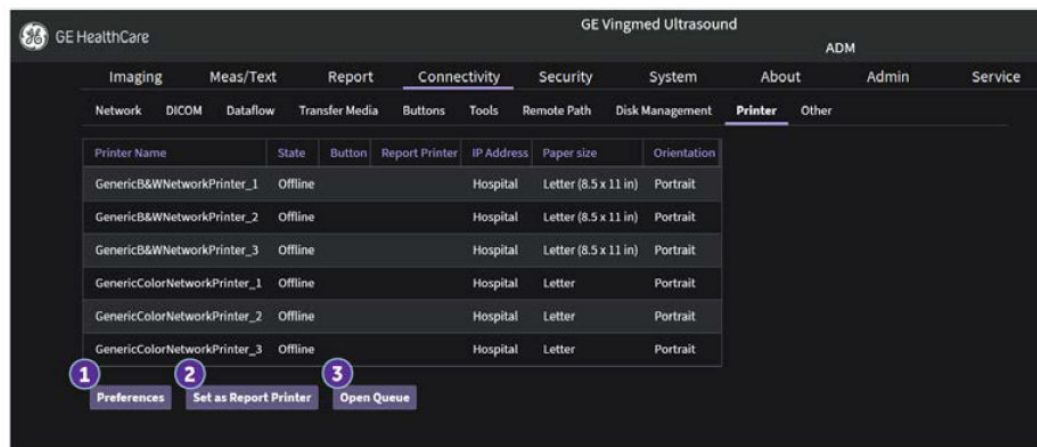


Figure 5-59 The Printer config sheet

To select a printer, highlight it in the table.

1. Click **Preferences** to adjust Paper size and Orientation.
2. Click **Set as Report printer** to use it for printing reports.
3. Click **Open Queue** to view jobs in the printer spooler.

5.3.2.11 Other sheet

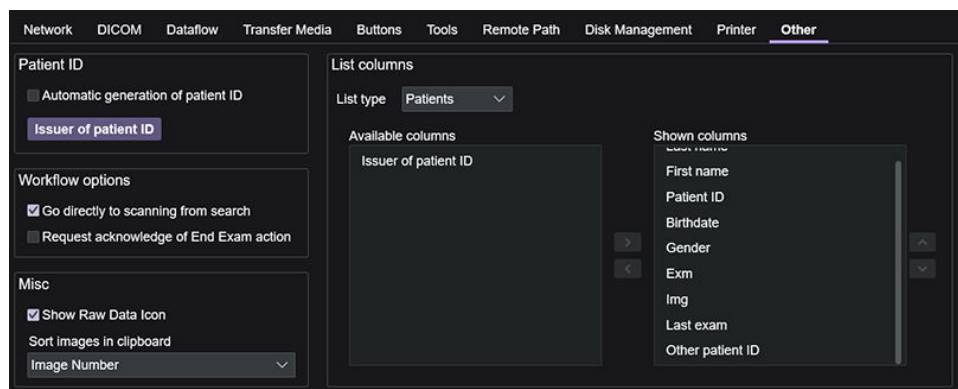


Figure 5-60 The Other sheet

5.3.2.11.1 Patient ID

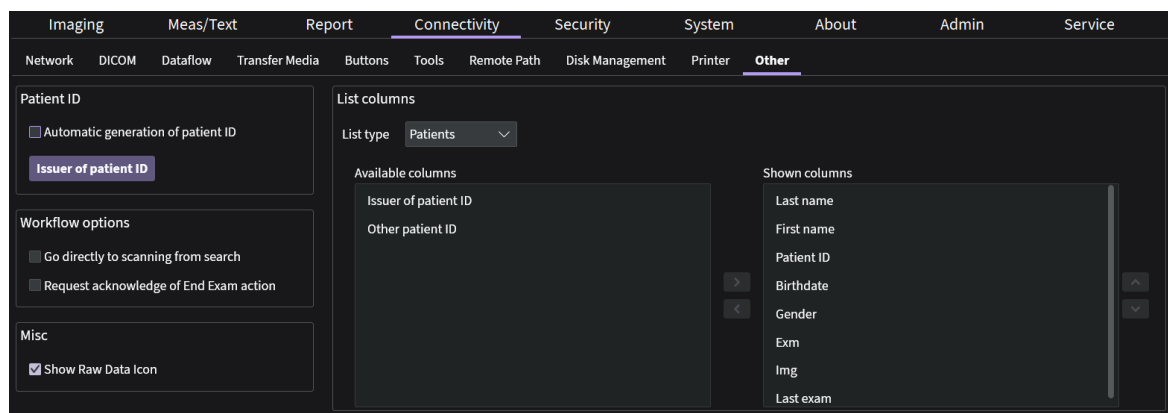


Figure 5-61 The Patient ID sheet

The following settings related to patient id can be adjusted:

Setting	Description
Automatic generation of patient ID	In the Archive screen ☒ : A Patient ID is automatically generated by the system. ☐ : A Patient ID is required and must be entered by the user when creating a new patient record in the archive.
Issuers of patient ID	In the Archive screen , the issuer of a patient ID may be specified for a patient ID. An issuer may be added, modified, or deleted. The issuer set as default will be used as issuer for all new patients that are created. In addition, in transfer between remote and local archive, two patients with same patient ID, one with empty issuer and one with default issuer, are considered the same patient.

5.3.2.11.2 Patient management presets

The following settings related to patient management can be adjusted:

Setting	Description
Request acknowledge of End Exam action	☒ : The user is asked to confirm action when ending an examination.
Go directly to scanning from search	☒ : The system goes directly to the Scanning screen after creating a patient record. ☐ : The system displays the Patient info and exam screen after creating a patient record for further information entry. The user must press Patient or one of the scanning keys on the Control panel to enter the Scanning screen.

5.3.2.12 InSite Setup

For instructions, see: [InSite](#) on page 213.

5.3.3 Security tab

5.3.3.1 Users and Security

5.3.3.1.1 Local System Users

The screenshot displays the 'Users' configuration interface. On the left, a 'User List' shows 'ADM' and 'USR'. The 'ADM' user is selected. The main panel shows the 'Identity' section with fields for 'Id' (ADM), 'Last Name' (System Administrator), 'First Name', 'Email', 'Address', 'Title', and 'Phone Number'. There are 'New' and 'Delete' buttons. Below this is the 'Member of Group(s)' section with checkboxes for 'Cardiologist', 'ConsultingPhys', 'DiagPhys', 'Fellow', 'GEAdmin', 'HospAdmin', 'Operator', 'Physician', 'RefDoc', 'Sonographer', and 'SysAdmin'. The 'Operator' and 'SysAdmin' checkboxes are checked. The 'Operator Rights' section has checkboxes for 'Admin', 'Create', 'Delete', 'Diagnose', 'PresetAdmin', 'PrintRep', 'ReceiveStreaming', 'SecureWipe', and 'StoreRep'. The 'Admin', 'Create', 'Delete', 'Diagnose', 'PresetAdmin', 'PrintRep', 'ReceiveStreaming', 'SecureWipe', and 'StoreRep' checkboxes are checked. At the bottom, there is an 'Autologin' dropdown set to 'Disable', an 'Auto screenlock (min)' field set to 0, and a 'Prohibit default password' checkbox which is checked.

Figure 5-62 The Users sheet

The ultrasound system requires operator registration.

The users are divided in groups with different rights as shown below.

Groups	Rights (see definitions below)									
	Admin	Create	Delete	Diagnose	PreseAdmin	PrintReport	Receive Streaming	Remote Control	Secure Wipe	Service
Cardiologist		+	+			+				Activated with a Dongle
Fellow		+				+				
Operator										
RemoteUsr							+	+		
ConsultingPhys							+			
GEAdmin	+	+	+			+				
Physician		+				+				
Sonographer		+				+				
DiagPhys				+						
HospAdmin					+	+			+	
RefDoc										
SysAdmin	+	+	+			+				

The rights associated to the user groups are:

Rights	Definitions
Create	Create and update patient record, examination, user and referring members. Transfer patient records and examinations. Move examinations.
Delete	Delete patient record, examination, user and referring members.
Diagnose	Make the Diagnosing physician/veterinarian available in the Patient info and exam screen. Sign off report.
Preset admin	Make application presets protected. Delete protected application presets.
Print report	Print a report.
Admin	System administration.
Receive streaming	The user will be allowed receiving Vivid images on a remote client, streamed from the system by the Data streaming feature.
Service	Access to the service platform.
Secure Wipe	Access to erasing patients securely.
Remote Control	

NOTE

When adding or modifying local users, the changes will not take effect until the system is restarted.

1. Press **Utility > Config** on the Touch Panel and log on as administrator if required.
2. Select the **Security** category and **Users** subgroup.

The **Users** sheet is displayed.

5.3.3.1.2 User Policies

Imaging Meas/Text Report Connectivity **Security** System

Users **User policies** LDAP System password Disk encryption Login banner Audit

☒ Enable policies

User policy

Block user after consecutive failing login attempts 3

Blocking time 10 minutes.

User name policy

Minimum user name length 6

Password policy

Minimum password length 8

Minimum number of character sets required in the password 3

Upper case characters 0

Lower case characters 0

Digits (0-9) 0

Non-alphanumeric characters (e.g. !\$#,%) 0

Maximum password age (days) 365

Minimum password age (hours) 1

Password history (no reuse of the last n used password) 10

Password cannot contain user name ☒

Figure 5-63 The User Policies sheet

User Policies can be enabled to enforce user account policies for all users on the system. The rules can be configured to set requirements for user name length, password complexity and for blocking of users. By default, users policies are disabled.

5.3.3.1.3 LDAP overview

Vivid Ultrasound System and EchoPAC Software Only support integration with LDAP compliant directory servers. If the system is configured to use LDAP, the access control and user authentication is performed with use of services from the connected directory server. The authentication will be done by the directory server, while the access control to the system is done with data received from the directory server.

5.3.3.1.4 LDAP

The screenshot shows the 'LDAP' configuration sheet. At the top, there are tabs: Imaging, Meas/Text, Report, Connectivity, **Security**, and System. Under the 'Security' tab, there are sub-tabs: Users, User policies, **LDAP**, System password, Disk encryption, Login banner, and Audit.

Enable LDAP authentication (checkbox)

Connection configuration

- Directory server URI: nohoreng01.corporate.ge.com
- Port: 389
- SSL: (checkbox)
- Buttons: Lookup..., Connect..., ...

User caching

- Enable caching: (checkbox)
- Remember user: 180 days
- Buttons: Sync. cache

Attribute mapping

Attribute	LDAP attribute
firstname	givenName
lastname	sn
prefix	personalTitle
phone	telephoneNumber
email	mail
address	postalAddress

Group mapping

Group	LDAP group(s)
Cardiologist	
ConsultingPhys	
DiagPhys	
Fellow	
GEAdmin	
HospAdmin	
Operator	
Physician	
RefDoc	

Buttons at the bottom: Default attributes, Advanced configuration..., Certificates..., Reset.

Figure 5-64 The LDAP sheet

Enable "LDAP authentication" to utilize services from an external Directory Server for authenticating users when logging in to the system. If enabled, it will not be possible to login to the system with users defined as local users on the system, except for the ADM user.

NOTE

If LDAP is configured, the log-in dialog indicates the log-in domain.

The system can be configured to use authentication services from a Microsoft Active Directory server or from another LDAP compatible Directory Server.

	Definition
Connection configuration	Set directory server, domain, and DN for users.
User caching	Set number of days user will be remembered without needing to log in with network access. When disabling this option, the cached user data will be deleted.
Field mapping	Map LDAP attributes to system user attributes.
Group mapping	Map LDAP groups to system groups. An LDAP group can be mapped to zero or more system groups. Several LDAP groups can be mapped to the same system group.

To define the LDAP properties:

1. Enter the configuration properties.
2. Press **Connect** to test the connection and enter a valid user name and password for the LDAP server in the dialog that appears.
3. Define group mapping for the LDAP user groups that shall be grant users access to the system.

The user will be assigned one or more system groups according to the group mapping and which of these LDAP groups the user is a member of.

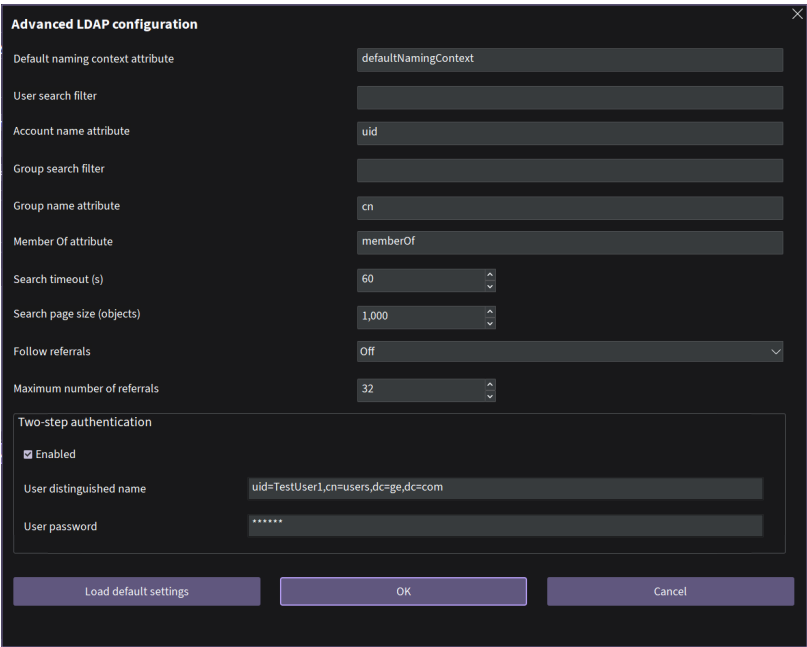


Figure 5-65 The Advanced LDAP configuration screen of the LDAP sheet

If the Directory Server does not support anonymous connection for the authentication service, two step authentication is needed. Then valid user credentials for a user with access to the Directory Server must be entered here.

NOTE

The user credentials will be stored on the system.

5.3.3.2 System password

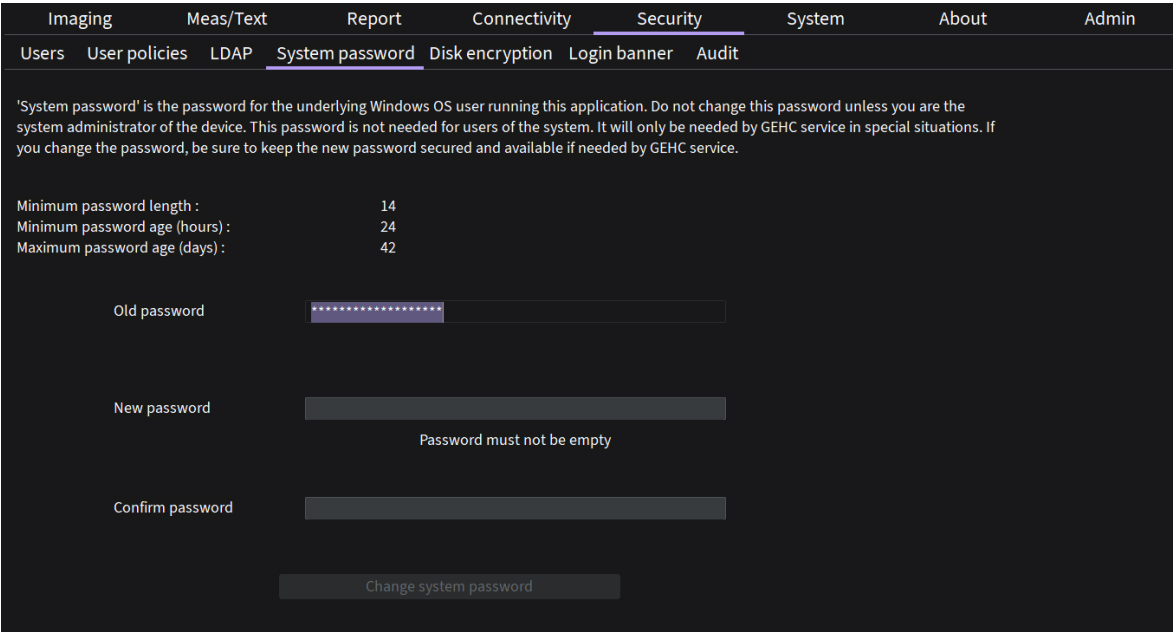
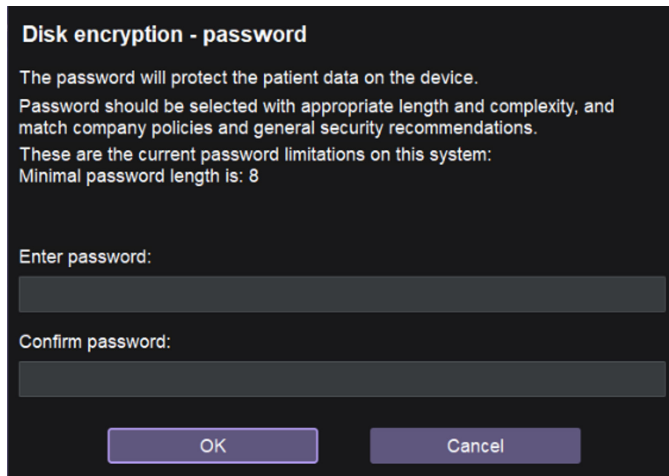


Figure 5-66 The System Password sheet

The System Password is the password for the underlying Windows OS user running this application. Do not change this password unless you are the system administrator of the device. This password is not needed for users of the system. It will only be needed for GE HealthCare service in special situations. If you change the password, be sure to keep the new password secured and available if needed by GE HealthCare service.

5.3.3.2.1 Password

When the “Password” policy is selected, a dialog shown in [Figure 5-72 Question dialog](#) on page 189 pops up to display important information before proceeding. The “Password” dialog then pops up to let the user enter the password. The password policy is compliant with the system password policy, which you can configure on the **Security > User policies** tab. The Figure below shows only one password policy active and valid: “Minimum password length is: 8”.



Disk encryption - password

The password will protect the patient data on the device.

Password should be selected with appropriate length and complexity, and match company policies and general security recommendations.

These are the current password limitations on this system:
Minimal password length is: 8

Enter password:

Confirm password:

OK Cancel

Figure 5-67 Password dialog

After the password is configured, the same encryption process is launched as described in section [Auto Unlock](#) on page 188.

NOTE

Under the “Password” policy, the user needs to enter password every time the system boots up.

5.3.3.3 Disk Encryption

The disk encryption feature is enabled by default for all partitions to better protect customer from privacy and security related risks and is set up before shipping the equipment to customer. However, the customer still can configure disk encryption on the **Configuration** page shown in [Figure 5-68](#) on page 187 below. The customer can only configure encryption of the Archive partition, while other partitions will be controlled by the system and automatically lock/unlock when the system is shut down or booted up.

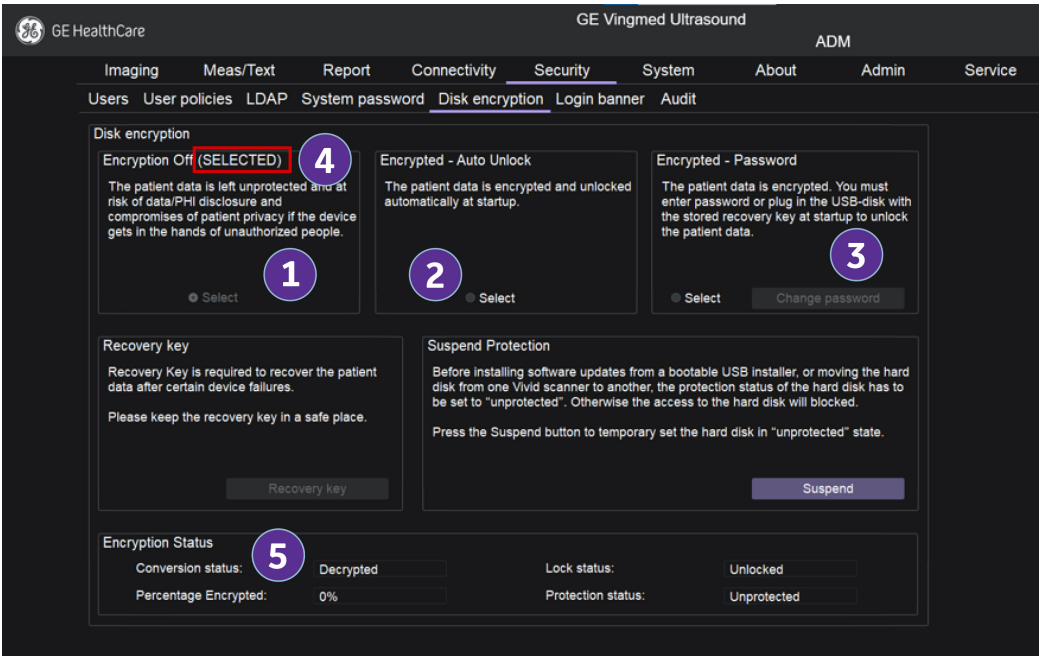


Figure 5-68 The Disk Encryption Sheet

Three disk encryption policies are supported by the system: Decryption, Auto Unlock, and Password. The explanation and supported operations of encryption policies are displayed in 3 cards indicated above as items 1, 2, and 3. The area indicated by item 1 displays the name of policy in use, with its title having text “Selected” next to it (see item 4).

The encryption status displays in area 5.

5.3.3.3.1 Decryption

Press the **Decrypt** button to proceed to the decryption process. The following dialog displays.

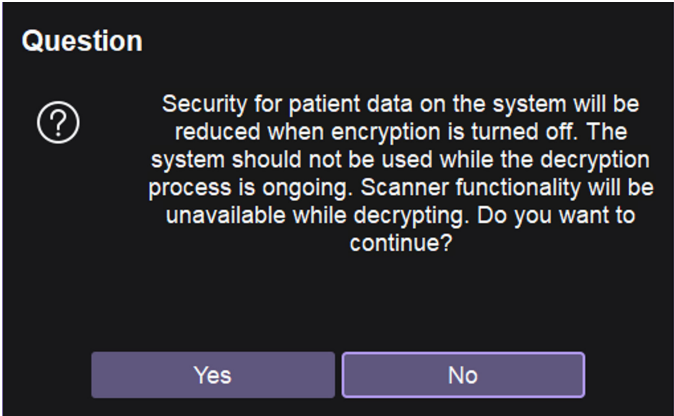


Figure 5-69 Decryption information dialog

Press **Yes** to decrypt the Archive partition.

CAUTION
Decryption will make the Archive partition unprotected, and the patient related data will be at a higher risk of leakage.

A decryption dialog will pop up to show the decryption progress.

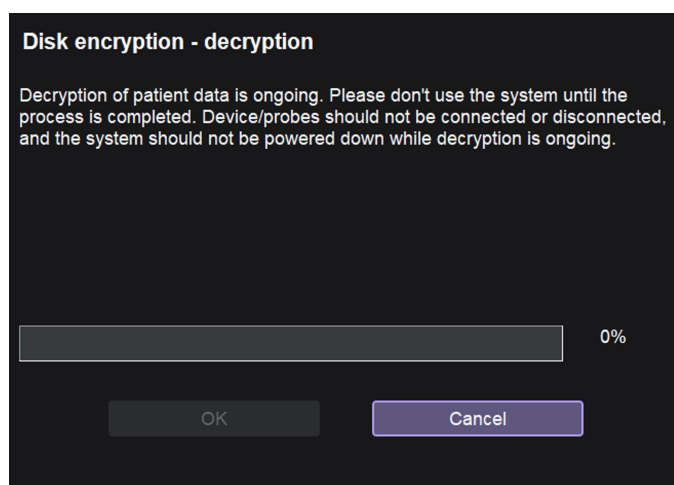


Figure 5-70 Decryption in progress

This dialog will not close itself automatically. Instead, the **OK** button becomes active when decryption is finished.

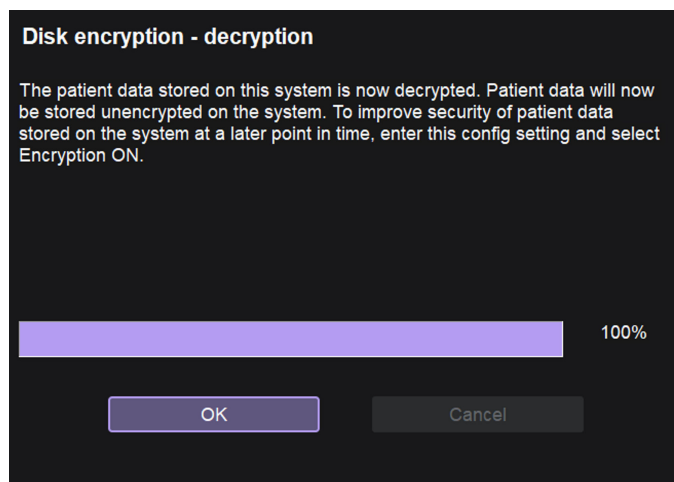


Figure 5-71 Decryption finished

5.3.3.3.2 Auto Unlock

This is the default encryption policy on the system. Under this policy, the Archive partition is encrypted, and it unlocks the Archive partition automatically at bootup.

When "Auto Unlock" is selected, a question dialog pops up that displays information the user needs to know before proceeding and inquires if the user wants to continue (see below).

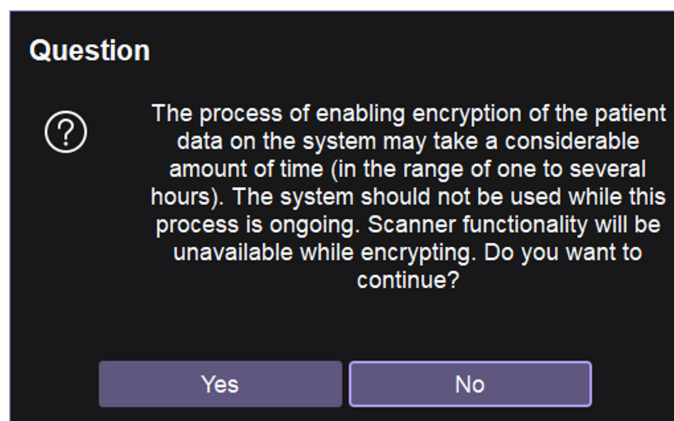


Figure 5-72 Question dialog

Press **Yes** to continue. If a recovery key already exists in the disk, a dialog will pop up to inquire if the user wants to reuse it or create a new one (see [Figure 5-73](#) on page 189below). This dialog will not display if there is no existing recovery key on the disk.

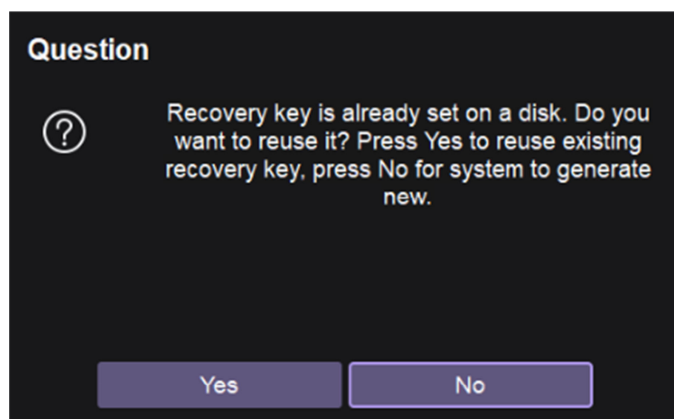


Figure 5-73 Recovery key reuse dialog

Press **Yes** to reuse and **No** to create a new one. Note that either button will direct user to the recovery key backup process shown as follows.

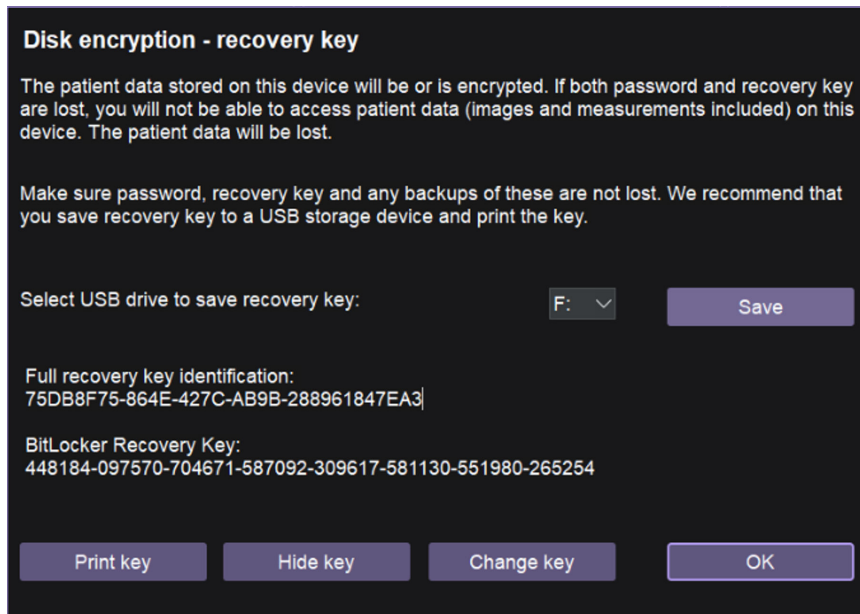


Figure 5-74 Recovery key backup dialog

To back up the recovery key, insert a USB stick, select the driver letter of the USB stick in the dropdown list, and then press **Save**.

Press **Print key** to prompt the respective dialog and print the recover key on paper. Press **Show key** to display the recovery key on the screen. Press **Change key** to generate a new recovery key.

Press **OK** to proceed to the encryption process. In the encryption dialog, a progress bar indicates the encryption progress as shown below.

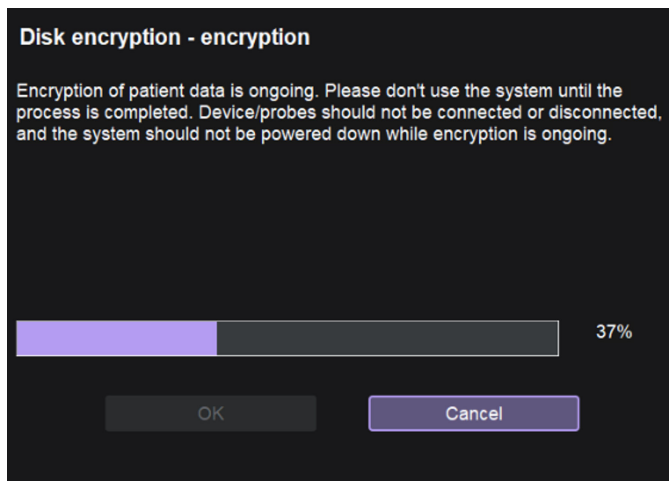


Figure 5-75 Encryption in progress

NOTE

Please consider that the encryption process takes a long time. When the process is complete, the encryption dialog will not close automatically. Instead, the **OK** button will become active as shown below. Press it to close the dialog.

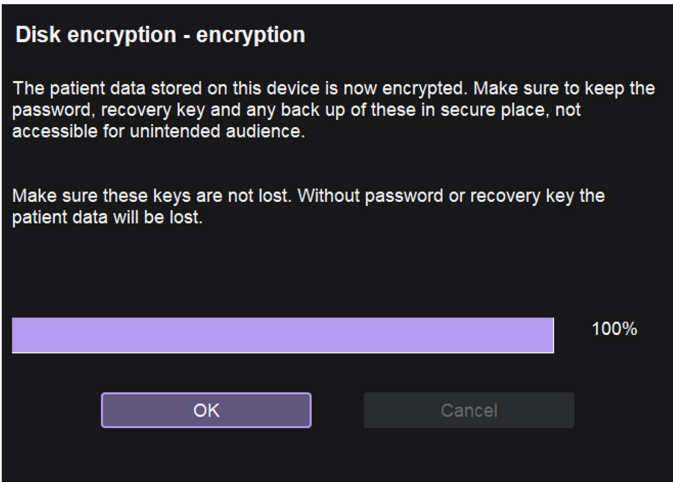


Figure 5-76 Encryption complete

5.3.3.4 Login Banner

The Login Banner provides an option to display information to the user when logging in to the system. When enabled, the user is required to accept the message to complete the login process. The login banner is configurable for an administrator of the system.

5.3.3.4.1 Enable the Login Banner

1. Press **Utility** > **Config** on the Touch Panel as administrator.
2. Select **Security** category and the **Login banner** subgroup.
3. Check off the **Enable login banner**.

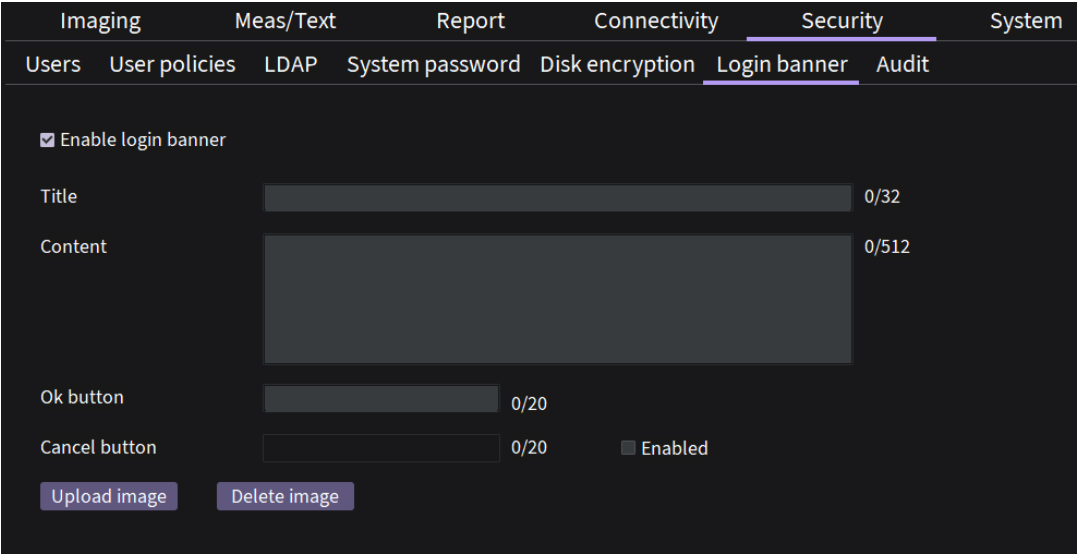


Figure 5-77 Login banner window

4. Enter the title text to be displayed into **Title** field.
5. Enter the text to be displayed into **Content** field.
6. Enter text to be displayed on the **Ok** button into the **Ok button** field.
7. (Optionally) Check off **Enable** for showing a Cancel button and enter the text to be displayed on the **Cancel** button.
8. (Optionally) Upload an image to be displayed on the Login Banner. Insert the USB Flash Drive with the image file to use. Select **Upload image**.

NOTE

Acceptable image formats are .jpg, .jpeg, .png and .bmp.

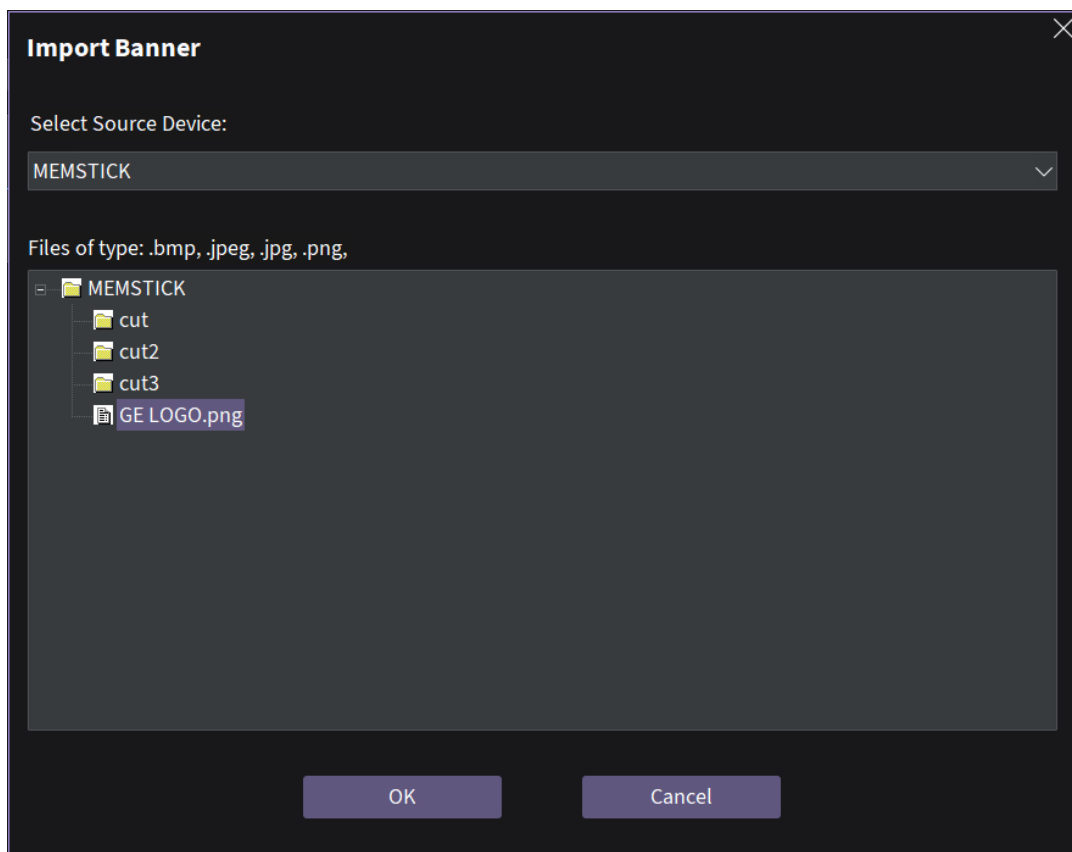


Figure 5-78 Upload image to login banner

At next login to the system, the login banner is shown to the authenticated user. Press the **OK** button to complete the login.

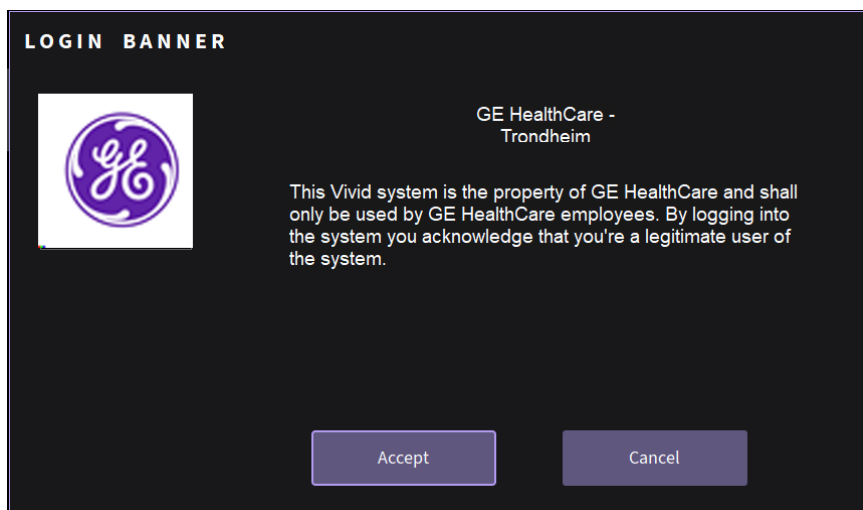


Figure 5-79 Example of Login banner

5.3.3.5 Database maintenance

1. Press **Utility** > **Configon** the Touch panel and log on as ADM.

2. Select the **Security** category and the **Audit** subgroup.
3. Press **View Audit Log**.
The **Audit Log Viewer** window is displayed.
4. Adjust the filters as required and press **Search**.
A log is generated showing the activity on the system.

Audit Log Viewer

Filter

Start date 31/08/2024 End date 05/11/2024 User ID Patient ID

Clear filter Search Save results Close

Type	User ID	Timestamp	Scanner name	Dataflow name	Patient ID	Issuer of patient ID	Study or series UID	SOP instance UID	Affected User ID
TimeZone	***	9/2/2024 8:07:02 PM	V-PIO-000110	UNKNOWN					
TimeSynchronizationNtpServerList	***	9/2/2024 8:07:03 PM	V-PIO-000110	UNKNOWN					
TimeSynchronizationNtpServerList	***	9/2/2024 8:07:03 PM	V-PIO-000110	UNKNOWN					
TimeSynchronizationSyncType	***	9/2/2024 8:07:03 PM	V-PIO-000110	UNKNOWN					
TimeSynchronizationNTPServer	***	9/2/2024 8:07:03 PM	V-PIO-000110	UNKNOWN					
User disconnected	***	9/2/2024 8:07:14 PM	V-PIO-000110	No Archive					
TimeZone	***	9/2/2024 8:08:48 PM	V-PIO-000110	UNKNOWN					
TimeSynchronizationNtpServerList	***	9/2/2024 8:08:48 PM	V-PIO-000110	UNKNOWN					
TimeSynchronizationNtpServerList	***	9/2/2024 8:08:48 PM	V-PIO-000110	UNKNOWN					

Figure 5-80 The Audit log viewer

5. Press **Save results** to save the log file.
The **Save as** window is displayed.
6. Select the destination media and enter a file name.
7. Press **Save** to save the log file.

5.3.4 System tab

5.3.4.1 General System Settings

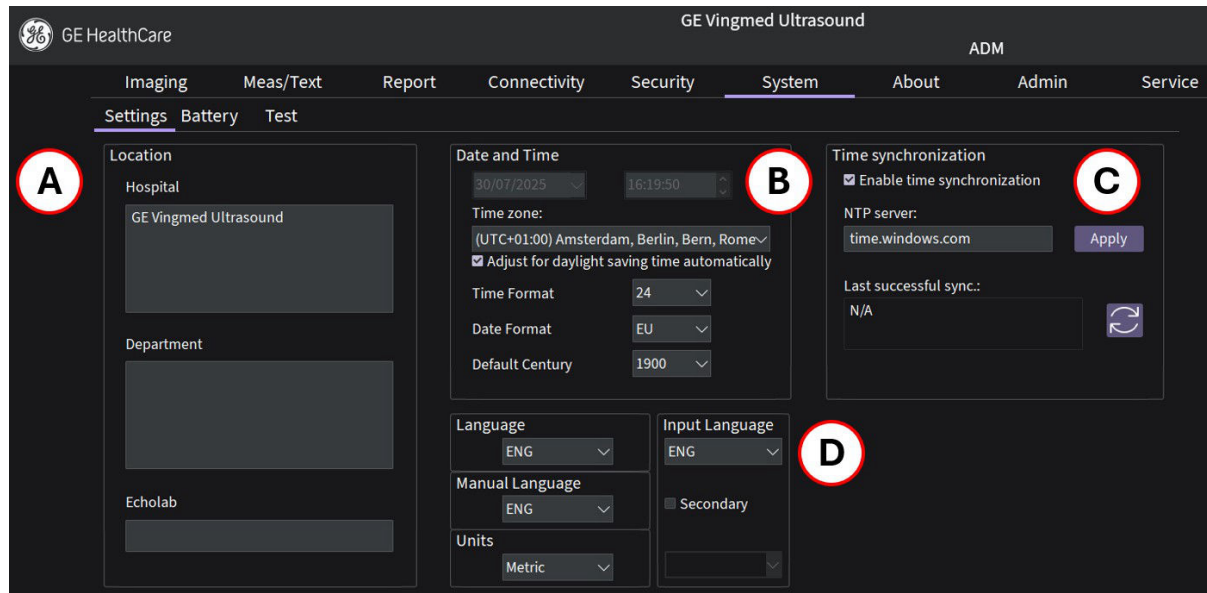


Figure 5-81 The Settings sheet

1. Press **Utility > Config** on the Touch panel and log on as administrator if required.
2. Select the **System** category and **Settings** subgroup.
The **Settings** sheet is displayed.

NOTE

Changes will be effective only after rebooting the system.

Location (A)

1. **Hospital:** Enter the hospital name. This information is displayed on the scanning screen's Title bar and on the image properties of all saved images.
2. **Department:** Enter the department name. This information is displayed on the image properties of all saved images.

Date and Time (B)

1. **Date:** Select the correct date from the pop-up window.
(Not editable if Time synchronization is enabled.)
2. **Time:** Select either hour, minute or second, and then press the arrowhead buttons to set the time.
(Not editable if Time synchronization is enabled.)
3. **Time zone:** Select the correct time zone from the drop-down menu.
4. **Adjust for daylight saving time automatically:** Check to automatically adjust for daylight saving time. (Only available for some time zones.)
5. **Time Format:** Select the desired format (24 or 12 AM/PM) from the drop-down menu.
6. **Date Format:** Select the desired format (EU or US) from the drop-down menu.

Time Synchronization (C)

1. **Enable time synchronization:** Check to synchronize the system clock with an external time synchronization source.
Uncheck to set the date and time manually.
2. **NTP server:** The URL for time synchronization source. Press **Apply** to validate the connection to the entered URL.
Default server: time.windows.com
3. Press the **Synchronize** button, see table below, to synchronize immediately with selected time source. The button indicates synchronization status:

	The Synchronize button
	Synchronization is in progress
	Synchronization succeeded
	Synchronization failed

4. **Last successful sync.** displays the date, time & source of the last time a successful time synchronization was performed.

NOTE

If the time and date is more than 24 hours off when enabling time synchronization, the synchronization might fail. To avoid this, adjust the date and time manually before enabling time synchronization.

Language and Units (D)

- **Language:** Select the desired language for the system from the drop-down menu.
- **Manual Language:**
 - The user manuals are delivered with the system on a manual media (USB Flash Drive). To be able to access them on the ultrasound system, they must first be installed and then the requested language must be selected.
 - Select the desired language for the Online manual from the drop-down menu. If not available, the English manual will be displayed as default.
- **Input Language:** Select the default alphanumeric keyboard language configuration on the Touch panel.
- **Units:** Select the desired units (Metric or US) from the drop-down menu.

5.3.4.2 Battery Settings

Shipping and Long-term Storage

Prior to shipping, the absolute charge level of the system battery must be less than 30%

Prior to long-term storage, the absolute charge level of the system battery must be between 25% and 30%.

To verify if the system is currently ready for shipping or long-term storage, navigate to **Config > System > Battery** and select **Shipping / storage** charging mode.

When **Shipping / storage** charging mode is selected, information boxes indicate whether the system is ready for shipping or long term storage. On the same page, the absolute battery charge level is indicated per battery pack.

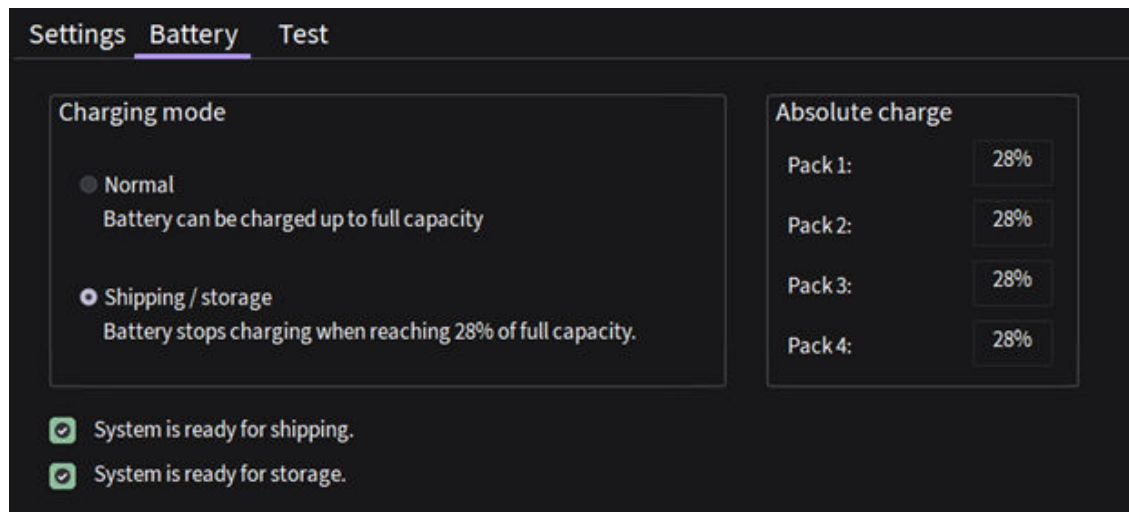


Figure 5-82 **Config > System > Battery** page with **Shipping / storage** charging mode.

When **Shipping / storage** charging mode is selected, the battery will not charge above 28%. For the system to be ready for shipping or long-term storage, the user can perform the following sequence:

1. Make sure the system is powered with AC.
2. Navigate to **Config > System > Battery**.
3. Select **Shipping / storage** charging mode.
4. Disconnect the AC cable and let the battery discharge until shutdown.
5. Reconnect the AC cable and let it charge to above 25%, if it is below that level.

Charging stops when the charge level reaches 28%, which is appropriate for shipping or long-term storage.

When **Shipping / storage** charging mode is selected, the battery status bare icon includes a warning indicator and the tooltip states that **Charging mode is set to shipping / storage**.

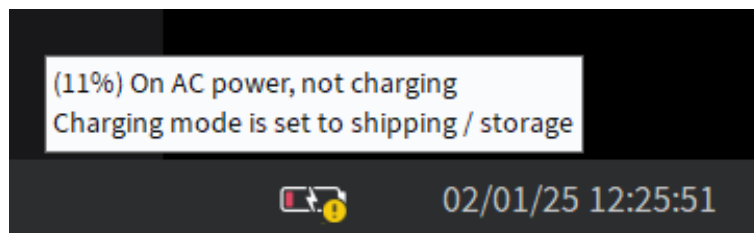


Figure 5-83 **Battery icon and tooltip** when **Shipping /storage** mode is set

Upon reception of the shipped system, or after long-term storage, the **Shipping / storage** charging mode must be unselected:

1. Make sure the system is powered with AC.
2. Navigate to **Config > System > Battery**.
3. Select **Normal** charging mode.
4. Let the battery charge fully.

5.3.5 About tab

The **About** tab gives you access to the following system information:

- System Version
- SW version

- Firmware Version
- HW Version
- Probes
- Battery

5.3.6 Admin tab

The **Admin** tab gives you access to the following data management and licenses information:

- Backup
- Restore
- Licenses
- Activity report

5.3.6.1 Archiving and Loading Presets

Archive Presets to Removable Media

1. Select **Utility > Config > Admin > Backup**
2. Select (check) **System Configuration**.
3. Select the removable media from the **Destination Device** pulldown menu.
4. Click **Start backup** to start the archiving of the presets-

Loading Presets from Removable Media

1. **Utility > Config > Admin > Restore**
2. Select (check) **System Configuration** to select all the presets.
3. Click **Restore now**.

5.3.6.1.1 Data Backup and restore

The Backup/Restore function enables the user to:

- Copy/Restore the patient archive.
- Copy/Restore the system configuration. The Copy/Restore system configuration feature enables the user to configure several units with identical presets, providing that the units have the same software version.

To minimize accidental loss of data, perform backup of the patient archive stored on the local harddrive at least **once a week**.



WARNING

GE HealthCare is not responsible for lost data if the suggested backup procedures are not followed and will not aid in the recovery of lost data.

There is no backup function for the images or reports (no creation of a safety copy). For long-term storage, images and reports should be moved to a USB HD or to a network shared folder using the Disk management procedure (see [Disk management](#) on page 174).



CAUTION

DO NOT use the local harddrive for long-term image storage.

Only users with administration rights have access to the backup/restore function.

Backup procedure

1. Press **Patient**.
2. In the **Archive** screen, select the dataflow **Local Archive - Int. HD**.

3. Press **Utility > Config** on the Touch panel.
4. Select the category **Admin**.
5. Select the **Backup** sheet.

Data to backup	Result	Last successful backup
☒ Patient Archive ☐ Validate Patient Archive		No record
☒ System Configuration		No record

Destination Device: USB HD/Memstick (20220922_01) (G:) ▼

Local Path: G:\

Start backup

Go to Remote Path tab to set up credentials if needed:

Remote Path

Figure 5-84 The Backup sheet

6. In the **Backup** sheet, select as needed:
 - **Patient archive** to backup the patient records.
 - **System configuration** to copy system settings and user presets.
7. Select a removable media or a shared network folder as destination.

NOTE

To be able to select a network share folder, the path (of type: \\server-name\share-name) must have been entered in the **Remote Path** field.

8. If the backup is done to a removable media, insert a dedicated media in the drive.
9. Select **Start backup**.

The following situations may occur:

 - The system is checking that the removable media is inserted. If not, a dialogue window is displayed prompting the user to insert a media.

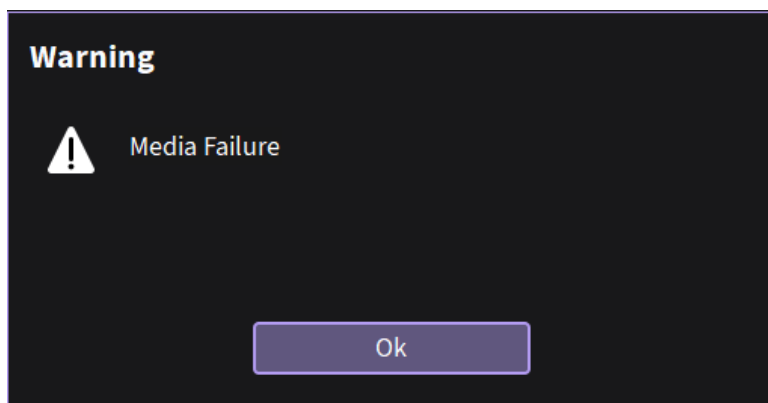


Figure 5-85 The Insert media window

Insert the media and select **OK**.

- If using a CD/DVD, the system is checking if the media needs to be formatted. If yes, a dialogue window is displayed prompting the user to enter a media label.

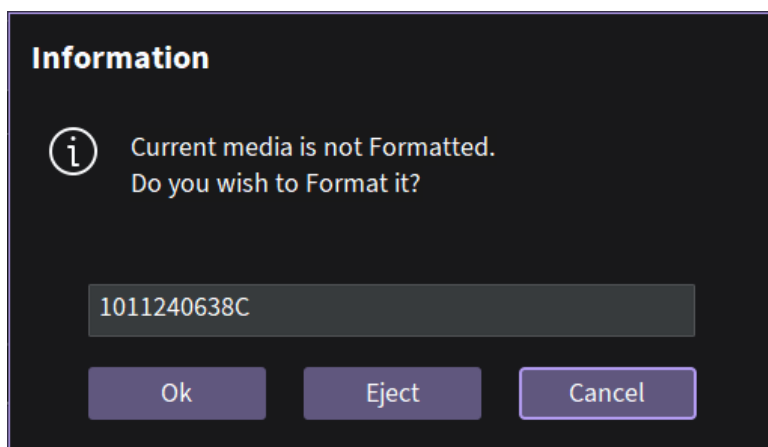


Figure 5-86 The Enter media label window

Type in a label for the media and select **OK**.

NOTE

Only the following characters and signs can be used when labeling a media: A - Z, a - z, 0 - 9, “ _ ” and “ - ”. Do not use more than 11 characters or signs. Do not use space.

NOTE

Select **Eject** to perform the backup using another removable media. Select **Cancel** to stop the backup operation.

- The system is checking if there is already a backup or a Disk management copy on the media. If the following error message is displayed, the disk is ejected and the user is asked to use a new media that does not contain any backup or Disk management data.